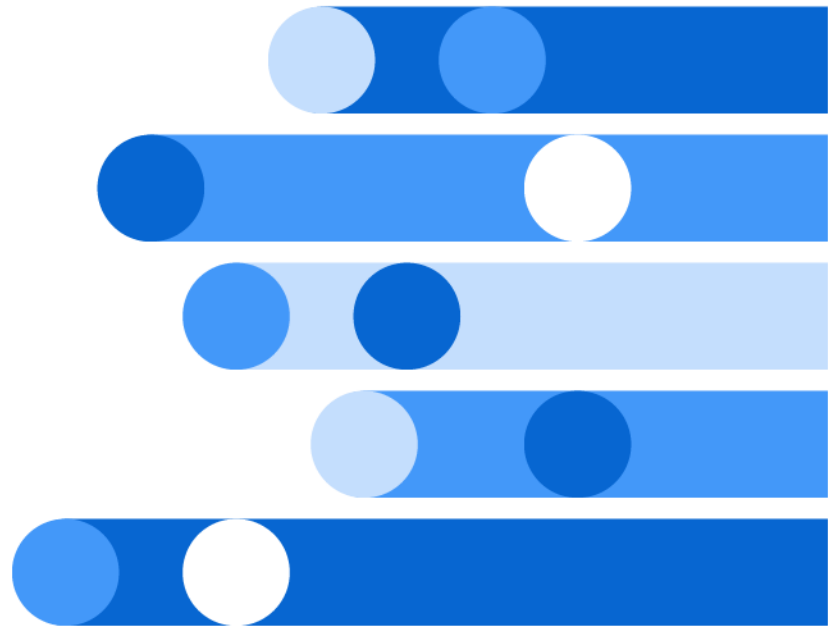




SAS[®] Viya[®]: Machine Learning User's Guide

2026.04 - 2026.06*



* This document might apply to additional versions of the software. Open this document in [SAS Help Center](#) and click on the version in the banner to see all available versions.



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SAS® Viya®: Machine Learning User's Guide

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Overview

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About SAS Viya: Machine Learning

SAS Viya Machine Learning enables you to create predictive models that utilize machine learning and data mining techniques. Pipelines consist of nodes, where each node represents either a data preprocessing technique, a statistical modeling method, or a machine learning method. The examples in the following sections use several of these nodes and explore scenarios including risk modeling applications, open source code, and integration with SAS Visual Analytics. For more information about each node, see [SAS Viya: Machine Learning Node Reference](#).

Installing SAS Viya as a Progressive Web App

You can install SAS Viya as a Progressive Web App (PWA). The two primary benefits of using a PWA are session persistence and service worker caching.

- When you access SAS Viya as a PWA from the Windows Start menu, your session will never time out. When you access SAS Viya through a web browser, your session will time out after a period of inactivity and you will be logged out.
- When installed as a PWA, SAS Viya initializes faster. The PWA can use the service worker cache, which is separate from the browser cache. Using the service worker cache reduces the number of cold starts needed for SAS Viya Machine Learning to initialize.

To install SAS Viya Machine Learning as a PWA:

- 1 Sign into SAS Viya.
- 2 Open SAS Viya in a Chromium-based web browser.
- 3 Open the web browser's menu and select **Install SAS**. If you are using the Microsoft Edge browser, select **Apps** ⇒ **Install SAS** from the web browser's menu.

SAS Viya is now installed as a desktop program named SAS. After one web application has been installed as a PWA, all other SAS applications are accessible in the PWA.

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Charitable Giving Example

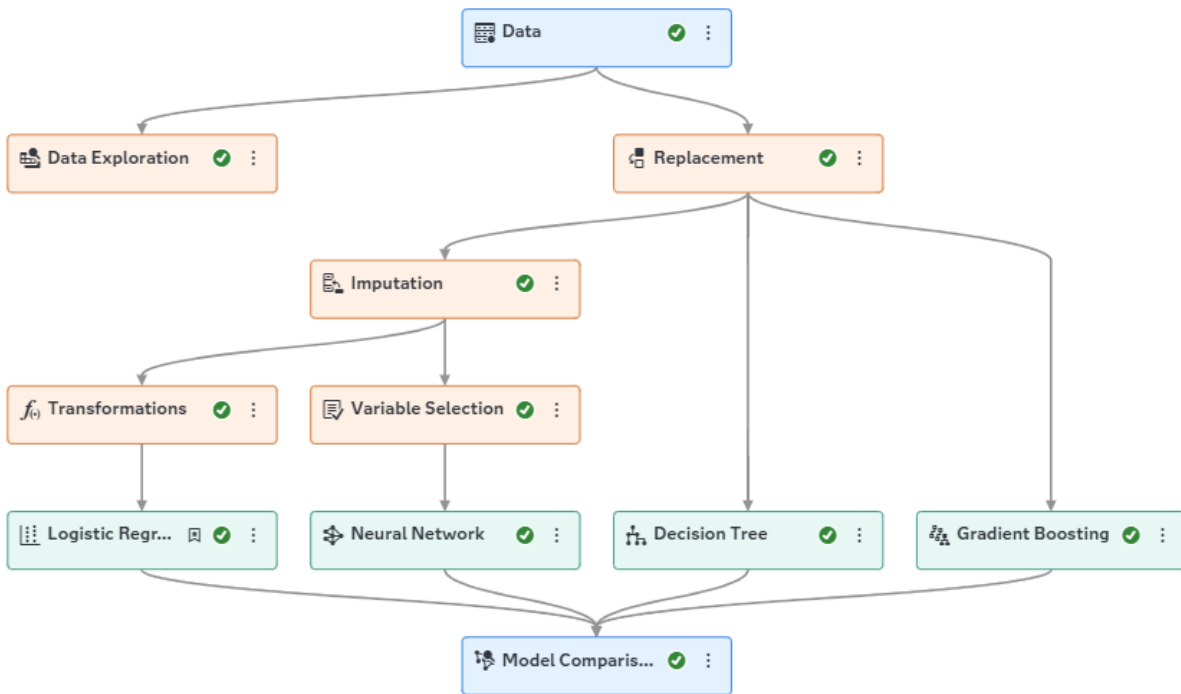
Tutorial Scenario

The Charitable Giving tutorial is intended for new SAS Model Studio users. Although prior data mining experience is beneficial, any user can follow the discussion and complete the steps. The tutorial defines the problem, explores and visualizes the input data, performs data preparation, specifies model fit criteria, and then creates, configures, and trains multiple competing statistical modeling algorithms. A champion model is selected, and then score code is generated. The score code performs the trained champion model's analytic task on new data.

The narrative requires sequential data mining steps. Follow the steps in the order of presentation so that your results match the example. Later, you can modify your final pipeline model settings and add new competing models to see how your changes would affect predicted results.

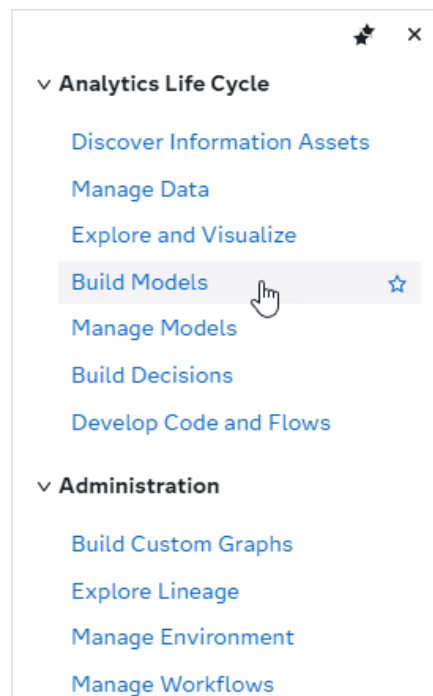
The tutorial uses a training data set with known target variable values named DONOR_DATA. You can download DONOR_DATA from the [SAS Viya Machine Learning product page](#). Click **Example Data for Getting Started with SAS Viya Machine Learning in SAS Model Studio**. Download the ZIP file and extract DONOR_DATA.SAS7BDAT to a directory that your SAS Viya Machine Learning server can access.

In the tutorial, you are a data analyst at a national charitable organization. Your organization wants to use the results of a previous solicitation for donations to better target its next one. You want to determine which of the individuals in your mailing database are the most generous donors. By approaching only these people, your organization can spend less money on the solicitation effort and more money on charitable concerns. The finished pipeline diagram will resemble the one shown here:



Create the Project and Import the Input Data

This example assumes that you are signed in to the Welcome to SAS Viya application. In the upper left corner of the Welcome to SAS Viya page, click  and select **Build Models**.



You are directed to the Projects page. To create the project that you will use in this example:

- 1 Click **New Project** in the upper right corner of the Projects page.
- 2 Enter **Charitable Giving Example** for **Name** in the New Project window.

New Project [X]

Name: *
Charitable Giving Example

Type: *
Data Mining and Machine Learning ▼

Template:
Blank template ▼ Browse

Data: *
Browse

Location:
My Folder ▼ Browse

Description:

Advanced

Save Cancel

- 3 Select **Data Mining and Machine Learning** for **Type**.

Note: **Forecasting** and **Text Analytics** are additional options if you have licensed those products.

- 4 Ensure that **Blank template** is specified for **Template**.

- 5 In the **Data** field, click **Browse**. The Choose Data window appears.
- 6 In the upper right corner of the Choose Data window, click **Import data**. The Import Data window appears.
- 7 In the left pane of the Import Data window, click **Local files** and navigate to the folder where DONOR_DATA is stored. Select **DONOR_DATA.sas7bdat** and click **Open**.
- 8 In the Import Data window, click **Import item** in the upper right corner. When the data is successfully imported, a note appears, saying that the data is ready for use.

 The table was successfully imported on Oct 7, 2019 10:33 AM and is ready for use.

- 9 Once the data set is successfully imported, click **Add** in the lower right corner of the Import Data window. This returns you to the Choose Data window.
- 10 In the Choose Data window, select **DONOR_DATA** in the list of available tables and click **OK**. This brings you back to the New Project window.
- 11 In the New Project window, click **Save**. This directs you to the **Data** tab, where you can modify the variables in your data set.

Modify Variables

The **Data** tab shows the properties of the variables in the input data. To modify the properties of a variable, select the variable by clicking the check box to the left of the variable name, and then specify the new properties in the Options pane. Changes that you make to each variable are automatically applied and saved.

CAUTION

To avoid making unwanted changes to variable properties, you must manually deselect each variable that you modify when you are finished making changes to its properties.

For this example, modify the variables as follows:

- 1 On the Data tab, note the role for each variable in the **Role** column. Ensure that the variables roles are set to the following values:
 - CLUSTER_CODE is set to **Rejected**
 - CONTROL_NUMBER is set to **ID**
 - TARGET_B is set to **Target**
 - TARGET_D is set to **Rejected**
 - _dmIndex_ is set to **Key**.
 - _PARTIND_ is set to **Partition**.

In data mining, a strategy for assessing the quality of model generalization is to partition the data source. A portion of the data, called the *training data set*, is used for preliminary model fitting. The rest is reserved for empirical validation and is often split into two parts: validation data and test data. The validation data set is used to prevent a modeling node from overfitting the training data and to compare models. The test data set is used for a final assessment of the model. In this example, the data has been pre-partitioned where 60% of the data is assigned for training, 30% of the data is assigned for validation, and 10% of the data is assigned for testing.

- All other variable roles are set to **Input**.

If you need to change a variable role, do the following:

- a Ensure that no variables are currently selected.
 - b Select the variable that you want change by clicking the check box to the left of the variable name.
 - c In the Options pane, expand **Role** and select the new variable role.
- 2 Select **_PARTIND_** and click **Map Partition Levels** on the right side of the window. Ensure that the partition levels are set to the following values:
- **Training Level** is set to **1**.
 - **Validation Level** is set to **0**.
 - **Test Level** is set to **2**.
 -

The observations of your data are assigned to a partition based on the values shown.

Click **Cancel**.

- 3 Select the following variables:
 - FILE_AVG_GIFT
 - LAST_GIFT_AMT
 - LIFETIME_AVG_GIFT_AMT

In the options pane, set the property **Transform** to **Log10** for these variables.

TIP It is possible to select multiple variables for editing at one time by selecting the box in front of each variable name. By simultaneously selecting FILE_AVG_GIFT, LAST_GIFT_AMT, and LIFETIME_AVG_GIFT_AMT, you can change the **Transform** property for all three variables at once.

Create a Pipeline

- 1 Select the **Pipelines** tab in the upper left corner.
- 2 Right-click the **Data** node and select **Run**.
- 3 Once the **Data** node has run successfully, continue with the following sections to build your pipeline.

Generate Descriptive Statistics

To see a statistical summary of the input data:

- 1 Right-click the **Data** node and select **Add child node** ⇒ **Miscellaneous** ⇒ **Data Exploration**.
- 2 Right-click the **Data Exploration** node and select **Run**.
- 3 Once the pipeline has run successfully, right-click the **Data Exploration** node and select **Results** from the menu that appears. The following pieces of information about the input data appear either as graphic tiles or rows in a table of projects:
 - **Data Partition Summary**
 - **Important Inputs**
 - **Class Variable Summaries**
 - **Class Variable Distributions**
 - **Interval Variable Moments**
 - **Interval Variable Summaries**
 - **Interval Variable Distributions**
 - **Missing Values**
 - **Target by Input Correlations**
 - **Target by Input Crosstabulations**
 - **Pearson Correlations (Absolute Value)**
 - **Input Correlations**
 - **Properties**
 - **Output**
- 4 Click **Close** to return to the pipeline.

Replace Missing Values

In this example, the variables SES and URBANICITY are class variables for which the value ? denotes a missing value. SAS denotes a missing value using a blank for character variables and a period for numeric variables. Because this data uses a question mark instead, SAS Model Studio sees it as an additional level of a class variable. However, the knowledge that these values are missing will be useful later in the model-building process. To implement a **Replacement** node:

- 1 Right-click the **Data** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Replacement**.
- 2 Once created, select the **Replacement** node.
- 3 In the options pane, complete the following:
 - a Set **Replacement value for unknown class levels** to **Missing value**.
 - b Expand **Interval Inputs**.
 - c Ensure that the **Default limits method** property is set to **Standard deviation from the mean**.
- 4 Right-click the **Replacement** node and select **Run**.
- 5 Once the node has run successfully, right-click the node and select **Results** to view detailed information in each of the following sections of interest:
 - **Class Variables** — Displays the replacement values for class variables.

Name	Variable Label	Replace Variable	Type
CARD_PROM_12		REP_CARD_PROM_12	
DONOR_GENDER		REP_DONOR_GENDER	
FREQUENCY_STATUS_97NK		REP_FREQUENCY_STATUS_97NK	
HOME_OWNER		REP_HOME_OWNER	
INCOME_GROUP		REP_INCOME_GROUP	
IN_HOUSE		REP_IN_HOUSE	
OVERLAY_SOURCE		REP_OVERLAY_SOURCE	
PEP_STAR		REP_PEP_STAR	
SUBSIDIES_SUM		REP_SUBSIDIES_SUM	

- **Interval Variables** — Displays the replacement values for interval variables.

Name	Variable Label	Replace Variable	Limits
DONOR_AGE		REP_DONOR_AGE	STD
FILE_AVG_GIFT		REP_FILE_AVG_GIFT	STD
FILE_CARD_GIFT		REP_FILE_CARD_GIFT	STD
LAST_GIFT_AMT		REP_LAST_GIFT_AMT	STD
LIFETIME_AVG_GIFT_AMT		REP_LIFETIME_AVG_GIFT_AMT	STD
LIFETIME_CARD_PROM		REP_LIFETIME_CARD_PROM	STD
LIFETIME_GIFT_AMOUNT		REP_LIFETIME_GIFT_AMOUNT	STD
LIFETIME_GIFT_COUNT		REP_LIFETIME_GIFT_COUNT	STD
LIFETIME_GIFT_RANGE		REP_LIFETIME_GIFT_RANGE	STD

- **Replacement Counts** — Displays the number of observations replaced for each variable.

Name	Variable Label	Train	Role	Variable Level
CARD_PROM_12		0	INPUT	NOMINAL
DONOR_AGE		59	INPUT	INTERVAL
DONOR_GENDER		0	INPUT	NOMINAL
FILE_AVG_GIFT		143	INPUT	INTERVAL
FILE_CARD_GIFT		148	INPUT	INTERVAL
FREQUENCY_STATUS_97NK		0	INPUT	NOMINAL
HOME_OWNER		0	INPUT	BINARY
INCOME_GROUP		0	INPUT	NOMINAL
INCOME_YEAR		0	INPUT	BINARY

- 6 Click **Close** to close the Results window.

Automatically Train and Prune a Decision Tree

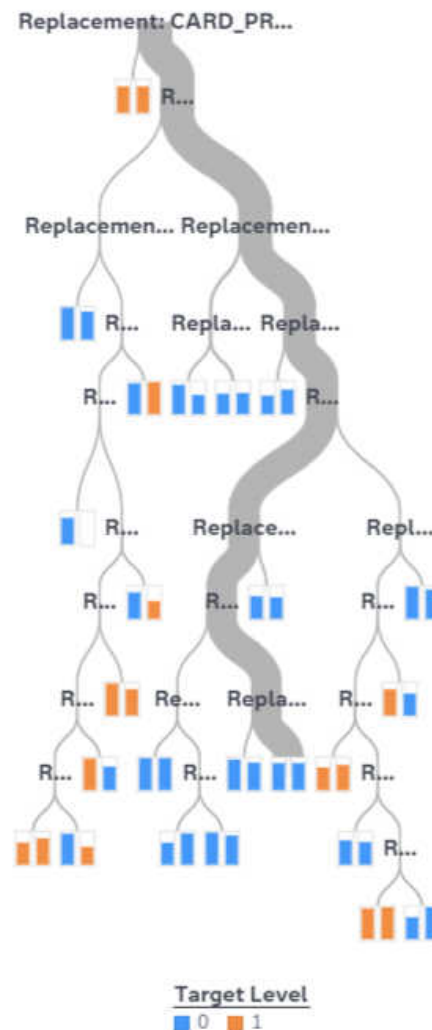
Decision tree models are advantageous because they are conceptually easy to understand, yet they readily accommodate nonlinear associations between input variables and one or more target variables. They also handle missing values without the need for imputation. Therefore, you decide to first model the data using decision trees. You will compare decision tree models to other models later in this example.

Note: A **Model Comparison** node is automatically created when you add a **Supervised Learning** node to your pipeline. Therefore, a **Model Comparison** node will be added to your pipeline when you add the **Decision Tree** node.

To insert a **Decision Tree** node:

- 1 Right-click the **Replacement** node and select **Add child node** ⇒ **Supervised Learning** ⇒ **Decision Tree**.
- 2 Once created, select the **Decision Tree** node.

- 3 In the options pane, complete the following:
 - a Expand **Splitting Options**.
 - b Set **Minimum leaf size** to **8**.
 - c Expand **Pruning Options**.
 - d Set **Subtree method** to **Reduced error**.
- 4 Right-click the **Decision Tree** node and select **Run**.
- 5 Right-click the **Decision Tree** node and select **Results** to view components such as the following:
 - **Tree Diagram** — Displays the decision tree. The line width of the tree is proportionally given by the ratio of the number of observations in the branch to the number of observations in the root node. When you move your pointer over a node of a tree, a text box displays information about the node. Zoom in to display additional details about each node.

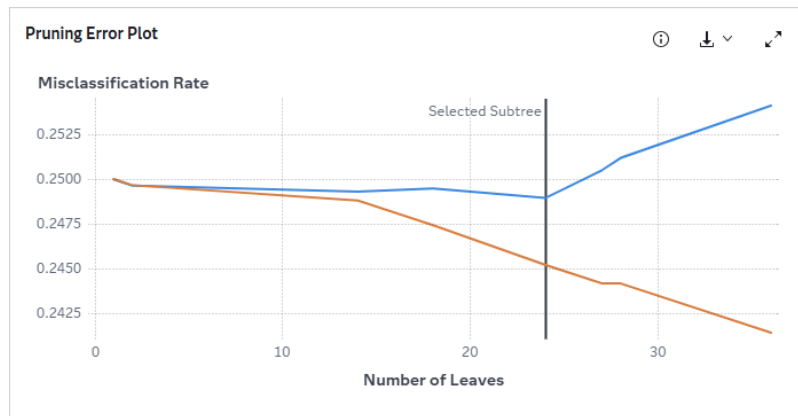


- **Treemap** — The rectangular regions that represent the tree nodes are colored by probability of event or proportion correctly classified for a class target or the average of an interval target. The node width is proportional to

the number of observations in the node. Select a node to display additional details about that node.



- **Pruning Error Plot** — An explanation of this plot is provided.



- **Fit Statistics** — Located on the **Assessment** tab.

Target Name	Data Role	Partition Indicator	Formatted Partition	N
TARGET_B	TEST	2	2	
TARGET_B	TRAIN	1	1	
TARGET_B	VALIDATE	0	0	

- Click **Close** to close the Results window.

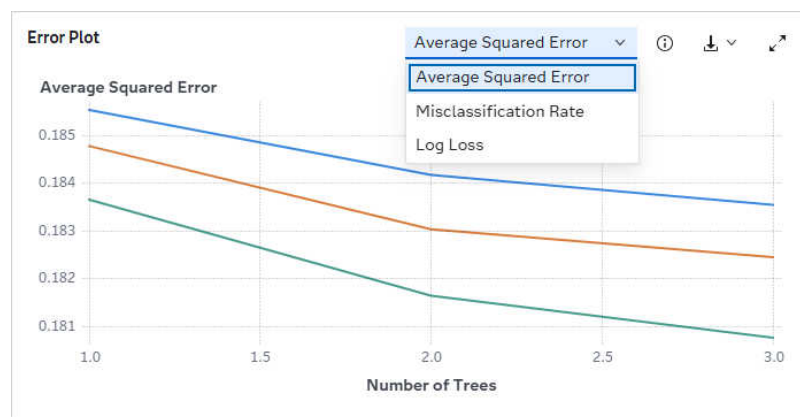
Create a Gradient Boosting Model

The Gradient Boosting node uses a partitioning algorithm to search for an optimal partition of the data for a single target variable. Gradient boosting is an approach that resamples the analysis data several times to generate results that form a weighted average of the resampled data set. Tree boosting creates a series of decision trees that form a single predictive model. Like decision trees, boosting makes no assumptions about the distribution of the data. Boosting is less prone to overfit the data than a single decision tree. If a decision tree fits the data fairly well, then boosting often improves the fit. For more information, see [Overview of Gradient Boosting](#).

To create a gradient boosting model of the data:

- 1 Right-click the **Replacement** node and select **Add child node** ⇒ **Supervised Learning** ⇒ **Gradient Boosting**.
- 2 Once created, select the **Gradient Boosting** node.
- 3 In the options pane, expand **Tree-splitting Options**. Set **Maximum depth** to **10**.
- 4 Enable the **Perform Autotuning** option. Autotuning automatically adjusts the hyperparameters that govern the model training process to the best values. For more information, see [Overview of Autotuning](#).
- 5 Right-click the **Gradient Boosting** node and select **Run**.
- 6 Right-click the **Gradient Boosting** node and select **Results** to view components such as the following:

■ Error Plot



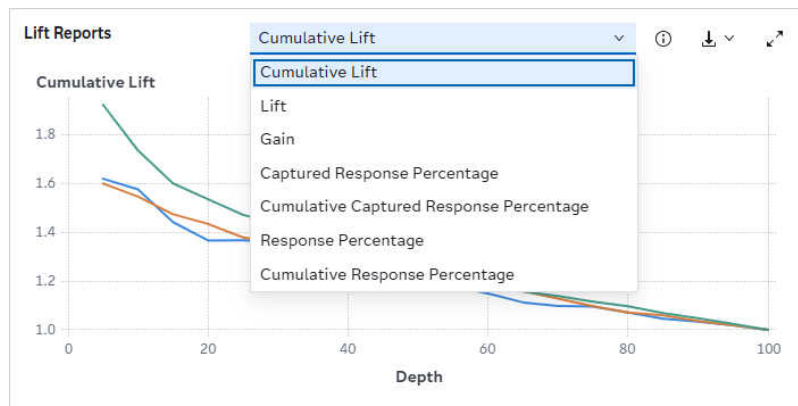
■ Variable Importance

Variable Importance				
Variable Label	↑	Role	Variable Name	Training Importance
Replacement: FILE_AVG_GIFT		INPUT	REP_FILE_AVG_GIFT	5.4107
Replacement: FILE_CARD_GIFT		INPUT	REP_FILE_CARD_GIFT	2.7296
Replacement: FREQUENCY_STATUS_97NK		INPUT	REP_FREQUENCY_STATUS_97NK	8.4270
Replacement: INCOME_GROUP		INPUT	REP_INCOME_GROUP	2.7286
Replacement:		INPUT	REP_MEDIAN_HO	2.1298

■ Autotune Best Configuration

Autotune Best Configuration	
Parameter	Value
Evaluation	4
Number of Variables to Try	36
Learning Rate	0.56
Sampling Rate	0.5
Lasso	3.33333333
Ridge	10
Number of Bins	100
Maximum Tree Levels	4
Kolmogorov-Smirnov	0.1828401009

■ Lift Reports — Located on the **Assessment** tab.



Because of the nondeterministic behavior of SAS Viya, your results might not match these results.

- 7 Click **Close** to close the Results window.

Impute Missing Values

For decision trees, missing values are not problematic. Surrogate splitting rules enable you to use the values of other input variables to perform a split for observations with missing values. In SAS Model Studio, however, models such as regressions and neural networks ignore observations that contain missing values, which reduces the size of the training data set. Less training data can substantially weaken the predictive power of these models. To overcome this obstacle of missing data, you can impute missing values before you fit the models.

TIP Impute missing values before fitting a model that ignores observations with missing values if you plan to compare those models with a decision tree. Model comparison is most appropriate between models that are fit with the same set of observations.

To impute missing values:

- 1 Right-click the **Replacement** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Imputation**.
- 2 Once created, select the **Imputation** node.
- 3 In the options pane, under **Interval Inputs**, set **Default method** to **Median**.
- 4 Right-click the **Imputation** node and select **Run**.
- 5 Once the **Imputation** node has successfully run, right-click the node and select **Results**. Explore the following:
 - **Input Variable Statistics**

Input Variable Statistics				
Input Variable	Variable ...	Number of Miss...	Percent Missing	Imputab
REP_CARD_PROM_12	NOMINAL	0	0	
REP_DONOR_AGE	INTERVAL	2,925	25.1656	
REP_DONOR_GENDE R	NOMINAL	0	0	
REP_FILE_AVG_GIFT	INTERVAL	0	0	
REP_FILE_CARD_GIFT	INTERVAL	0	0	
REP_FREQUENCY_ST ATUS_97NK	NOMINAL	0	0	
REP_HOME_OWNED	BINARY	0	0	

■ Imputed Variables Summary

Imputed Variable	Method	Input Variable	Value	Numeric Value
IMP_REP_DONOR_AGE	MEDIAN	REP_DONOR_AGE		
IMP_REP_INCOME_GROUP	COUNT	REP_INCOME_GROUP		
IMP_REP_MONTHS_SINCE_LAST_PROM_R	MEDIAN	REP_MONTHS_SINCE_LAST_PROM_RESP		
IMP_REP_WEALTH_RATING	COUNT	REP_WEALTH_RATING		

- 6 Click **Close** to close the Results window.

Transform Variables

At the beginning of this example in “[Charitable Giving Example](#)” on page 4, you were instructed to set property **Transform** to **Log10** for variables FILE_AVG_GIFT, LAST_GIFT_AMT, and LIFETIME_AVG_GIFT_AMT. This enables the transformation of these variables by using the **Log10** methodology. To execute the transformation of these variables:

- 1 Right-click the **Imputation** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Transformations**.
- 2 Right-click the **Transformations** node and select **Run**.

Create a Logistic Regression

As part of your analysis, you want to include some parametric models for comparison with the decision trees that you built earlier in this example. Because it is familiar to the management of your organization, you have decided to include a logistic regression as one of the parametric models. To do so:

- 1 Right-click the **Transformation** node and select **Add child node** ⇒ **Supervised Learning** ⇒ **Logistic Regression**.

- 2 Right-click the **Logistic Regression** node and select **Run**.
- 3 Once the node has successfully run, right-click the **Logistic Regression** node and select **Results**. Explore the following:

■ **Parameter Estimates**

Parameter Estimates					
Effect	Parameter	↑ t Value	Sign	Estimate	A
REP_PEP_STAR	REP_PEP_STAR 1	.	+	0	
REP_FREQUENCY_ STATUS_97NK	REP_FREQUENCY_ STATUS_97NK 4	.	+	0	
Intercept	Intercept	0.9169	-	-0.1947	
REP_FREQUENCY_ STATUS_97NK	REP_FREQUENCY_ STATUS_97NK 3	2.2488	-	-0.1664	
REP_NUMBER_PR OM_12	REP_NUMBER_PR OM_12	3.5180	-	-0.0231	
REP MONTHS SIN	REP MONTHS SIN	3.9328	+	0.0028	

■ **Regression Fit Statistics**

Regression Fit Statistics				
Statistic	Description	Training	Validation	Testing
M2LL	-2 Log Likelihood	12,708.6535	6,342.1266	2,113.7133
AIC	AIC (smaller is better)	12,728.6535	6,362.1266	2,133.7133
AICC	AICC (smaller is better)	12,728.6725	6,362.1645	2,133.8275
SBC	SBC (smaller is better)	12,802.2609	6,428.8017	2,189.4022
ASE	Average Square Error	0.1814	0.1813	0.1809

- **Fit Statistics** – Located on the **Assessment** tab.

Fit Statistics				
Target Name	Data Role	Partition Indicator	Formatted Partition	Numb
TARGET_B	TEST	2	2	
TARGET_B	TRAIN	1	1	
TARGET_B	VALIDATE	0	0	

- Click **Close** to close the Results window.

Create a Neural Network

A neural network model is a stronger predictive model than the other models that you have generated up to this point. Neural networks are a class of parametric models that can accommodate a wider variety of nonlinear relationships between a set of predictors and a target variable than can logistic regression.

You know that the management of your organization prefers a stronger predictive model. You choose to add a neural network model even though it will be more complicated to explain than a regression model or a decision tree.

Building a neural network model involves two main phases. First, define the network configuration. You can think of this step as defining the structure of the model that you want to use. Then, iteratively train the model. The **Neural Network** node trains a specific neural network configuration, and is best used when you know a lot about the structure of the model that you want to define.

Before creating a neural network, you will reduce the number of input variables with the **Variable Selection** node. Performing variable selection reduces the number of input variables and saves computer resources. To use the **Variable Selection** node followed by the **Neural Network** node:

- Right-click the **Imputation** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Variable Selection**.
- Once created, right-click the **Variable Selection** node and select **Run**.
- Right-click the **Variable Selection** node and select **Add child node** ⇒ **Supervised Learning** ⇒ **Neural Network**.

- 4 Once created, select the **Neural Network** node.
- 5 Take these actions in the options pane:
 - a Set **Number of hidden layers** to **5**.
 - b Expand **Target Layer Options** and select **Direct connections**.

▼ Target Layer Options

☒ Direct connections

Interval target standardization:

Midrange ▼

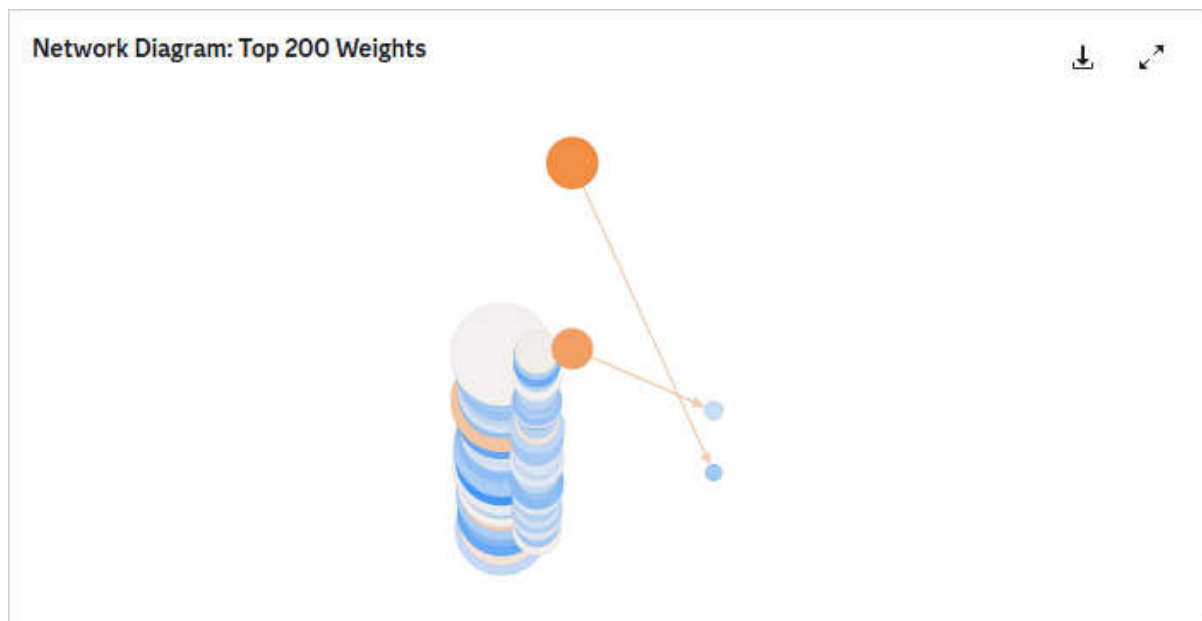
Interval target error function:

Normal ▼

Interval target activation function:

Identity ▼

- 6 Right-click the **Neural Network** node and select **Run**.
- 7 Once the **Neural Network** node has successfully run, right-click the node and select **Results** to view components such as the **Network Diagram**.



- 8 Click **Close** to close the Results window.

Compare Models

To use the **Model Comparison** node to compare the models that you have built in this example and to select one as the champion model:

- 1 Right-click the **Model Comparison** node that was created when you first created the **Decision Tree** node and select **Run**.
- 2 Right-click the **Model Comparison** node and select **Results**.
- 3 In the **Model Comparison** pane, you can see that the **Logistic Regression** model is selected as the champion model. In the Model Comparison node, SAS Model Studio selects the champion model based on the value of a single statistic. You can specify which statistic to use for selection in the options pane. Because you did not change the value of this property, the default statistic (Kolmogorov-Smirnov statistic) was used.

Note: The selection statistic can also be specified in the **Rules** section within the Project Settings window. To edit the project settings, click  in the upper right corner of the window and click **Project settings**. Modifications to the project settings are applied to all applicable areas throughout the project.

- 4 Click **Close** to close the Results window.

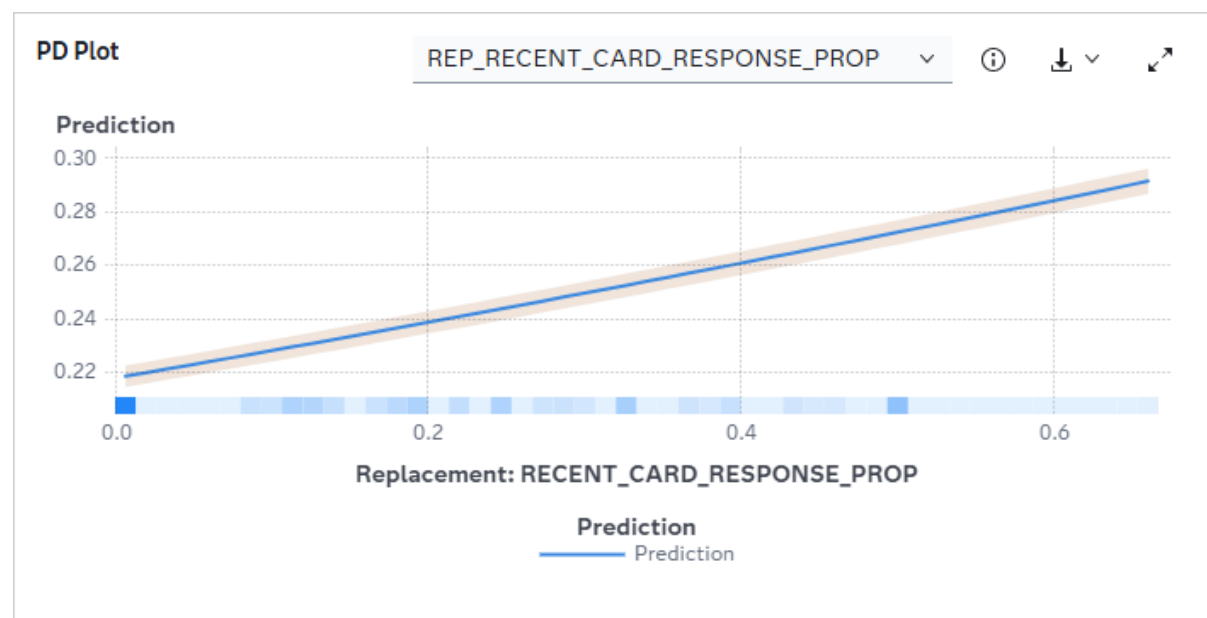
Interpretability of Champion Model

Now that you have selected the Logistic Regression model as the Champion Model, you want to bring to management more information about the interpretation of the model. There are several model interpretability plots and reports that can be generated. More details about the model interpretability plots available in SAS Model Studio can be found in [Model Interpretability Plots](#).

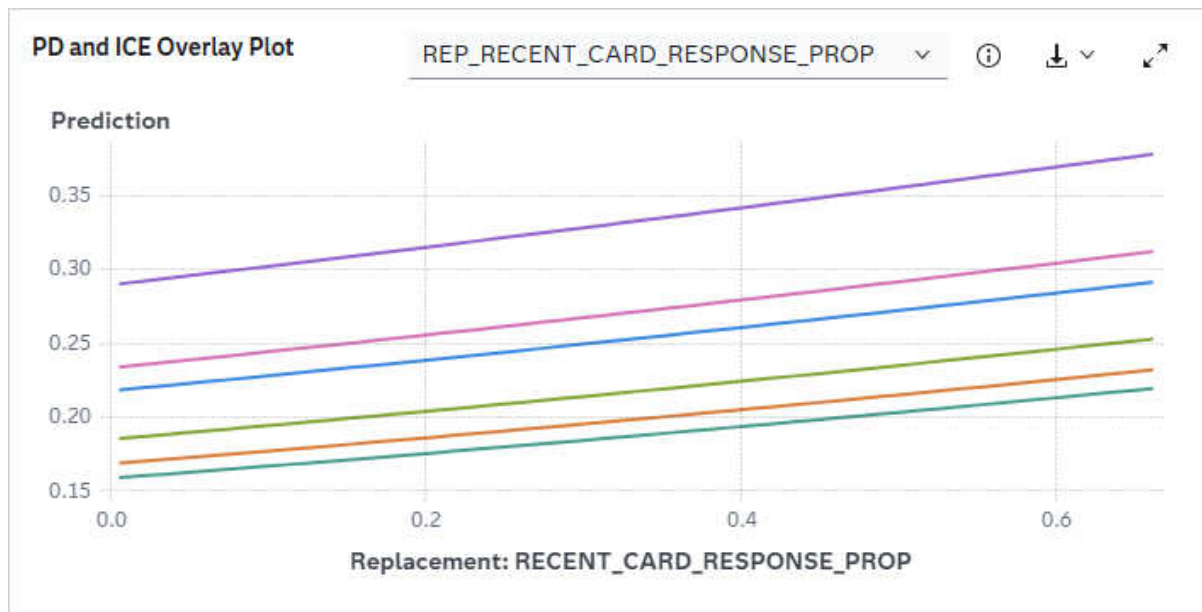
- 1 Click the **Logistic Regression** node.
- 2 In the options pane, expand the **Global Interpretability** options and select **Variable Importance** and **PD plots**.
- 3 Expand the **Local Interpretability** options and select **ICE plots** and **LIME**.
- 4 Right-click the **Logistic Regression** node and select **Run**.
- 5 Once the **Logistic Regression** node has successfully run, right-click the node and select **Results**. Explore the following on the **Model Interpretability** tab:
 - **Surrogate Model Variable Importance**

Variable Label	Variable Name	Relative Imp...	Role	Variable Level
Replacement: RECENT_CARD_ RESPONSE_PRO P	REP_RECENT_C ARD_RESPONSE _PROP	1	INPUT	INTERVAL
Replacement: MEDIAN_HOME _VALUE	REP_MEDIAN_H OME_VALUE	0.4930	INPUT	INTERVAL
Replacement: FREQUENCY_ST ATUS_97N	REP_FREQUENC Y_STATUS_97N	0.2309	INPUT	NOMINAL

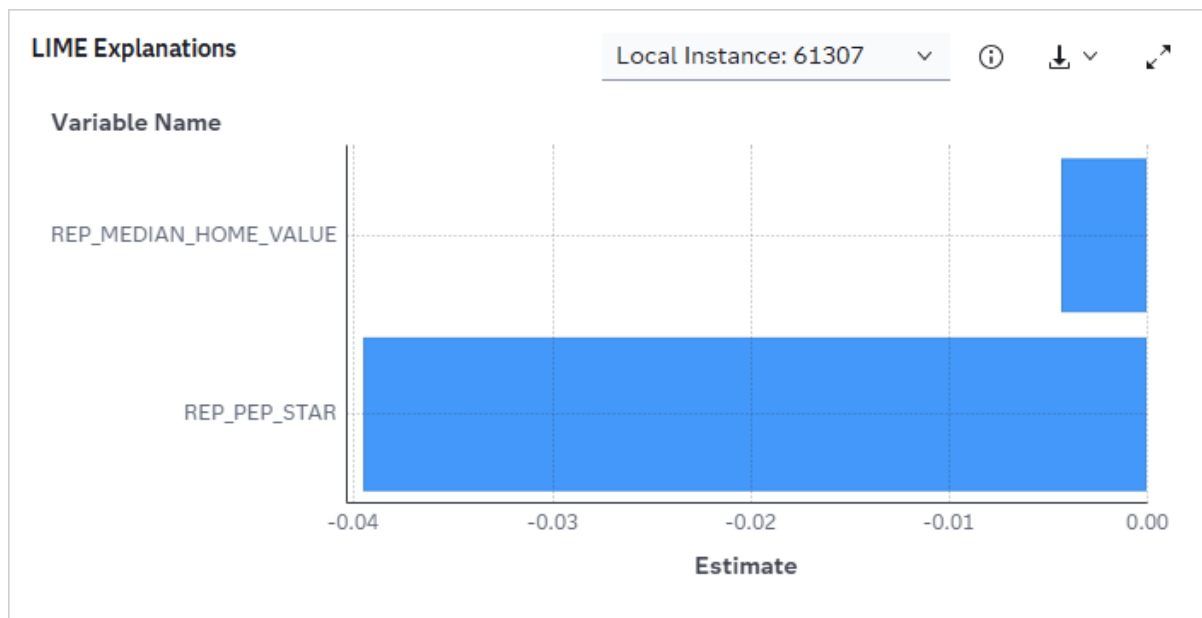
■ PD Plot



■ PD and ICE Overlay Plot



■ **LIME Explanations**



For more information about the model interpretability plots generated in the **Logistic Regression** node, see [Logistic Regression Results](#).

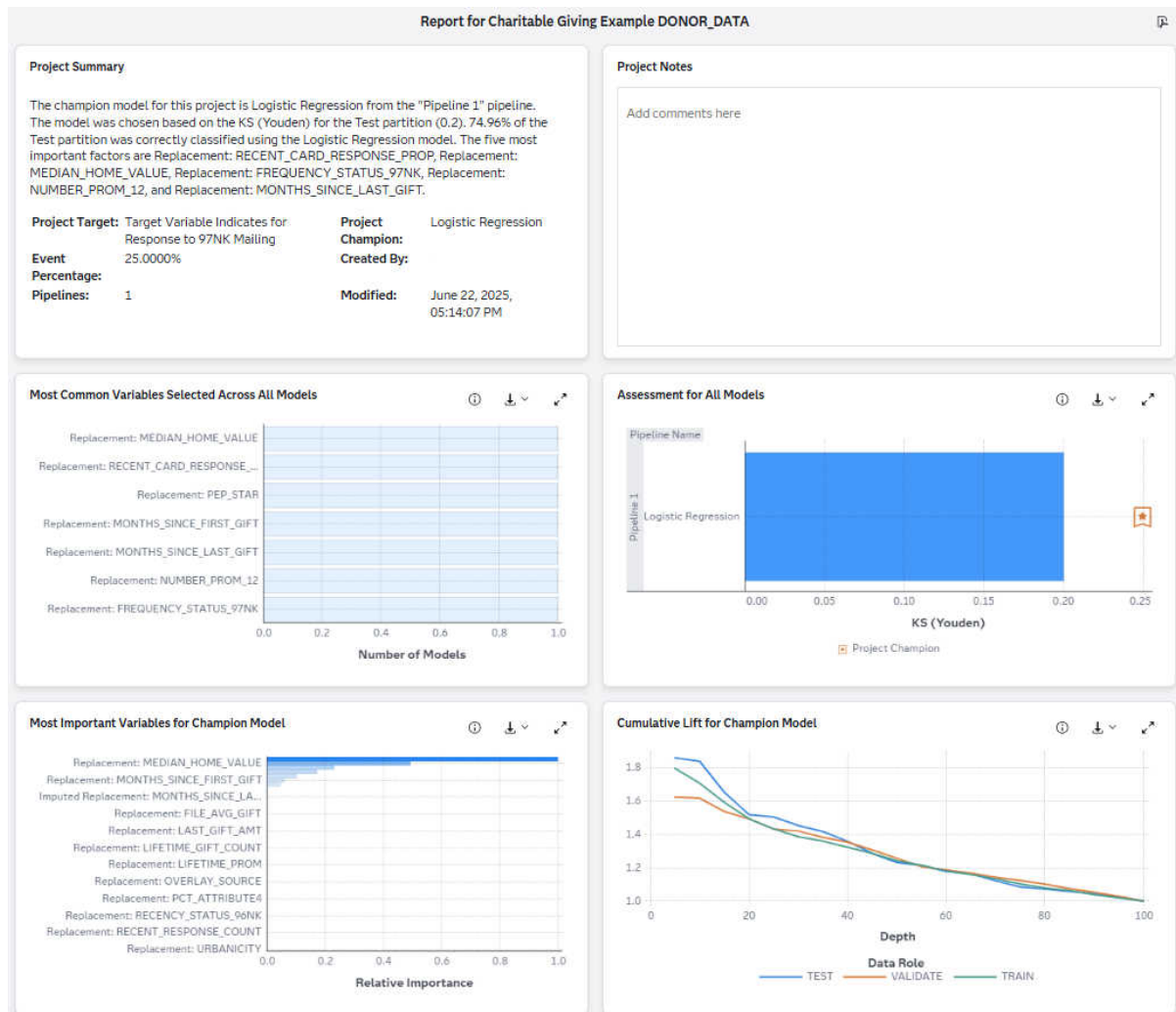
- 6 Click **Close** to close the Results window.

Project Insights

The **Insights** tab provides a summarized report of the project. The report includes the following information:

- the most commonly selected variables across all pipeline champion model

- a comparison of the assessment statistic across all pipeline champion models
- the most important variables for the selected champion model
- the cumulative lift plot for the selected champion model



In the image above, the **Forward Logistic Regression** model was created by the **Advanced template for class target** that is included with SAS Visual Data Mining and Machine Learning. The **Ensemble** model used automated machine learning to dynamically build a pipeline that is based on your data.

Publish the Champion Model

Before completing this section, ensure that an administrator has set up a location where you can publish your model. For more information, see [SAS Viya Platform: Publishing Destinations](#).

Publishing a model enables you to execute the model in various run-time engines. You can also manage a published model in SAS Model Manager, if it is licensed. For more information, see [Publish Models on page 154](#).

To publish a model:

- 1 Navigate to the **Pipeline Comparison** tab.
- 2 Ensure that the champion model is selected at the top of the **Pipeline Comparison** tab.
- 3 In the upper right corner of the **Pipeline Comparison** tab, click  and select **Publish models**.
- 4 The Publish Models window appears. Select the destination where you want your model to be published.
- 5 Select **Publish**.
- 6 The Publishing Results window appears. This window shows the name and published name of your model, as well as the status of your model (publishing, published successfully, and so on).
- 7 Close the **Publishing Results** window.

SAS Code Node Examples

Overview

In this section, you use the **SAS Code** node included with SAS Model Studio to perform two different tasks. In the first example, you create a gradient boosting model with the GRADBOOST procedure. In the second example, you use the FOREST procedure to perform variable selection.

Both examples use the DONOR_DATA data set that is found on the [SAS Viya Machine Learning product page](#). Click **Example Data for Getting Started with SAS Viya Machine Learning**. Download the ZIP file and extract the contents to a directory that your SAS Viya Machine Learning server can access. This data includes names of those who were solicited for charitable donations and indicates whether they donated.

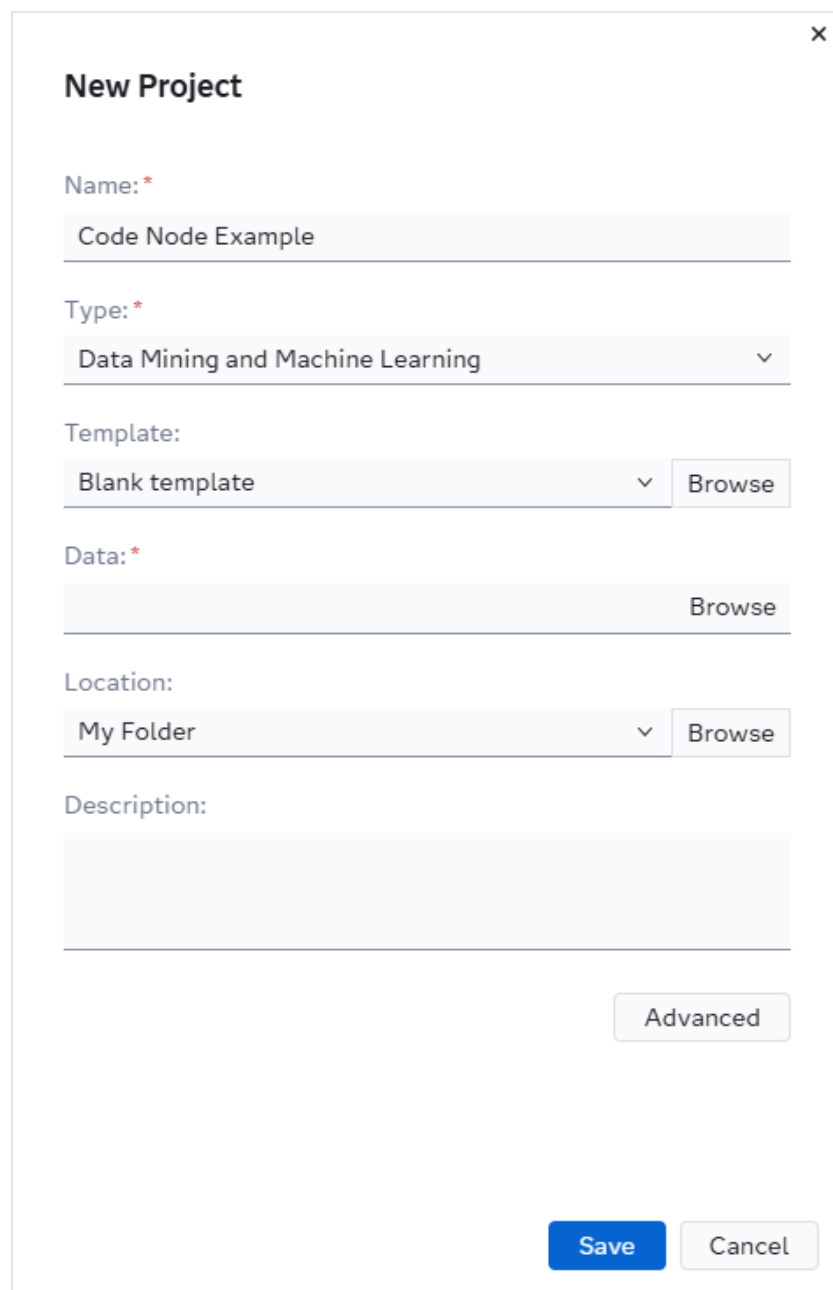
Download the ZIP file and extract the contents to a directory that your SAS Viya Machine Learning server can access.

Both examples require you to complete the steps in the [Create the Project and Import the Input Data](#) and [Modify Variables](#) sections. After completing these sections, you can complete the examples in any order.

Create the Project and Import the Input Data

This example assumes that you are signed in to the Welcome to SAS Viya application. To create the project that you will use in this example:

- 1 In the upper left corner of the Welcome to SAS Viya page, click , and select **Build Models**.
- 2 Click **New Project** in the upper right corner of the page.
- 3 Enter **Code Node Example** for **Name** in the New Project window.



The "New Project" dialog box is shown with a close button (X) in the top right corner. It contains the following fields and controls:

- Name:** * A text input field containing "Code Node Example".
- Type:** * A dropdown menu showing "Data Mining and Machine Learning" with a downward arrow.
- Template:** A dropdown menu showing "Blank template" with a downward arrow, and a "Browse" button to its right.
- Data:** * A text input field with a "Browse" button to its right.
- Location:** A dropdown menu showing "My Folder" with a downward arrow, and a "Browse" button to its right.
- Description:** A large text area for entering a description.
- An "Advanced" button located below the description field.
- "Save" and "Cancel" buttons at the bottom right.

- 4 Select **Data Mining and Machine Learning** for **Type**.
- 5 Ensure that **Blank template** is specified for **Template**.
- 6 In the **Data** field, click **Browse**. The Choose Data window appears.
- 7 In the upper right corner of the Choose Data window, click **Import data**. The Import Data window appears.
- 8 In the left pane of the Import Data window, click **Local files**, and navigate to the folder where DONOR_DATA is stored. Select **DONOR_DATA.sas7bdat** and click **Open**.
- 9 In the Import Data window, click **Import item** in the upper right corner. When the data is successfully imported, a note appears, saying that the data is ready for use.

 The table was successfully imported on Oct 7, 2019 10:33 AM and is ready for use.

- 10 Once the data set is successfully imported, click **Add** in the lower right corner of the Import Data window. This returns you to the Choose Data window.
- 11 In the Choose Data window, select **DONOR_DATA** in the list of available tables and click **OK**. This returns you to the New Project window.
- 12 In the New Project window, click **Save**. This directs you to the **Data** tab, where you can modify the variables in your data set.

Modify Variables

The **Data** tab shows the properties of the variables in the input data. To modify the properties of a variable, select the variable by clicking the check box to the left of the variable name, and then specify the new properties in the Options pane. Changes that you make to each variable are automatically applied and saved.

CAUTION

To avoid making unwanted changes to variable properties, you must manually deselect each variable that you modify when you are finished making changes to its properties.

For this example, modify the variables as follows:

- 1 On the Data tab, note the role for each variable in the **Role** column. Ensure that the variables roles are set to the following values:
 - CLUSTER_CODE is set to **Rejected**
 - CONTROL_NUMBER is set to **ID**
 - TARGET_B is set to **Target**
 - TARGET_D is set to **Rejected**
 - _dmIndex_ is set to **Key**.

- **_PARTIND_** is set to **Partition**.

In data mining, a strategy for assessing the quality of model generalization is to partition the data source. A portion of the data, called the *training data set*, is used for preliminary model fitting. The rest is reserved for empirical validation and is often split into two parts: validation data and test data. The validation data set is used to prevent a modeling node from overfitting the training data and to compare models. The test data set is used for a final assessment of the model. In this example, the data has been pre-partitioned where 60% of the data is assigned for training, 30% of the data is assigned for validation, and 10% of the data is assigned for testing.

- All other variable roles are set to **Input**.

If you need to change a variable role, do the following:

- Ensure that no variables are currently selected.
 - Select the variable that you want change by clicking the check box to the left of the variable name.
 - In the Options pane, expand **Role** and select the new variable role.
- On the **Data** pane, select the **_PARTIND_** variable. In the options pane, select **Map Partition Levels** and ensure that the partition levels are set to the following values:
 - **Training Level** is set to **1**.
 - **Validation Level** is set to **0**.
 - **Test Level** is set to **2**.

Click **Cancel**.

- Select the following variables:
 - FILE_AVG_GIFT
 - LAST_GIFT_AMT
 - LIFETIME_AVG_GIFT_AMT

In the options pane, set the property **Transform** to **Log10** for these variables.

TIP It is possible to select multiple variables for editing at one time by selecting the box in front of each variable name. By simultaneously selecting FILE_AVG_GIFT, LAST_GIFT_AMT, and LIFETIME_AVG_GIFT_AMT, you can change the **Transform** property for all three variables at once.

Create a Gradient Boosting Model

This example requires you to complete the steps in the [Create the Project and Import the Input Data](#) and [Modify Variables](#) sections. This example also assumes that you have not created any other pipelines before starting this section. To create the gradient boosting model:

- 1 Select the **Pipelines** tab in the upper left corner.
- 2 Right-click the **Data** node and select **Run**.
- 3 Once the **Data** node has run successfully, right-click the **Data** node and select **Add child node** ⇒ **Miscellaneous** ⇒ **SAS Code**.
- 4 Right-click the **SAS Code** node and select **Move** ⇒ **Supervised Learning**. A **Model Comparison** node is automatically added to the pipeline.
- 5 Select the **SAS Code** node. In the options pane, click **Open Code Editor**.
- 6 In the Training code editor, enter the following code:

```
proc gradboost data=&dm_data
  numBin=20 maxdepth=6 maxbranch=2 minleafsize=5
  minuseinsearch=1 ntrees=10 learningrate=0.1 samplingrate=0.5 lasso=0
  ridge=0 seed=1234;
  %if &dm_num_interval_input %then %do;
    input %dm_interval_input / level=interval;
  %end;

  %if &dm_num_class_input %then %do;
    input %dm_class_input / level=nominal;
  %end;

  %if "&dm_dec_level"="INTERVAL" %then %do;
    target %dm_dec_target / level=interval ;
  %end;

  %else %do;

    target %dm_dec_target / level=nominal;
  %end;

  &dm_partition_statement;
  ods output
    VariableImportance = &dm_lib..VarImp
    Fitstatistics       = &dm_data_outfit
  ;
  savestate rstore=&dm_data_rstore;
run;

%dmcas_report(dataset=VarImp, reportType=Table,
  description=%nrbquote(Variable Importance));
```

```
%dmcas_report(dataset=VarImp, reportType=BarChart, category=Variable,
               response=RelativeImportance, description=%nrquote(Relative
Importance Plot));
```

This code uses the following SAS Model Studio macro variables and macros:

- **DM_DATA** — A macro variable that identifies the CAS training table. If partitioned, the table contains the `_PartInd_` variable that identifies which observations are used for training, validation, and test. This table is transient and is dropped when the node finishes running.
- **DM_NUM_INTERVAL_INPUT** — A macro variable that identifies the number interval input variables.
- **DM_NUM_CLASS_INPUT** — A macro variable that identifies the number of class input variables.
- **DM_DEC_LEVEL** — A macro variable that identifies the measurement level (binary, interval, ordinal, or nominal) of the target variable.
- **DM_PARTITION_STATEMENT** — A macro variable that identifies the partition statement. This variable is blank if the data is not partitioned.
- **DM_LIB** — A macro variable that identifies the SAS library where the variable importance table is saved. This table is named `VarImp`.
- **DM_DATA_OUTFIT** — A macro variable that identifies the fit statistics data set. This is the data set that is used by the model comparison node to select the best model in the pipeline.
- **DM_DATA_RSTORE** — A macro variable that identifies the remote analytic store that is created by the GRADBOOST procedure. This table is used by SAS Model Studio to score and assess the model.
- **%DM_INTERVAL_INPUT** — A macro that identifies the interval input variables.
- **%DM_CLASS_INPUT** — A macro that identifies the class input variables.
- **%DM_DEC_TARGET** — A macro that identifies the project target variable.
- **%DMCAS_REPORT** — A macro that enables the addition of more reports to the Results window.

7 In the upper right corner of the code editor, click .

8 Click **Close**.

9 Right-click the **SAS Code** node and select **Run**.

There are two `DMCAS_REPORT` calls to display the contents of the variable importance table `VarImp`.

- **reportType=Table** — This call adds the Variable Importance table to the results, as indicated in the `DESCRIPTION` argument.
- **reportType=BarChart** — This call adds a bar chart that contains Relative Importance for each input.

10 Right-click the **SAS Code** node and select **Results**. Review the following results:

- **Variable Importance**

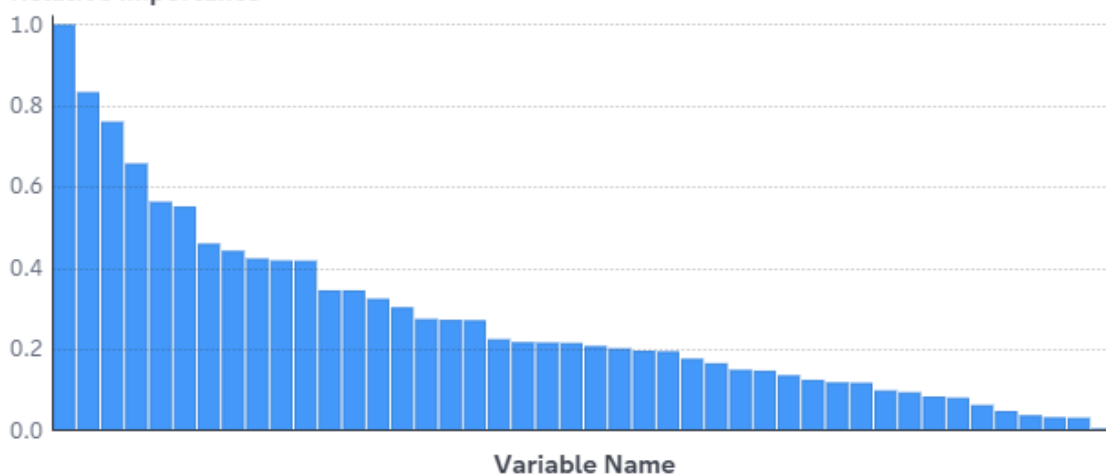
Variable Importance

Variable Name	Training Importance	Importance Standard Deviation	
WEALTH_RATING	19.8347	9.7111	
INCOME_GROUP	16.5277	4.3936	
RECENT_CARD_RESP ONSE_COUNT	15.0858	5.5896	
RECENT_RESPONSE_ COUNT	13.0410	9.6935	
MEDIAN_HOME_VAL UE	11.1665	6.0200	
FRFQUENCY STATUS	10.9336	12.8972	

■ Relative Importance Plot

Relative Importance Plot

Relative Importance



- **Path EP Score Code** — Because you specified that the **SAS Code** node is a supervised learning node, it automatically creates EP Score Code. More specifically, this code produces an analytic store as its score code.

Path EP Score Code

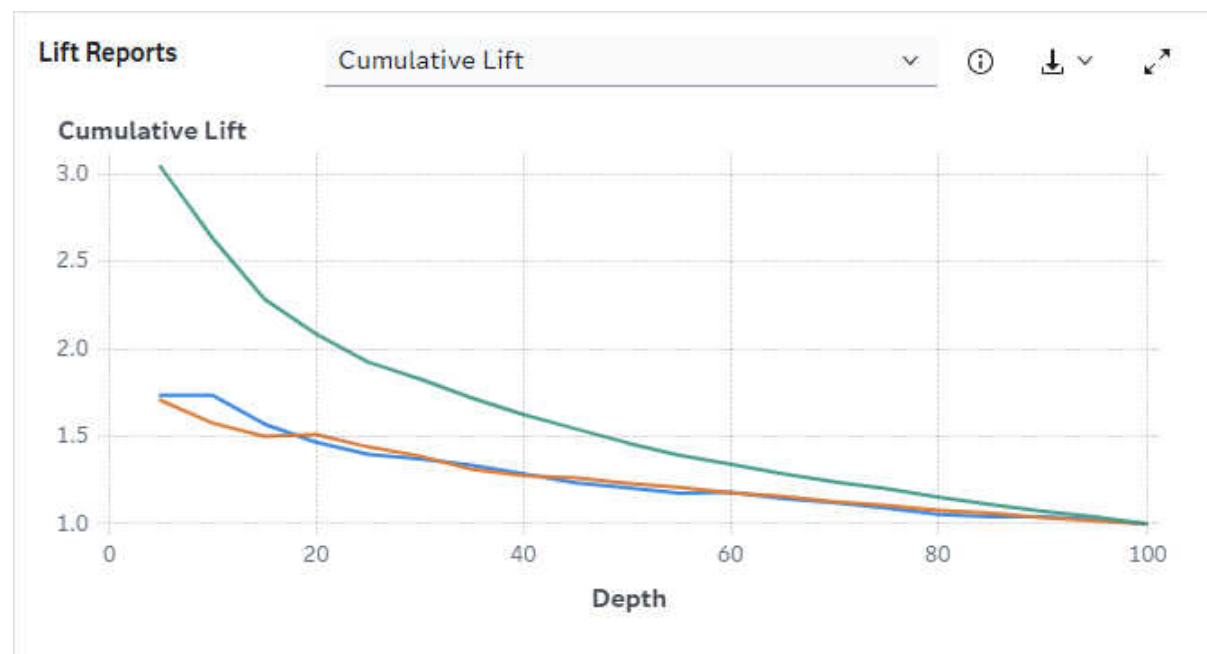
```

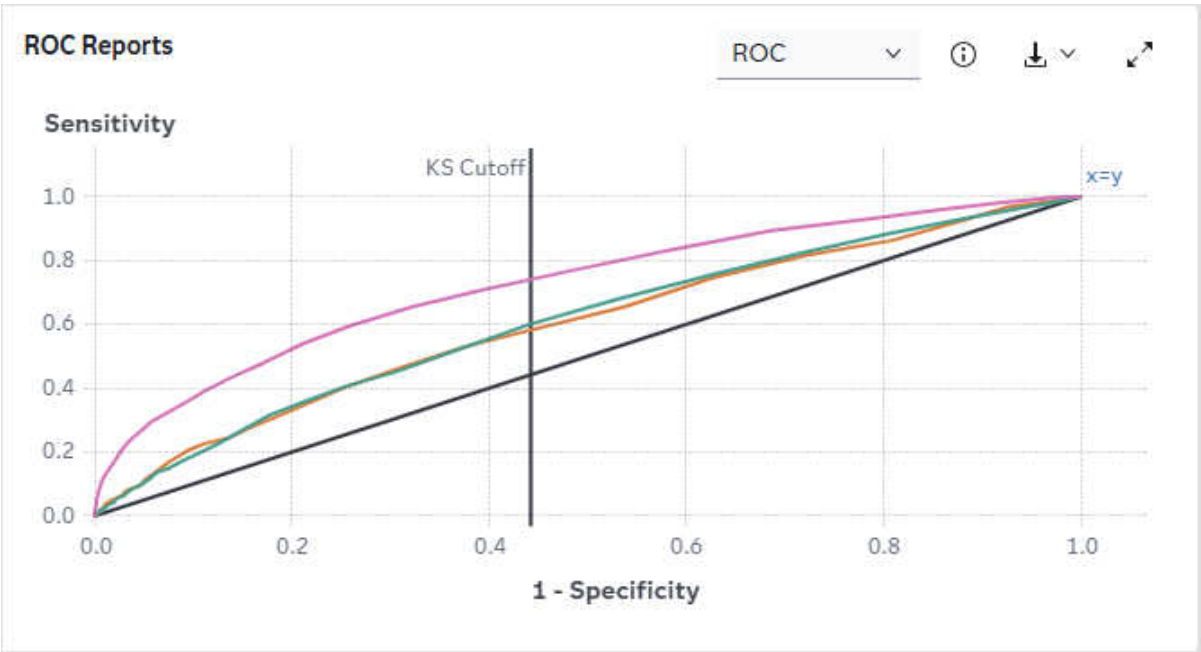
1  /*-----
2  /* Product:      Visual Data Mining and Machine Learning
3  /* Release Version: V2025.07
4  /* Component Version: V2025.06
5  /* CAS Version:    V.04.00M0P06192025
6  /* SAS Version:    V.04.00M0P061925
7  /* Site Number:    70180938
8  /* Host:          sas-cas-server-default-client
9  /* Encoding:       utf-8
10 /* Java Encoding:   UTF8
11 /* Locale:         en_US
12 /* Project GUID:    15d5ece8-59b4-4590-9796-b3be9fe460fa
13 /* Node GUID:       95d65b06-d0ea-4e4d-95a2-44bafddc2f16
14 /* Node Id:         8VCGW0I4W1I0DJ7WMMTU77DHY
15 /* Algorithm:       sascode
16 /* Generated by:    stengl
17 /* Date:            23JUN2025:17:50:03

```



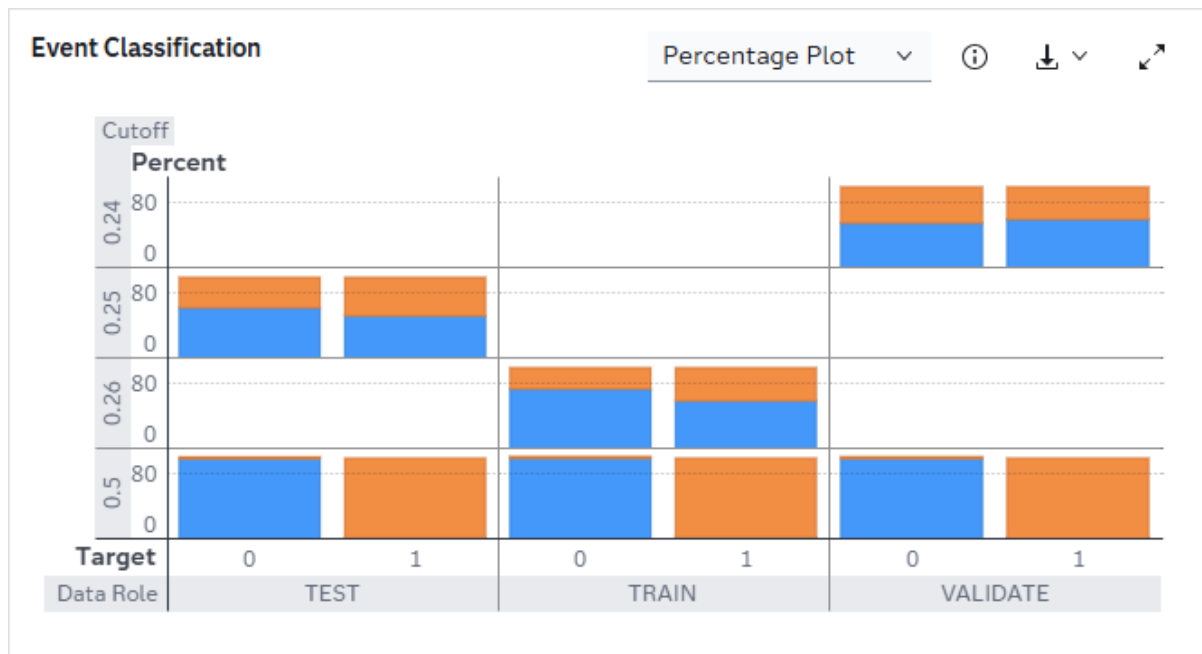
- **Lift Reports, ROC Reports, Fit Statistics, and Event Classification** — These plots are automatically created because the **SAS Code** node is a supervised learning node. Click **Assessment** in the upper left corner of the SAS Code Results window to access these plots.





Fit Statistics

Target ...	Data Role	Partitio...	Formatt...	Number...	Average...	
TARGET_B	TEST	2	2	1,937	0.1824	
TARGET_B	TRAIN	1	1	11,623	0.1698	
TARGET_B	VALIDATE	0	0	5,812	0.1825	



11 Click **Close** to close the Results window.

Perform Variable Selection

This example requires you to complete the steps in the [Create the Project and Import the Input Data](#) and [Modify Variables](#) sections. This example also assumes that you have not created any other pipelines before starting this section.

- 1 Select the **Pipelines** tab in the upper left corner.
- 2 Right-click the **Data** node and select **Run**.
- 3 Once the **Data** tab has run successfully, right-click the **Data** node and select **Add child node** ⇒ **Miscellaneous** ⇒ **SAS Code**.
- 4 Select the **SAS Code** node. In the options pane, click **Open Code Editor**.
- 5 In the Training code editor, enter the following code:

```
proc forest data=&dm_data
  minleafsize=5 minuseinsearch=1 seed=12345 loh=0 numbin=20
  ntrees=100 maxdepth=20 inbagfraction=0.6 ;
  partition fraction (valid=0.3 seed=12345);
  %if &dm_num_interval_input %then %do;
    input &dm_interval_input / level=interval;
  %end;

  %if &dm_num_class_input %then %do;
    input &dm_class_input / level=nominal;
  %end;

  %if "&dm_dec_level"="INTERVAL" %then %do;
```

```

        target %dm_dec_target / level=interval ;
    %end;

    %else %do;
        target %dm_dec_target / level=nominal;
    %end;
    grow IGR;
    ODS output VariableImportance = &dm_lib..forestvarimportance ;
run;

%dmcas_report(dataset=forestvarimportance, reportType=BarChart,
category=Variable, response=RelativeImportance,
sortDirection=descending, sortBy=RelativeImportance,
description=%nrquote(Relative Importance Plot));

filename _frf "&dm_file_deltacode";
data _null_;
    length string $200;
    file _frf;
    set &dm_lib..forestvarimportance ;
    where RelativeImportance < 0.3;
    string = 'if NAME = "'!!kstrip(Variable)!!'" then ROLE="REJECTED";';
    put string;
run;
filename _frf;

%dmcas_report(file=&dm_file_deltacode, reportType=CodeEditor,
description=%nrquote(Metadata Changes));

```

This code uses the following SAS Model Studio macros:

- **DM_FILE_DELTACODE** — A macro variable that identifies the file that contains the DATA step code to modify the columnsmeta information that is exported by the node.

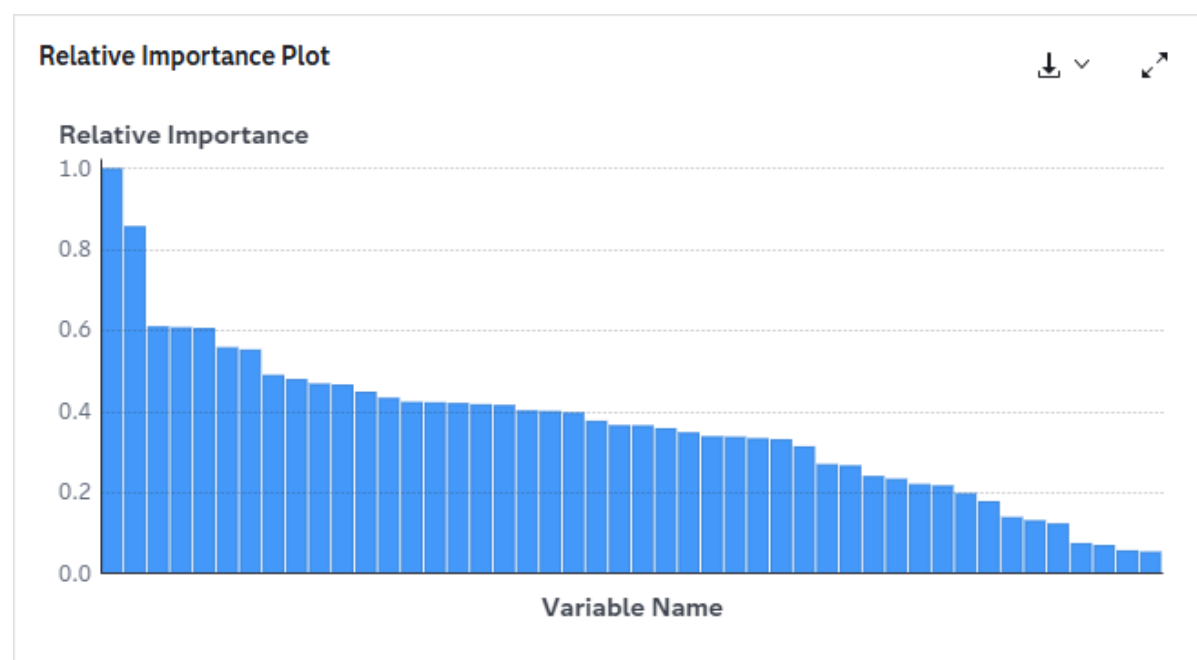
- 6 In the upper right corner of the code editor, click .

This code uses PROC FOREST to identify the relative variable importance for all variables in the input data. Those variables with a relative importance less than 0.3 are assigned the role **Rejected**. All other variables are kept.

- 7 Click **Close**.
- 8 Right-click the **SAS Code** node and select **Run**.
- 9 Right-click the **SAS Code** node and select **Results**.

There are two DMCAS_REPORT calls to display the contents of the variable importance table VarImp.

- **reportType=BarChart** — This call adds the Relative Importance Plot to the results.



- `reportType=CodeEditor` — This call adds the Metadata Changes information to the results. All of the variables that are dropped from the analysis are listed here.

Metadata Changes

```

1  if NAME = "SES" then ROLE="REJECTED";
2  if NAME = "LIFETIME_GIFT_RANGE" then ROLE="REJECTED";
3  if NAME = "LIFETIME_MIN_GIFT_AMT" then ROLE="REJECTED";
4  if NAME = "RECENT_AVG_CARD_GIFT_AMT" then ROLE="REJECTED";
5  if NAME = "REGENCY_STATUS_96NK" then ROLE="REJECTED";
6  if NAME = "OVERLAY_SOURCE" then ROLE="REJECTED";
7  if NAME = "MOR_HIT_RATE" then ROLE="REJECTED";
8  if NAME = "MONTHS_SINCE_ORIGIN" then ROLE="REJECTED";
9  if NAME = "PEP_STAR" then ROLE="REJECTED";
10 if NAME = "DONOR_GENDER" then ROLE="REJECTED";
11 if NAME = "PCT_ATTRIBUTE1" then ROLE="REJECTED";
12 if NAME = "IN_HOUSE" then ROLE="REJECTED";
13 if NAME = "RECENT_STAR_STATUS" then ROLE="REJECTED";
14 if NAME = "HOME_OWNER" then ROLE="REJECTED";
15 if NAME = "PUBLISHED_PHONE" then ROLE="REJECTED";
16

```

There is no score code available for this node because it is not a supervised learning node. There are also no assessment plots.

- 10 Click **Close** to close the Results window.

Integrating SAS Model Studio with SAS Visual Analytics

Overview

The examples in this section show the typical process of creating a model in SAS Visual Analytics, copying it to SAS Model Studio, and then continuing your analysis. The first example is an introduction to the integration between SAS Visual Analytics and SAS Model Studio, and uses a forest model to create inputs for a support vector machine. The second example is more advanced, and uses a generalized additive model (GAM) to highlight some of the differences and caveats that can arise when copying a model from SAS Visual Analytics to SAS Model Studio.

SAS Visual Analytics enables you to create many different types of reports that demonstrate the efficacy of various statistical models. For more information, see [Working with SAS Visual Statistics](#) in the SAS Visual Analytics documentation. Many of the reports created in SAS Visual Analytics can be sent to SAS Model Studio for comparison against other SAS Model Studio models.

In SAS Visual Analytics, you start by identifying the data that you want to model. Next, you can adjust certain characteristics of that data or create new data items. Then, you add one or more objects to the workspace and assign data items to those objects. Objects vary in complexity from simple tables to more complex statistical models.

Reports in SAS Visual Analytics can range from single-page reports that contain a single object to multi-page reports that contain several dependencies and inter-object connections. However, when you copy a model from SAS Visual Analytics to SAS Model Studio, a four-node pipeline is always created. The four nodes in the pipeline are as follows:

- Data node
- Visual Data Preparation node
- A model node
- Model Comparison node

The data preparation and model nodes are discussed in more detail in these examples. These examples use the HMEQ data set, which contains 5,960 mortgage applications and indicates whether the applicant defaulted on the loan. The HMEQ data set can be found at <http://support.sas.com/documentation/onlinedoc/viya/examples.htm>.

Support Vector Machine Integration Example

Download the Sample Data

- 1 In a web browser, navigate to the [SAS Viya Example Data Sets](#) page.
- 2 Download the file **hmeq.csv** to your local machine.

Create the Report

This example assumes that you have already signed in to SAS Model Studio.

- 1 In the upper left corner of the Projects window, click , and select **Explore and Visualize**. This opens SAS Visual Analytics, and enables you to choose a data source, create a new report, or load an existing report in the Explore and Visualize window.
- 2 Click **New report** in the upper right corner of the page.
- 3 Click  in the left pane to open the Data pane.
- 4 Click **Import data** to open the Import Data window.
- 5 Select **Local files**, and navigate to the location where you saved the hmeq.csv file.
- 6 Select file hmeq.csv and click **Open**. This returns you to the Import Data window.
- 7 Click **Import item** to import the data.
- 8 When the data is successfully imported, click **Add to report**. This creates the new report, which is named **Report 1** by default.
- 9 Click  in the upper right corner of the page and select **Save**. This opens the Save As window.
- 10 In the **My Folder** pane, navigate to and select a location where you have Write permission.
- 11 In the **Name** field, enter **SVM Integration Example**, and click **Save**.

Create Inputs Using a Forest

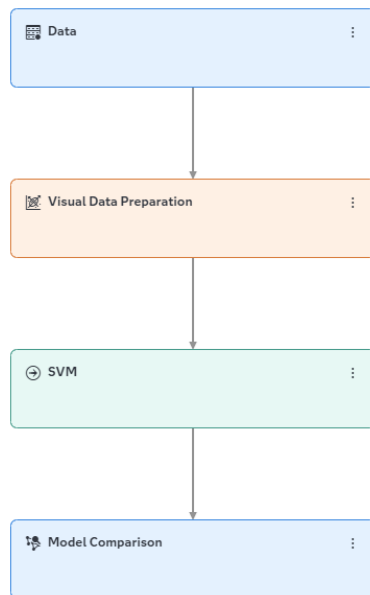
- 1 In the **Data** pane, right-click **BAD** and select **Convert to category**.
- 2 Click  in the upper right corner of the page and select **Save**.
- 3 From the left pane, click  to select an object. Navigate to the objects listed under **Machine Learning**, and drag the  **Forest** object onto the canvas to create a forest.
- 4 Click  in the right pane.
- 5 Click **Response** and select **LOAN**.
- 6 Click **Predictors** and select every variable except **BAD**. Click **Apply**.
- 7 Right-click the **Variable Importance** plot and select **Derive predicted**.
- 8 Review the new data items in the New Prediction Items window and click **OK**.
- 9 Click  to save the report.

Create a Support Vector Machine

- 1 Click  to add a new page to the report.
- 2 From the left pane, click  to select an object. Navigate to the objects listed under **Machine Learning**, and drag the  **Support vector machine** object onto the canvas to create a support vector machine.
- 3 Click  in the right pane.
- 4 Click **Response** and select **BAD**.
- 5 Click **Predictors** and select **CLAGE**, **DEBTINC**, **DELINQ**, **DEROG**, and **Predicted: LOAN**. Click **Apply**.
- 6 Click  and change the value for **Penalty value** from 1 to 10.
- 7 Click  to save the report.
- 8 Right-click on the report and select **Create pipeline** ⇒ **Add to new project**. This action copies the model and all data preparation steps to SAS Model Studio. SAS Model Studio automatically opens.

Continuing in SAS Model Studio

After creating the project and copying the necessary information from SAS Visual Analytics to SAS Model Studio, you should see a pipeline that resembles the following image. Your SAS Model Studio project is also named SVM Integration Example, as that title is inherited from SAS Visual Analytics.



This pipeline contains four nodes. The Visual Data Preparation node contains the score code necessary to re-create any data preparation that you performed in SAS Visual Analytics. In this example, this includes converting BAD to a category, and deriving the predicted variable from the Forest model that is used as an input to the SVM. From this point, you can modify the pipeline as if it were created in SAS Model Studio. The only restriction is that you cannot edit the properties of the Visual Data Preparation node.

- 1 In the upper left corner, click **Data** to open the **Data** tab. This tab displays all the information that SAS Model Studio knows about the input data set.
- 2 Notice that SAS Visual Analytics created two new variables: `_dmIndex_` and `_va_d__E_BAD`. The original target variable, BAD, has been assigned the role Rejected. SAS Visual Analytics creates temporary variables as needed to complete the tasks that you want to perform. In this case, changing BAD from a measure to a category necessitated the creation of the variable `_va_d__E_BAD`.

These name changes require careful consideration if you want to use a holdout data set. That holdout data set must have a target variable name that exactly matches the target variable name created by SAS Visual Analytics. The target variable is used for generating assessment statistics, not scoring.

- 3 Also, notice that all other variables are assigned the role Input. This does not match the support vector machine that you created in SAS Visual Analytics. The

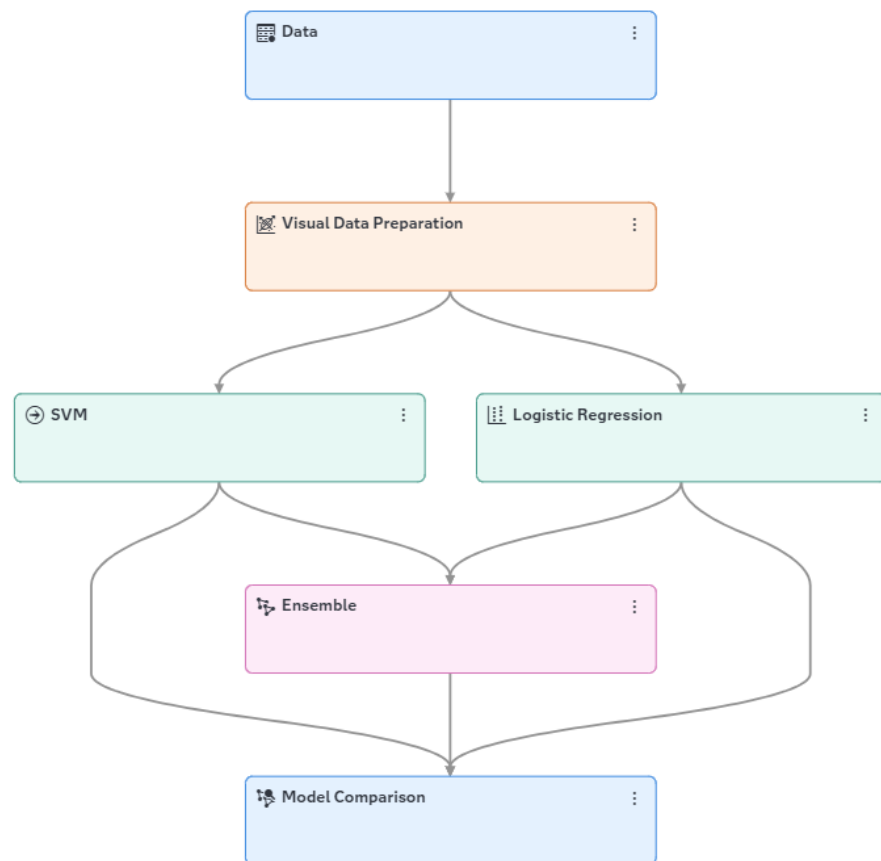
Visual Data Preparation node handles the role assignments when you run the pipeline.

Note: Changing the target variable information or the partition variable information on the **Data** tab causes the pipeline run to fail. The score code that is generated and applied in the Visual Data Preparation node requires the partition and variable information that was known in SAS Visual Analytics when you created the pipeline. After the project is created (the project advisor code is run), you cannot modify any data item on the **Data** tab. Therefore, if you altered a data item and your pipeline failed, you need to delete your SAS Model Studio project. Open your SAS Visual Analytics report, create a new pipeline, and try any modifications again.

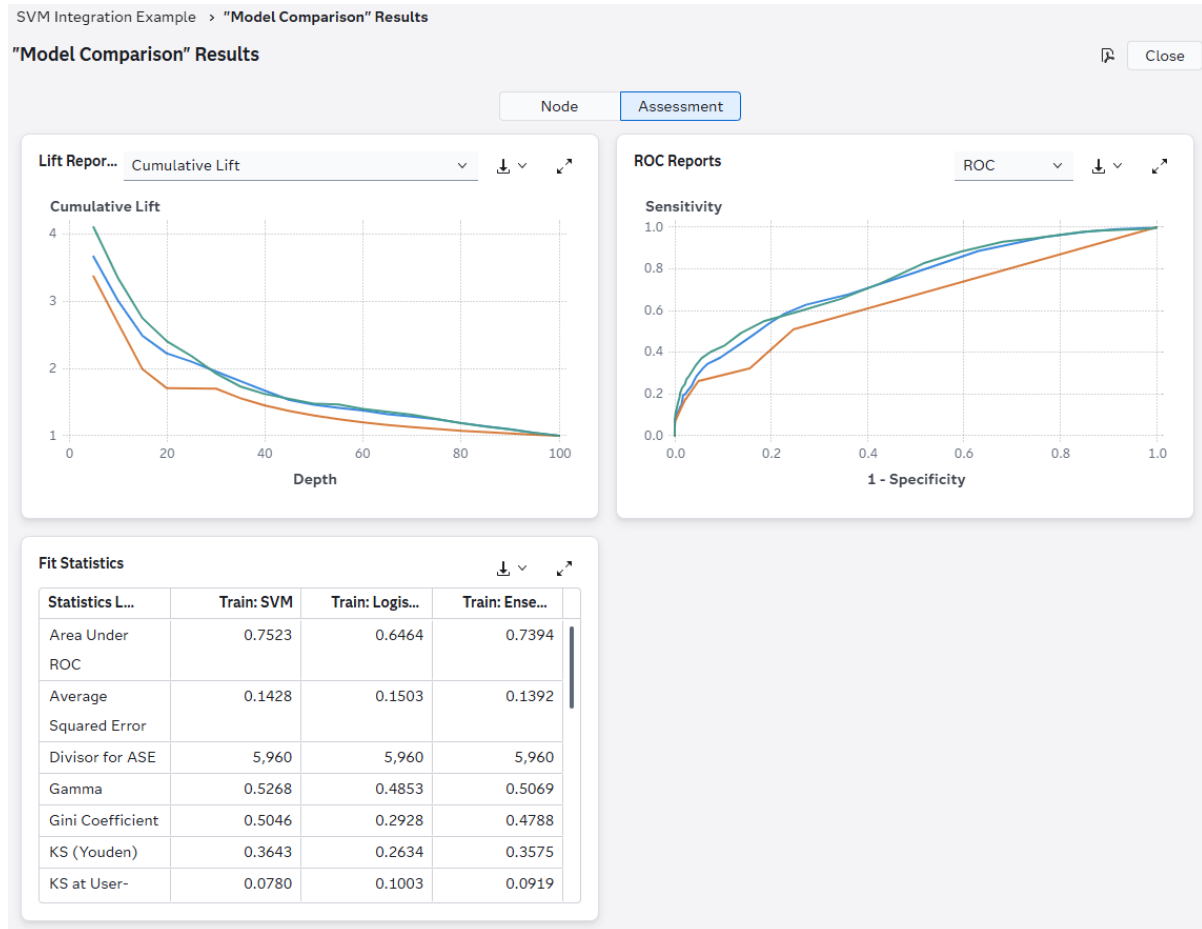
- 4 In the upper left corner, click **Pipelines** to open the **Pipelines** tab.

The Visual Data Preparation node ensures that all subsequent nodes see the data as it existed in SAS Visual Analytics. This node executes all of the model score code from all of the objects in SAS Visual Analytics that were used to create the copied model. This code can be DS1 code, analytic store code, or some combination of the two. The Visual Data Preparation node ensures that this code runs in the proper order and guarantees that subsequent nodes receive the proper information. This means that the Logistic Regression node that you add in the next step sees the data as it existed in SAS Visual Analytics.

- 5 Right-click the **Visual Data Preparation** node and select **Add child node** ⇒ **Supervised learning** ⇒ **Logistic Regression**. This adds a new supervised learning model to the pipeline. The data preparation steps used in SAS Visual Analytics are applied to the Logistic Regression node.
- 6 Right-click the **Logistic Regression** node and select **Add child node** ⇒ **Postprocessing** ⇒ **Ensemble**.
- 7 Right-click the **Ensemble** node and select **Select models** ⇒ **SVM**. At this point, your pipeline should resemble the following:



- 8 Right-click the **Model Comparison** node and select **Run**. This action runs all of the nodes preceding the **Model Comparison** node.
- 9 Right-click the **Model Comparison** node and select **Results**. This brings you to the **Node** tab, which displays the **Model Comparison** table. The table shows you which model is your champion model. Click **Assessment** to see the **Lift Reports** and **ROC Reports** charts, as well as the **Fit Statistics** table.



Review the results computed for each of the three models in the diagram. The Ensemble node was chosen as the champion model.

Click **Close** to close the Results window.

- 10 Select the SVM node, and look at the value specified for Penalty. Notice that the value is 10, which matches the value that you specified earlier in SAS Visual Analytics. The values and options that you set in the SAS Visual Analytics object are generally carried over to the corresponding node in SAS Model Studio.
- 11 Suppose you want to modify the support vector machine model to improve model performance. You can right-click the **SVM** node and select **Enable Properties**. However, if you enable the properties of a node created in SAS Visual Analytics, the model is retrained, which might yield new results. Once properties have been enabled, the model cannot be switched back to use the original SAS Visual Analytics score code.

In this example, you want to enable the properties of the **SVM** node. Right-click the **SVM** node, select **Enable Properties** and click **Yes**.

- 12 Click the **SVM** node. In the **Options** pane, enable **Perform Autotuning**.

Note: Any changes made to the **Autotune** options for an object in SAS Visual Analytics are not carried over to SAS Model Studio when that object is added to a SAS Model Studio project.

Autotuning automatically chooses the optimal values for the hyperparameters used to build your predictive model. For more information, see [Overview of Autotuning](#).

- 13 Right-click the **Model Comparison** node and select **Run**. This action runs all of the nodes preceding the Model Comparison node.
- 14 Right-click the **Model Comparison** node and select **Results**. Notice the improved misclassification rate of the SVM node model.
- 15 Click **Close** to close the Results window.

Generalized Additive Model (GAM) Integration Example

Download the Sample Data

Note: If you have already imported the data for the Support Vector Machine Integration example, skip these steps and proceed to [“Create the Report” on page 44](#).

- 1 In a web browser, navigate to the [SAS Viya Example Data Sets](#) page.
- 2 Download the file **hmeq.csv** to your local machine.

Create the Report

This example assumes that you have already signed in to SAS Model Studio.

- 1 In the upper left corner of the Projects window, click , and select **Explore and Visualize**. This opens SAS Visual Analytics, and enables you to choose a data source, create a new report, or load an existing report in the Explore and Visualize window.

Note: If you previously completed the Support Vector Machine Integration example, this step might return you to the report created in that example. If you

are returned to the report from the previous example, click  and select **Home** before proceeding with the next step.

- 2 Click **New report** in the upper right corner of the page.
- 3 Click  in the left pane to open the Data pane.

If you already imported the data for the Support Vector Machine Integration example, complete these steps:

- 1 Click **Add data** to open the Choose Data window.
- 2 Select the HMEQ table and click **Add**.
- 3 Click  in the upper right corner of the page and select **Save**. This opens the Save As window.
- 4 In the **My Folder** pane, navigate to and select a location where you have Write permission.
- 5 In the **Name** field, enter **GAM Integration Example**, and click **Save**.

If you did not import the data for the Support Vector Machine Integration example, complete these steps:

- 1 Click **Import data** to open the Import Data window.
- 2 Click **Local files**, and navigate to the location where you saved the hmeq.csv file.
- 3 Select the hmeq.csv file and click **Open**. This returns you to the Import Data window.
- 4 Click **Import item** to import the data.
- 5 When the data is successfully imported, click **Add to report**. This creates the new report, which is named **Report 1** by default.
- 6 Click  in the upper right corner of the page and select **Save**. This opens the Save As window.
- 7 In the **My Folder** pane, navigate to and select a location where you have Write permission.
- 8 In the **Name** field, enter **GAM Integration Example**, and click **Save**.

Create a Generalized Additive Model (GAM)

- 1 From the left pane, click  to select an object. Navigate to the objects listed under **Statistics**, and drag the  **Generalized additive model** object onto the canvas to create a GAM.
- 2 Click  in the upper left corner of the page, and select **+New data item**   **Spline effect**. The **New Spline Effect** window appears.

- 3 Select  **YOJ** and click **+**.
- 4 Click **OK** in the bottom right corner of the New Spline Effect window to create the new spline effect.
- 5 Click  in the right pane.
- 6 Click **Response** and select **BAD**.
- 7 Select **Spline effects** to open the Add Data Items window.
- 8 Select  **Spline(YOJ)** and click **Apply**.
- 9 Click **Continuous effects** and select **DEBTINC**, **DELINQ**, **DEROG**, and **MORTDUE**, and click **Apply**.
- 10 Click  in the right pane.
- 11 Under **Generalized Additive Model**, expand **General**.
- 12 Locate **Distribution** and select **Binary**.
- 13 Click  to save the report.
- 14 Right-click on the page and select **Create pipeline** ⇒ **Add to new project**. This action copies the model and all data preparation steps to SAS Model Studio. SAS Model Studio automatically opens.

Continuing in SAS Model Studio

The pipeline that was generated contains four nodes:

- Data node
- Visual Data Preparation node
- GAM node
- Model Comparison node

Complete the following steps:

- 1 Select the **GAM** node, and notice that the properties in the pane on the right side of the page are disabled. When you create a pipeline from an object in SAS Visual Analytics, the properties of the corresponding node are disabled by default. You can enable these properties, but doing so retrains the model, which in turn might yield different results even if none of the node properties are adjusted. This information is displayed in a warning message when you enable the node properties.
- 2 Right-click the **GAM** node and click **Run**.
- 3 When the pipeline run is complete, right-click the **GAM** node and select **Results**.
- 4 When viewing the results of the node run, there are six reports available on the **Node** tab and two reports available on the **Assessment** tab. Notice that the

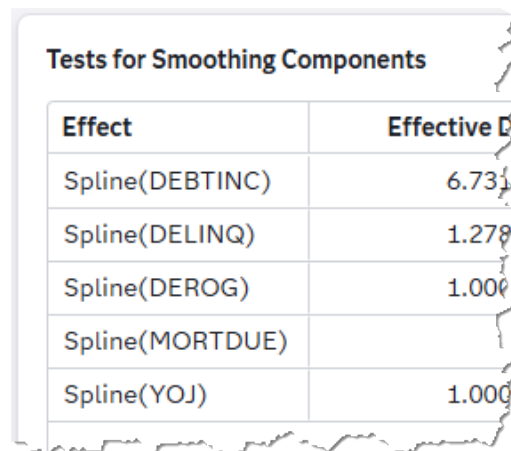
results do not contain any node-specific reports that would normally be available in SAS Model Studio. This is because the node properties were not enabled, and instead of retraining the model from SAS Visual Analytics, the pipeline run simply scored the model. Review these reports, and return to the **Pipelines** view.

- 5 Click **Close** to close the Results window.

In the first pipeline run, the properties of the GAM node are disabled. In the following steps, you enable the properties of the GAM node and explore the effects of doing so.

- 1 Right-click the **GAM** node and select **Enable Properties**. As previously mentioned, a warning message appears stating that enabling the node properties might change the results.
- 2 Click **Yes** in the Enable Properties window.
- 3 In the properties pane, notice that the interval target probability distribution is normal, and not binary as specified for the GAM object in SAS Visual Analytics. This is because the GAM node supports the binary distribution only when the target variable is binary, whereas SAS Visual Analytics supports the binary distribution for both binary targets and for interval targets with only two levels. In this example, the response (target) variable **BAD** was imported into SAS Visual Analytics as a Measure variable, and was not converted to a Category variable before training the GAM. As a result, SAS Model Studio treats **BAD** as an interval target, so the GAM node does not support the binary distribution that was specified in SAS Visual Analytics. Instead, the distribution is set to **Normal**, which is the default distribution for interval targets.
- 4 Right-click the **GAM** node and select **Run**.
- 5 When the pipeline run is complete, right-click the **GAM** node and select **Results**.
- 6 Notice that there are now additional reports that were not present when you ran the node without first enabling the node properties. When you enable the node's properties and run the node, the node retrains the model and behaves as it typically does in SAS Model Studio. Consequently, the results include reports that are specific to the node, such as the Smoothing Components report for the GAM node.

Another difference between the model created using SAS Visual Analytics and the retrained model in SAS Model Studio can be seen in the Tests for Smoothing Components report. Notice that the interval inputs that you specified as continuous effects in SAS Visual Analytics are now treated as spline effects.



Effect	Effective D
Spline(DEBTINC)	6.73
Spline(DELINQ)	1.278
Spline(DEROG)	1.000
Spline(MORTDUE)	1.000
Spline(YOJ)	1.000

The GAM node in SAS Model Studio treats all interval inputs as spline effects, whereas in SAS Visual Analytics, you can specify which variables you want as continuous effects or as spline effects. As a result of enabling the node properties, the continuous effects that you specified in SAS Visual Analytics are now treated as spline effects.

- 7 Click **Close** to close the Results window.

Open Source Code Node Example

Overview

In this section, you use the **Open Source Code** node to create a logistic regression model in Python and R. You also compare this model with a **Logistic Regression** node. Though a champion is chosen, the intent of the example is not to build the best model. Rather, this example demonstrates how models from Python or R are executed and compared in SAS Model Studio.

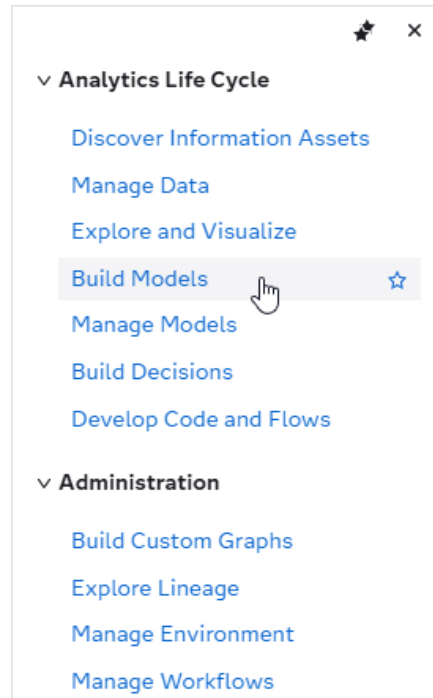
This example uses the HMEQ data set, which has a binary target that indicates whether a mortgage loan defaulted. The inputs include information such as loan amount, reason for loan, years at present job, and debt-to-income ratio.

If you are unsure about whether the HMEQ data set is available on your system, navigate to the [SAS Viya Example Data Sets](#) page and download **hmeq.csv**. Note where you saved this file.

Create the Project and Import the Input Data

This example assumes that you are signed in to the Welcome to SAS Viya application. To create the project that you use in this example:

- 1 In the upper left corner of the Welcome to SAS Viya page, click , and select **Build Models**.



- 2 Click **New Project** in the upper right corner of the Projects page.
- 3 Enter `Open Source Code Node Example` for **Name** in the New Project window.

New Project

Name: *

Open Source Code Node Example

Type: *

Data Mining and Machine Learning

Template:

Blank template

Browse

Data: *

Browse

Location:

My Folder

Browse

Description:

Advanced

Save Cancel

- 4 Select **Data Mining and Machine Learning** for **Type**.
- 5 Ensure that **Blank template** is specified for **Template**.
- 6 In the **Data** field, click **Browse**. The Choose Data window appears.
- 7 If **HMEQ** is in the **Available** list in the Choose Data window, select the HMEQ table and click **OK**.
If the HMEQ table is not listed, import it as follows:
 - a In the upper right corner of the Choose Data window, click **Import data**. The Import Data window appears.
 - b In the left pane of the Import Data window, click **Local files** and navigate to the folder where file hmeq.csv is stored. Select **hmeq.csv** and click **Open**.

- c In the upper right corner of the Import Data window, click **Import item**.
 - d After the data set is successfully imported, click **Add**. This returns you to the Choose Data window.
 - e In the Choose Data window, select **HMEQ** in the list of available tables and click **OK**. This returns you to the New Project window.
- 8 In the New Project window, click **Advanced** below the **Description** text box. The Project Settings window appears. Select **Partition Data** in the upper left corner of the window.
 - 9 Ensure that the **Create partition variable** option is selected, and click **Save** in the lower right corner of the window. This returns you to the New Project window.
 - 10 In the lower right corner of the New Project window, click **Save**. You are redirected to the **Data** tab, where you can modify the variables in your data set.

Modify Variable Roles

On the **Data** tab, variable roles are indicated in the **Role** column. To change the role of a variable, select the variable by clicking the check box to the left of the variable name. Once you have selected the variable, in the options pane, expand **Role** and select the new role for the variable. Changes that you make to each variable are automatically applied and saved.

CAUTION

To avoid making unwanted changes to variable properties, you must manually deselect each variable that you modify when you are finished making changes to its properties.

For this example, change the the variable roles as follows:

- 1 On the **Data** tab, select variable **Bad**.
- 2 On the Options pane, expand the drop-down list under **Role** and select **Target**.
- 3 Ensure that all other variables are set to **Input**.

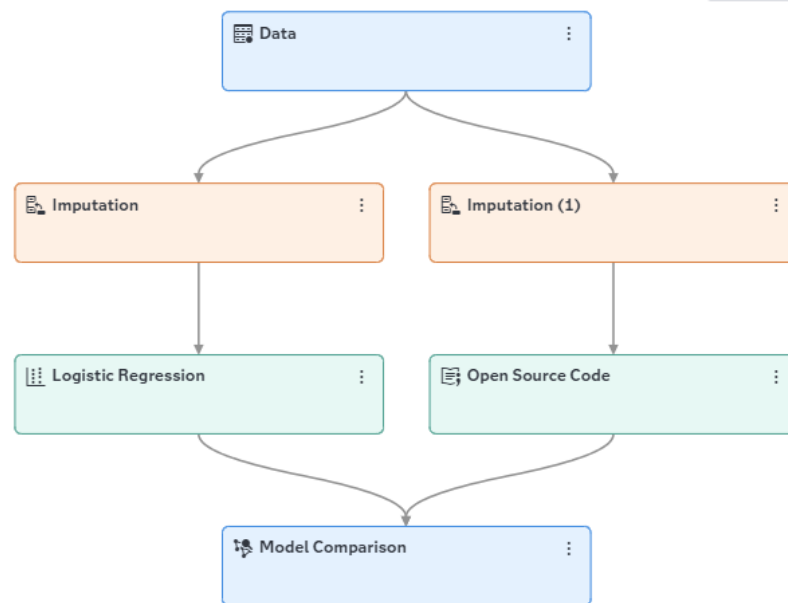
Create the Pipeline

This example requires you to complete the steps in the previous sections. This example also assumes that you have not created any other pipelines before starting this section. To create the pipeline that contains an open-source model:

- 1 Select the **Pipelines** tab in the upper left corner.

- 2 Right-click the **Data** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Imputation**.
- 3 Right-click the **Imputation** node and select **Add child node** ⇒ **Supervised Learning** ⇒ **Logistic Regression**.
- 4 Right-click the **Data** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Imputation**. This ensures that all missing data is imputed because some open-source packages cannot handle missing data.
- 5 Right-click the **Imputation** node that you added in step 4 and select **Add child node** ⇒ **Miscellaneous** ⇒ **Open Source Code**.
- 6 Right-click the **Open Source Code** node and select **Move** ⇒ **Supervised Learning**. This ensures that the node performs model assessment and can be compared to the **Logistic Regression** node.

Your current pipeline should resemble the following image.



- 7 Select the **Imputation** node that is above the **Open Source Code** node. In the options pane, select **Impute non-missing variables**.

After the pipeline is created, the open source code can be added to the **Open Source Code** node.

Add the Open Source Code

Use the **Open Source Code** node code editor to add your Python or R code as described in the following sections. For this example, follow the instructions for your preferred programming language.

Add Python Code

- 1 Select the **Open Source Code** node.
- 2 In the options pane, set the value of **Language** to **Python**.
- 3 Click **Open Code Editor**. Enter the following Python code in the training code editor:

```
from sklearn import ensemble

# Get full data with inputs + partition indicator
dm_input.insert(0, dm_partitionvar)
fullX = dm_inputdf.loc[:, dm_input]

# Dummy encode class variables
fullX_enc = pd.get_dummies(fullX, columns=dm_class_input,
drop_first=True)

# Create X (features/inputs); drop partition indicator
X_enc = fullX_enc[fullX_enc[dm_partitionvar] == dm_partition_train_val]
X_enc = X_enc.drop(dm_partitionvar, axis=1)

# Create y (labels)
y = dm_traindf[dm_dec_target]

#-----
# Training
#-----
# Fit Random Forest model w/ training data
params = {'n_estimators': 100, 'max_depth': 20, 'min_samples_leaf': 5}
dm_model = ensemble.RandomForestClassifier(**params)
dm_model.fit(X_enc, y)
print(dm_model)

#-----
# Scoring
#-----
# Score full data: posterior probabilities
fullX_enc = fullX_enc.drop(dm_partitionvar, axis=1)
dm_scoreddf_prob = pd.DataFrame(dm_model.predict_proba(fullX_enc),
columns=dm_predictionvar)

# Score full data: class prediction
dm_scoreddf_class = pd.DataFrame(dm_model.predict(fullX_enc),
columns=[dm_classtarget_intovar])

# Column merge posterior probabilities and class prediction
dm_scoreddf = pd.concat([dm_scoreddf_prob, dm_scoreddf_class], axis=1)
print(dm_scoreddf.head(10))

#-----
# Results
#-----
```

```
# Save VariableImportance to CSV
varimp = pd.DataFrame(list(zip(X_enc, dm_model.feature_importances_)),
                        columns=['Variable Name', 'Importance'])
varimp.to_csv(dm_nodedir + '/rpt_var_imp.csv', index=False)
```

Leave the scoring code editor empty.

Note: The scoring code editor is used when sending scoring code to SAS Model Manager, which does not apply in this example.

- 4 In the upper right corner of the code editor, click .
- 5 Click **Close**.
- 6 Right-click the **Model Comparison** node and select **Run**.
- 7 After the pipeline has successfully run, right-click the **Open Source Code** node and select **Results**.

Open Source Code Node Example Python > "Open Source Code" Results

"Open Source Code" Results

Summary Output Data

Node Assessment

rpt_var_imp.csv

Variable Name	Training Importance
IMP_CLAGE	0.0929
IMP_CLNO	0.0613
IMP_DEBTINC	0.3322
IMP_LOAN	0.0923
IMP_MORTDUE	0.0621
IMP_VALUE	0.0714
IMP_YOJ	0.0495
IMP_DELTINQ_1	0.0231
IMP_DELTINQ_2	0.0216

Python Code

```
1 #-----
2 # Language: PYTHON
3 #-----
4 # Variable declarations
5 dm_nodedir = '/opt/sas/viya/config/var/tmp/comprsv/default/6248da98-4685-49e7-a508'
6 dm_dec_target = 'BAD'
7 dm_partitionvar = '_PartInd_'
8 dm_partition_train_val = 1
9 dm_key = '_dmIndex_'
10
11 dm_pkiname = 'BIV4M63VRUDSSNE382ERKWFQ_pk1'
12 dm_pkipath = dm_nodedir + '/' + dm_pkiname
13 dm_scoreinputs_path = dm_nodedir + '/dm_opensrcScoreInputs.csv'
14 dm_scoreoutputs_path = dm_nodedir + '/dm_opensrcScoreOutputs.csv'
15
16 dm_class_input = ["IMP_DELTINQ","IMP_DEROG","IMP_30B","IMP_NINQ","IMP_REASON"]
17 dm_interval_input = ["IMP_CLAGE","IMP_CLNO","IMP_DEBTINC","IMP_LOAN","IMP_MORTDUE"]
```

Python Output

```
1 RandomForestClassifier(max_depth=20, min_samples_leaf=5)
2
3 0 0.207618 0.752382 1
4 1 0.268159 0.739841 1
5 2 0.198429 0.801571 1
6 3 0.428862 0.571138 1
7 4 0.538285 0.461715 0
8 5 0.319836 0.680164 1
9 6 0.173596 0.826404 1
10 7 0.346816 0.653184 1
11 8 0.298636 0.701370 1
12 9 0.412312 0.587688 1
13
```

Properties

Property Name	Property Value
codeLocation	mllearning
dataMiningVersion	V2025.06
dframe	true
envVar	DM_PYPATH
exactPctLift	true
exportWithFormat	true
fullDatasetReconstitution	false
osCode_Language	py
osCode file uri	/files/files/cf952c85-e31f-49c4-afed-

Output

Predicted Variables

Obs	Predicted Variable Name	Variable Name	Level Frequency	Level Frequency Percent	Raw Numeric Value	Raw Character Value	Formatted Value
1	I_BAD	BAD	-	-	-	-	-
2	P_BAD0	BAD	4771	80.0503	0		0
3	P_BAD1	BAD	1189	19.9497	1		1

Top 10 observations from the Input Data

- a Expand the **Python Code** results to view the actual code that was generated by SAS Model Studio and submitted. Notice that this code is a combination of precursor, user, and posterior code. The precursor and posterior code is added based on the node properties and whether the node is in the **Preprocessing** or **Supervised Learning** group.

- b On the **Assessment** tab, notice that assessment measures such as lift and ROC were computed for the open-source model.
 - c Click **Close** to close the Results window.
- 8 Right-click the **Model Comparison** node and select **Results**. The open-source model was chosen as the champion model based on having a better Kolmogorov-Smirnov statistic.

The **Assessment** tab lets you compare the results of the open-source model against the SAS logistic regression model .

- 9 Click **Close** to close the Results window. The example is now complete.

TIP You can find additional Python examples for the **Open Source Code** node in the GitHub repository at the following URL:

https://github.com/sassoftware/sas-viya-dmml-pipelines/tree/master/open_source_code_node

Add R Code

- 1 Select the **Open Source Code** node. In the options pane, set the value of **Language** to **R**.
- 2 In the options pane, click **Open Code Editor**. Enter the following code in the code editor:

```
# Build logistic regression model
dm_model <- glm(BAD ~ IMP_CLAGE + IMP_CLNO + IMP_DEBTINC +
  IMP_LOAN + IMP_MORTDUE + IMP_VALUE + IMP_YOJ,
  data=dm_traindf, family=binomial(link="logit"))

# View model output
summary(dm_model)

# Score
pred <- plogis(predict(dm_model, dm_inputdf))
dm_scoreddf <- data.frame(pred)
colnames(dm_scoreddf) <- c("P_BAD1")
dm_scoreddf$P_BAD0 <- 1 - dm_scoreddf$P_BAD1
```

- 3 In the upper right corner of the code editor, click .
- 4 Click **Close**.
- 5 Right-click the **Model Comparison** node and select **Run**.
- 6 After the pipeline has successfully run, right-click the **Open Source Code** node and select **Results**.

Open Source Code Node Example > "Open Source Code" Results

"Open Source Code" Results Close

Summary **Output Data**

Node **Assessment**

R Code

```

1 #-----
2 # Language: R
3 #-----
4 # Set R-work to node-work directory
5 dm_nodedir <- '/opt/sas/viya/config/var/tmp/compsrv/default
6 setwd(dm_nodedir)
7
8 # Variable declarations
9 dm_dec_target <- 'BAD'
10 dm_partitionvar <- '_PartInd_'
11 dm_partition_train_val <- 1
12
13 dm_class_input <- c("IMP_DELIQ", "IMP_DEROG", "IMP_JOB", "IMP
14 dm_interval_input <- c("IMP_CLAGE", "IMP_CLNO", "IMP_DEBTINC"
15 dm_input <- c(dm_class_input, dm_interval_input)
16 dm_model_formula <- as.formula(paste(dm_dec_target, '~', pa
17 dm_predictionvar <- c('P BAD0', 'P BAD1')

```

R Output

```

1
2 Call:
3 glm(formula = BAD ~ IMP_CLAGE + IMP_CLNO + IMP_DEBTINC + IM
4     IMP_MORTDUE + IMP_VALUE + IMP_YOJ, family = binomial(lf
5     data = dm_traindf)
6
7 Coefficients:
8             Estimate Std. Error z value Pr(>|z|)
9 (Intercept) -1.528e+00  2.623e-01  -5.827 5.63e-09 ***
10 IMP_CLAGE   -5.234e-03  6.359e-04  -8.231 < 2e-16 ***
11 IMP_CLNO     9.728e-03  4.735e-03   2.055 0.03992 *
12 IMP_DEBTINC  3.793e-02  6.757e-03   5.613 1.98e-08 ***
13 IMP_LOAN    -2.112e-05  4.637e-06  -4.554 5.27e-06 ***
14 IMP_MORTDUE -5.119e-06  1.751e-06  -2.923 0.00347 **
15 IMP_VALUE    3.535e-06  1.298e-06   2.723 0.00647 **
16 IMP_YOJ     -9.025e-03  6.505e-03  -1.387 0.16534
17 ---

```

Properties

Property Name	Property Value
codeLocation	mllearning
dataMiningVersion	V2025.06
dframe	true
envVar	DM_PYPATH
exactPctlLift	true
exportWithFormat	true
fullDatasetReconstitution	false
osCode_Language	r
osCode_file_uri	/files/files/9e02d33a-eab9-4204-

Output

Top 10 observations from the Input Data

Obs	BAD	_dmindex	Partition Indicator	Imputed CLAGE	Imputed CLNO	Imputed DEBTINC	Impu DEL
1	1	1000001	0	94.367	9.0000	33.7292	
2	1	1000002	1	121.833	14.0000	33.7292	
3	1	1000003	0	149.467	10.0000	33.7292	
4	1	1000004	1	180.990	21.4115	33.7292	
5	0	1000005	1	93.333	14.0000	33.7292	
6	1	1000006	1	101.466	8.0000	37.1136	
7	1	1000007	1	77.100	17.0000	33.7292	

- a Expand the **R code** results to view the actual code that was generated by SAS Model Studio and submitted. Notice that this code is a combination of precursor, user, and posterior code. The precursor and posterior code is added based on the node properties and whether the node is in the **Preprocessing** or **Supervised Learning** group.
 - b On the **Assessment** tab, notice that assessment measures such as lift and ROC were computed for the open-source model.
 - c Click **Close** to close the Results window.
- 7 Right-click the **Model Comparison** node and select **Results**. The SAS logistic regression model was chosen as the champion model based on having a better Kolmogorov-Smirnov statistic.

The **Assessment** tab lets you compare the results of the open-source model against the SAS logistic regression model .

- 8 Click **Close** to close the Results window. The example is now complete.

TIP You can find additional R examples for the **Open Source Code** node in the GitHub repository at the following URL:

https://github.com/sassoftware/sas-viya-dmml-pipelines/tree/master/open_source_code_node

Risk Modeling Example

Risk Modeling Overview

Note: The Risk Modeling Add-on is not included with SAS Viya. If your site has not licensed Risk Modeling Add-on for SAS Viya, the risk modeling nodes do not appear in your SAS Viya software.

The Risk Modeling Add-on for SAS Viya is used to evaluate the level of risk associated with applicants or customers. It provides statistical odds, or probability, that an applicant with any given score is good or bad. In its simplest form, a scorecard is built from a number of characteristics (that is, input or predictor variables). Characteristics are initially grouped into attributes. For example, age is a characteristic, and 25-33 is an attribute. Each attribute is associated with a number of scorecard points. These scorecard points are statistically assigned to differentiate risk, based on the predictive power of the characteristic variables, correlation between the variables, and business considerations.

The accepts data set is used to train the initial scorecard model. It consists of just the credit applicants who have been previously extended credit. The accepts data set labels the applicants as event or nonevent (that is, good or bad). The rejects data set is used to remedy selection bias in your data. The rejects data contains credit applicants that were denied credit. The rejects data is scored using the initial scorecard to infer whether the rejected applicants would have been good or bad credit risks. These observations are added to the accepts data and this augmented data set is used to build the final scorecard model.

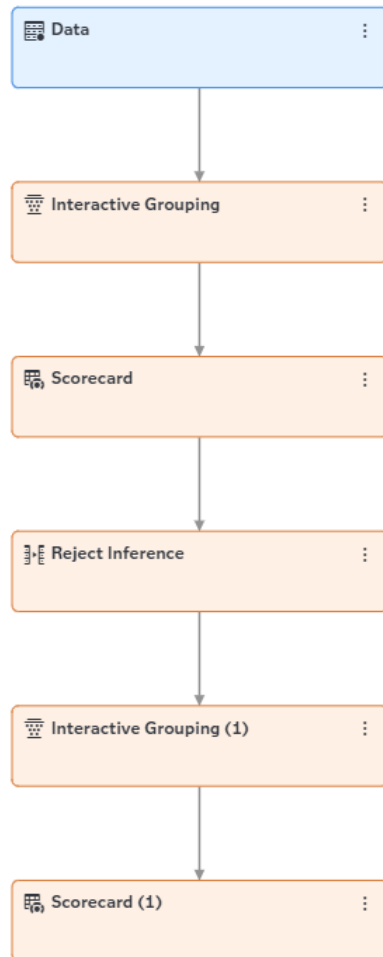
This example uses accepts and rejects data sets that are called `rm_accepts` and `rm_rejects`. You can download the data for this example from the [SAS Viya Machine Learning product page](#). Click **Data for Risk Modeling Example**. Download the ZIP file and extract `rm_accepts.SAS7BDAT` and `rm_rejects.SAS7BDAT` to a directory that your SAS Visual Data Mining and Machine Learning server can access.

To build a scorecard model in this example:

- 1 Group the characteristic variables into attributes.
- 2 Use a logistic regression model to create an initial scorecard.

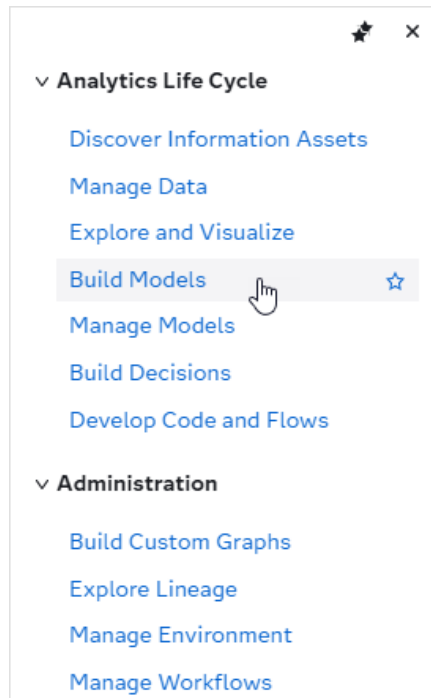
- 3 Perform reject inference on the initial model.
- 4 Create the final scorecard using the information that is obtained in the previous steps.

The finished pipeline diagram should resemble the one that is shown here:



Create the Project and Import the Input Data

This example assumes that you are signed in to the Welcome to SAS Viya application. In the upper left corner of the Welcome to SAS Viya page, click  and select **Build Models**.



You are directed to the Projects page. Here is how to create the project that is used in this example:

- 1 Click **New Project** in the upper right corner of the Projects page.
- 2 Enter **Risk Modeling Example** for **Name** in the New Project window.

New Project

Name: *
Risk Modeling Example

Type: *
Data Mining and Machine Learning

Template:
Blank template Browse

Data: *
Browse

Location:
My Folder Browse

Description:

Advanced

Save Cancel

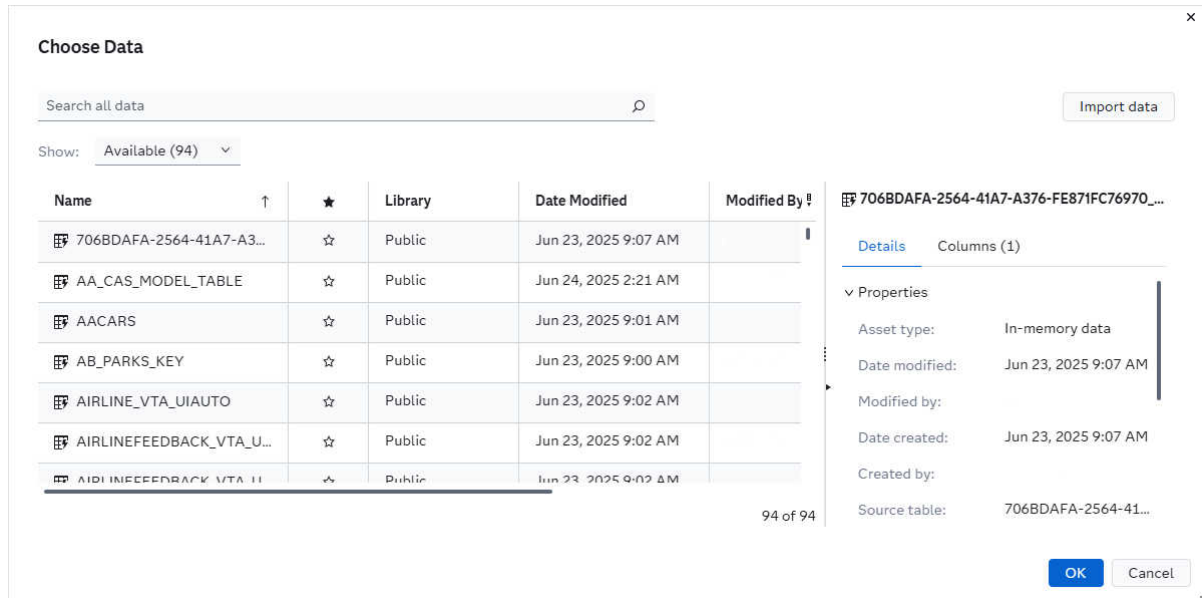
- 3 Select **Data Mining and Machine Learning** for **Type**.

Note: **Forecasting** and **Text Analytics** are additional options if you have licensed those products.

- 4 In the **Template** field, click **Browse**. The Browse Templates window appears.
- 5 In the Browse Templates window, select **Basic template for risk modeling** and click **OK**.

Note: When adding a Scorecard node to a pipeline, the Scorecard node is specified as a miscellaneous node. However, when using the basic template for risk modeling, the Scorecard node is specified as a supervised learning node.

- 6 In the **Data** field, click **Browse**. The Choose Data window appears.



- 7 In the upper right corner of the Choose Data window, click **Import data**. The Import Data window appears.
- 8 In the left pane of the Import Data window, click **Local files** and navigate to the folder where `rm_accepts.sas7bdat` is stored. Select `rm_accepts.sas7bdat` and click **Open**.
- 9 Click **Import item** in the upper right corner of the Import Data window. When the data is successfully imported, a note appears, saying that the data is ready for use.

✓ The table was successfully imported on Oct 7, 2019 10:33 AM and is ready for use.

- 10 Once the data set is successfully imported, click **Add** in the lower right corner of the Import Data window. This returns you to the Choose Data window.
- 11 In the Choose Data window, select **RM_SELECTS** in the list of available tables and click **OK**. This returns you to the New Project window.
- 12 In the New Project window, click **Save**. This directs you to the **Data** tab, where you can modify the variables in your data set.

Modify Variable Roles

On the **Data** tab, variable roles are indicated in the **Role** column. To change the role of a variable, select the variable by clicking the check box to the left of the variable name. Once you have selected the variable, in the options pane, expand **Role** and select the new role for the variable. Changes that you make to each variable are automatically applied and saved.

CAUTION

To avoid making unwanted changes to variable properties, you must manually deselect each variable that you modify when you are finished making changes to its properties.

For this example, change the the variable roles as follows:

- 1 On the **Data** tab, select variable **GB**.
- 2 On the Options pane, expand the drop-down list under **Role** and select **Target**.
- 3 Ensure that all other variables are set to **Input**.

Group the Characteristic Variables into Attributes

You first want to group the characteristic values into attributes that can be used as input variables for your scorecard model:

- 1 Click the **Pipelines** tab in the upper left corner.
- 2 Right-click the **Interactive Grouping** node and select **Run**.
- 3 When the node has run successfully, right-click the node and select **Results** to view detailed information about the variable groups that were created:
 - **Output Variables** — Displays all characteristics and their corresponding information value (IV) and Gini statistic. A report is created for both the training partition and the validation partition. IV is a measure of the predictive value of a characteristic and the Gini statistic is used to measure how equal the event rates are across the attributes of a characteristic. In this example, characteristics with an IV greater than 0.1 are used as input variables in subsequent modeling steps. All other attributes are rejected. In the options pane, the **Variable Selection Options** enable you to adjust the variable selection method and cutoff values.

Training Output Variables

Variable	Gini St...	Inform...	Level f...	Calcula...	New Role	Pre-De...	Level	M
AGE	33.3240	0.3780	INTERVAL	INPUT	DEFAULT		INTERVAL	
STATUS	23.9680	0.2330	NOMINAL	INPUT	DEFAULT		NOMINAL	
INC1	24.0280	0.2040	NOMINAL	INPUT	DEFAULT		NOMINAL	
TMJOB1	22.3940	0.1950	INTERVAL	INPUT	DEFAULT		INTERVAL	
CARDS	18.3330	0.1680	NOMINAL	INPUT	DEFAULT		NOMINAL	
INCOME	22.4510	0.1650	INTERVAL	INPUT	DEFAULT		INTERVAL	
INC	20.9420	0.1560	NOMINAL	INPUT	DEFAULT		NOMINAL	
PERS_H	19.8870	0.1530	NOMINAL	INPUT	DEFAULT		NOMINAL	
EC CARD	16.5560	0.1430	BINARY	INPUT	DEFAULT		BINARY	

Validation Output Variables

Variable	Gini St...	Inform...	Level f...	Calcula...	New Role	Pre-De...	Level	M
AGE	34.5410	0.4330	INTERVAL	INPUT	DEFAULT		INTERVAL	
INCOME	30.5550	0.3170	INTERVAL	INPUT	DEFAULT		INTERVAL	
INC1	29.1380	0.3060	NOMINAL	INPUT	DEFAULT		NOMINAL	
INC	28.1790	0.2910	NOMINAL	INPUT	DEFAULT		NOMINAL	
STATUS	24.1150	0.2330	NOMINAL	INPUT	DEFAULT		NOMINAL	
PERS_H	25.5720	0.2310	NOMINAL	INPUT	DEFAULT		NOMINAL	
TMJOB1	24.2730	0.2280	INTERVAL	INPUT	DEFAULT		INTERVAL	
CARDS	19.7780	0.2030	NOMINAL	INPUT	DEFAULT		NOMINAL	
EC CARD	17.5560	0.1750	BINARY	INPUT	DEFAULT		BINARY	

- **Event Rate Plot** — Displays the event rate for each variable group. A report is created for both the training partition and the validation partition.



- **Statistics Table** — Displays several statistics such as the event count, nonevent count, and weight of evidence (WOE) for each variable group. A report is created for both the training partition and the validation partition.

Training Statistics Table

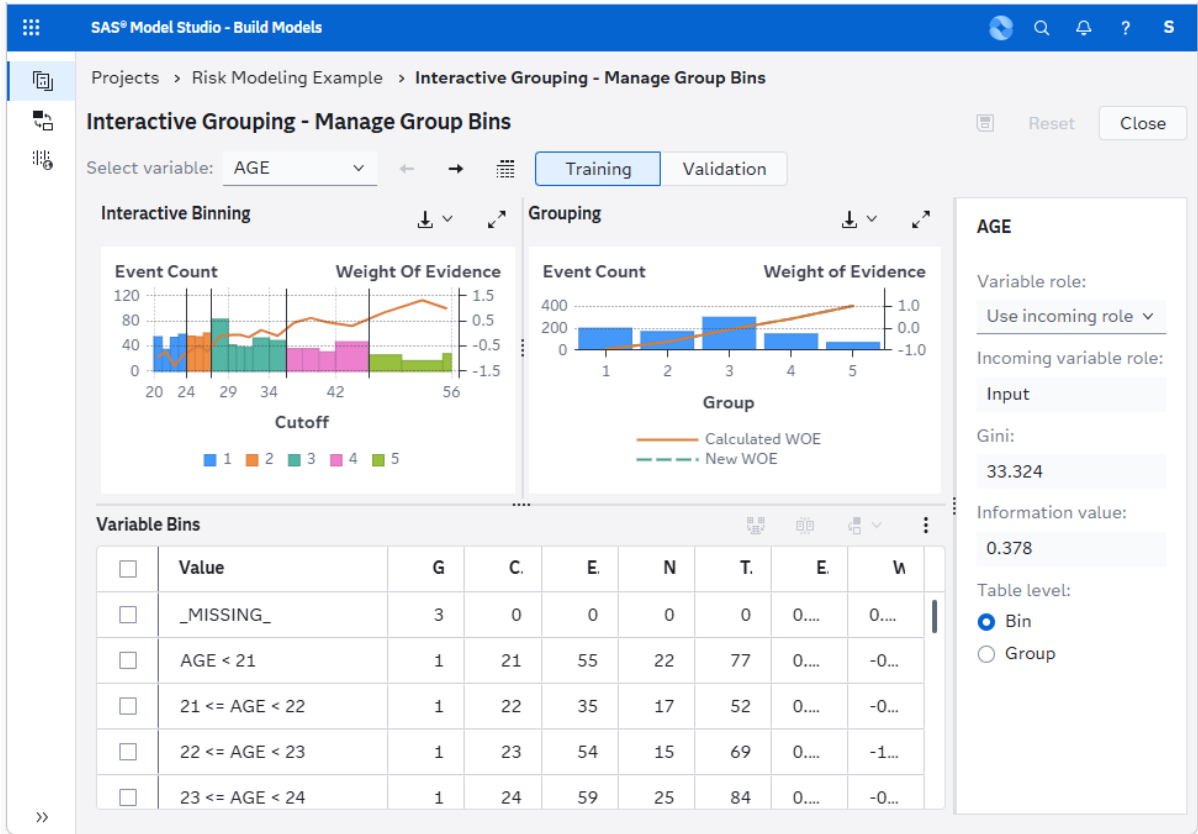
Variable	Group	Label	Event C...	Non-Ev...	Weight...	Manual...	Group ...	G
AGE	1	AGE < 24	203	79	-0.9438	.	0.7200	
AGE	2	24 <= AGE	172	93	-0.6149	.	0.6490	
AGE	3	27 <= AGE	304	291	-0.0437	.	0.5110	
AGE	4	36 <= AGE	150	236	0.4532	.	0.3890	
AGE	5	46 <= AGE	71	201	1.0406	.	0.2610	
BUREAU	1	1.00 2.00	610	592	-0.0300	.	0.5070	
BUREAU	2	3.00	290	308	0.0602	.	0.4850	
CAR	1	Without	245	150	-0.4906	.	0.6200	
CAR	2	Car Car	655	750	0.1354	.	0.4660	

Validation Statistics Table

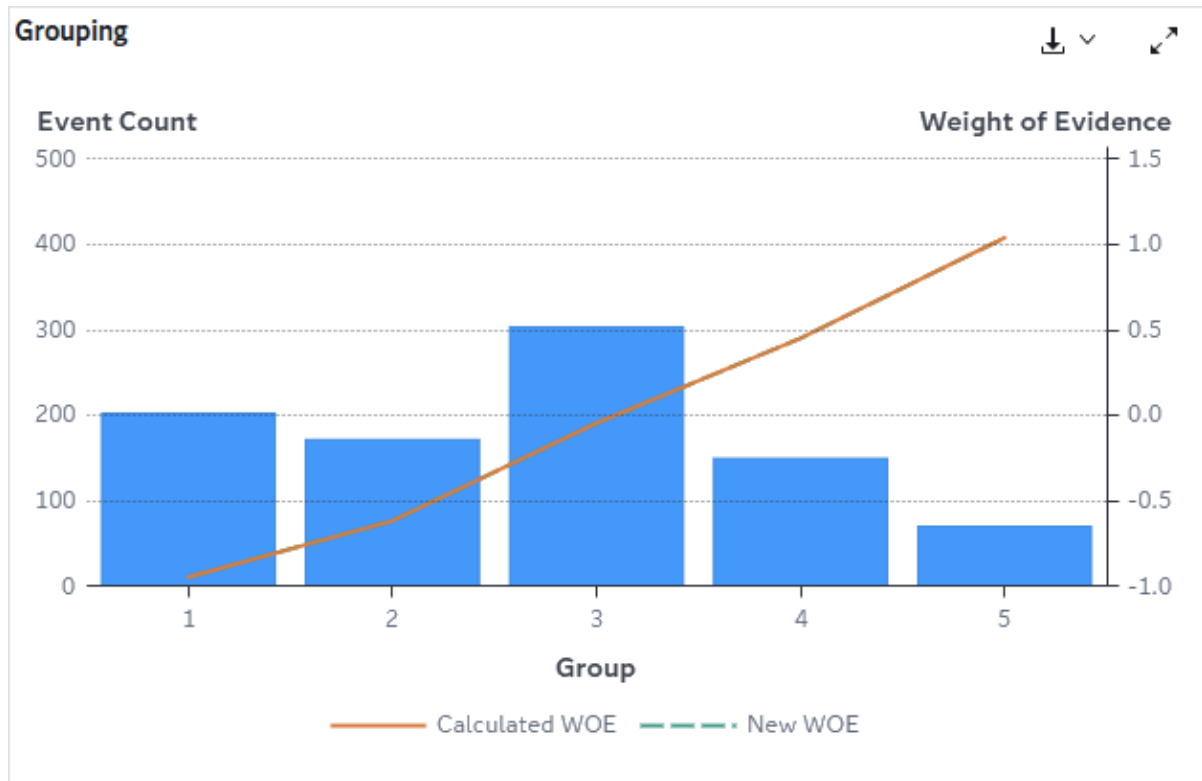
Variable	Group	Label	Event C...	Non-Ev...	Weight...	Manual...	Group ...	G
AGE	1	AGE < 24	107	29	-1.3055	.	0.7870	
AGE	2	24 <= AGE	72	41	-0.5631	.	0.6370	
AGE	3	27 <= AGE	140	142	0.0142	.	0.4960	
AGE	4	36 <= AGE	84	123	0.3814	.	0.4060	
AGE	5	46 <= AGE	47	115	0.8948	.	0.2900	
BUREAU	1	1.00 2.00	301	282	-0.0652	.	0.5160	
BUREAU	2	3.00	149	168	0.1200	.	0.4700	
CAR	1	Without	118	100	-0.1655	.	0.5410	
CAR	2	Car Car	332	350	0.0528	.	0.4870	

- 4 Click **Close** to close the Results window.
- 5 After running the node and viewing the results, you might decide that you want to adjust the variable groups that were generated for operational or business reasons. To open the Interactive Grouping editor and manually adjust variable groups, right-click the **Interactive Grouping** node, and select **Manage Group Bins**.

The Interactive Grouping editor opens:



The Grouping chart displays the event count and WOE for each group of the characteristic AGE. WOE measures the strength of an attribute of a characteristic in differentiating good risk and bad risk applicants. Negative values indicate that the particular group contains a higher proportion of bad risk applicants than good risk applicants. That is, negative WOE values indicate that the applicants in the group present a greater risk.



The Interactive Binning chart displays the event count and WOE for each bin of the variable AGE.



- The Interactive Grouping editor enables you to split, merge, or change the groups of the selected bins. Each time you adjust a variable bin, the Gini statistic and IV of the characteristic are updated in the right pane.

To split a bin:

- a Right-click the bar that corresponds with the group **46<= AGE <50**, and select **Split bin**. The Split Bin window appears.
- b Enter **48** for the **New Cutoff Value** and click **Save**. Group 5 now has an additional bin with a cutoff value of 48.

To merge bins:

- a Select the three bins in group 2 by clicking on the first bin and dragging the cursor across the other two bins.
- b Right-click the selected bins and select **Merge bins**. Group 2 now contains just one bin.

To change the group of selected bins:

- a Select the last two bins in group 3.
- b Right-click on the selected bins and select **Change group** ⇒ **Create new group**. There are now six groups.

Note: When you click **Change group**, you can also merge the selected bins into existing groups.

- 7 Click  and then click **Close** to exit the Interactive Grouping editor.

Build an Initial Scorecard

Next, you want to use the grouped characteristics to build your initial scorecard. You can choose to use the WOE or GRP variables that were generated by the **Interactive Grouping** node for the analysis. The WOE variables contain the weight of evidence of each binned variable and the GRP variables contain the group ID.

The **Scorecard** node generates the scorecard with logistic regression and scaling. The regression coefficients are used to scale the scorecard. Scaling makes the scorecard conform to a particular range of scores. The two main elements of scaling a scorecard are determining the odds at a certain score and determining the points that are required to double the odds. The **Odds**, **Scorecard Points**, and **Points to Double the Odds** properties control the scaling.

This example uses the default settings for the logistic regression and scaling:

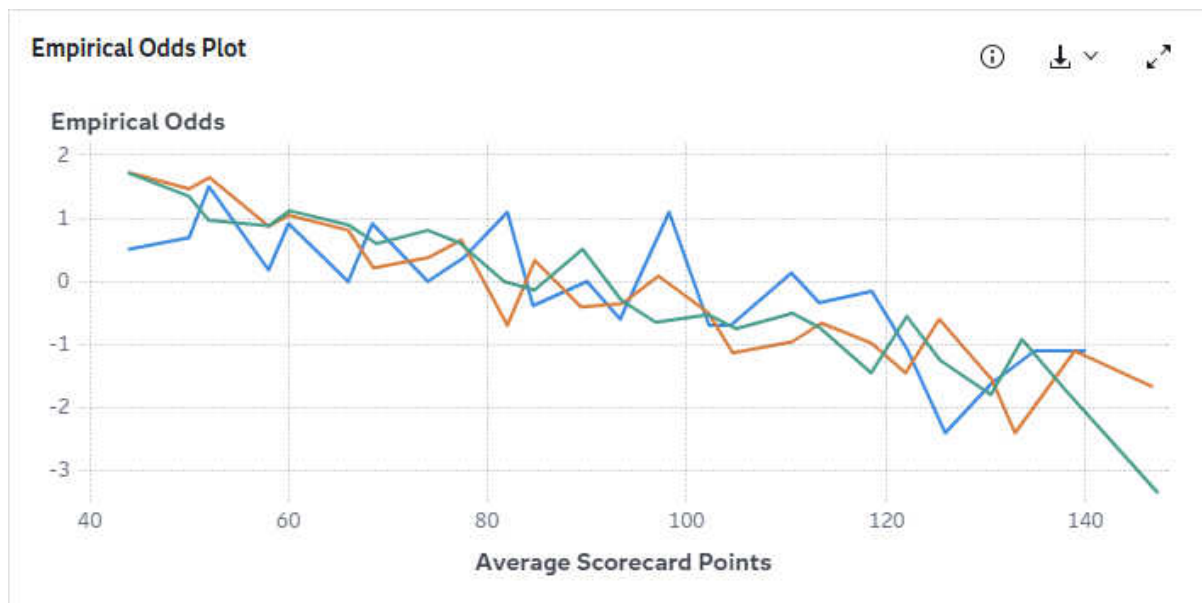
- 1 Select the **Scorecard** node.
- 2 In the options pane, set **Scorecard type** to **Detailed**.
- 3 Right-click the **Scorecard** node and select **Run**.
- 4 When the node has run successfully, right-click the node and select **Results** to view detailed information in each of the following sections of interest:

- **Scorecard** — Displays the scorecard points, WOE, event rate (percentage of bad risk applicants in that score range), percent (percentage of bad risk applicants that have a score higher than the lower limit of the score range), and the regression coefficient for each attribute.

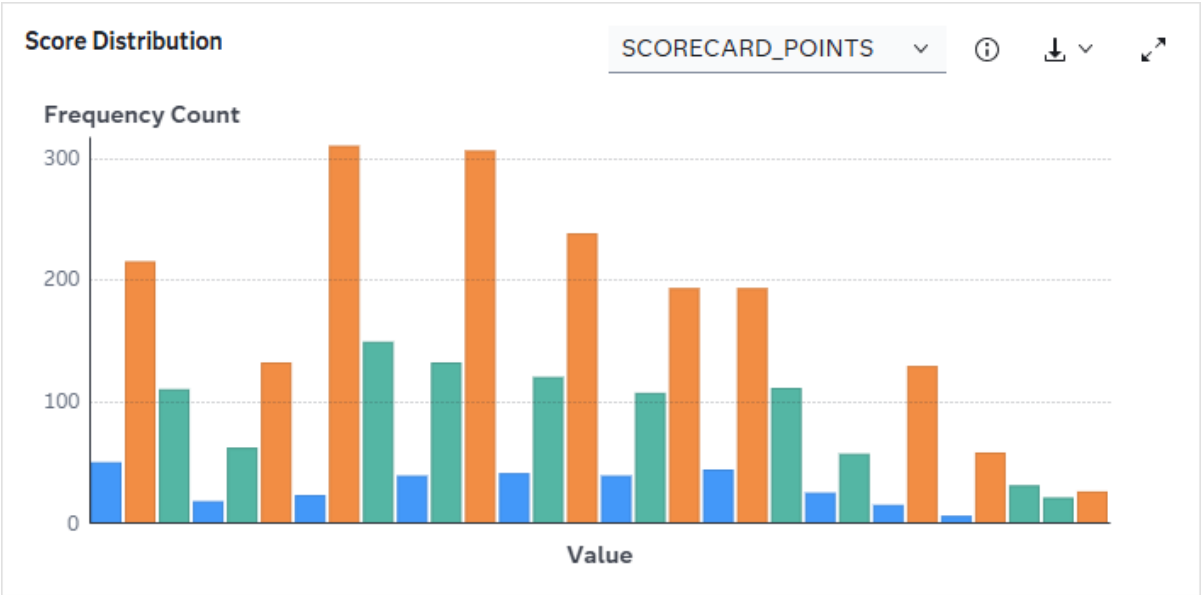
Scorecard

		Group	Scorecard Points	Weight of Evidence	Event Rate GB = 1	Percent	Coefficient
Age	AGE < 24	1	5	-0.94	72.75	15.90	-0.63
	24 <= AGE < 27	2	11	-0.61	63.88	13.93	-0.63
	27 <= AGE < 36 _MISSING_	3	21	-0.04	50.36	32.57	-0.63
	36 <= AGE < 40	4	32	0.54	38.58	10.80	-0.63
	40 <= AGE < 46	5	29	0.37	41.18	11.33	-0.63
	46 <= AGE	6	41	1.04	27.80	15.47	-0.63

- **Empirical Odds Plot** — Displays the empirical odds plotted against the average values of the scorecard points. The negative slope implies that good risk applicants have larger scores than bad risk applicants.



- **Score Distribution** — Located on the **Assessment** tab. Displays the distribution of the scorecard points, the event odds, and the log of the event odds.

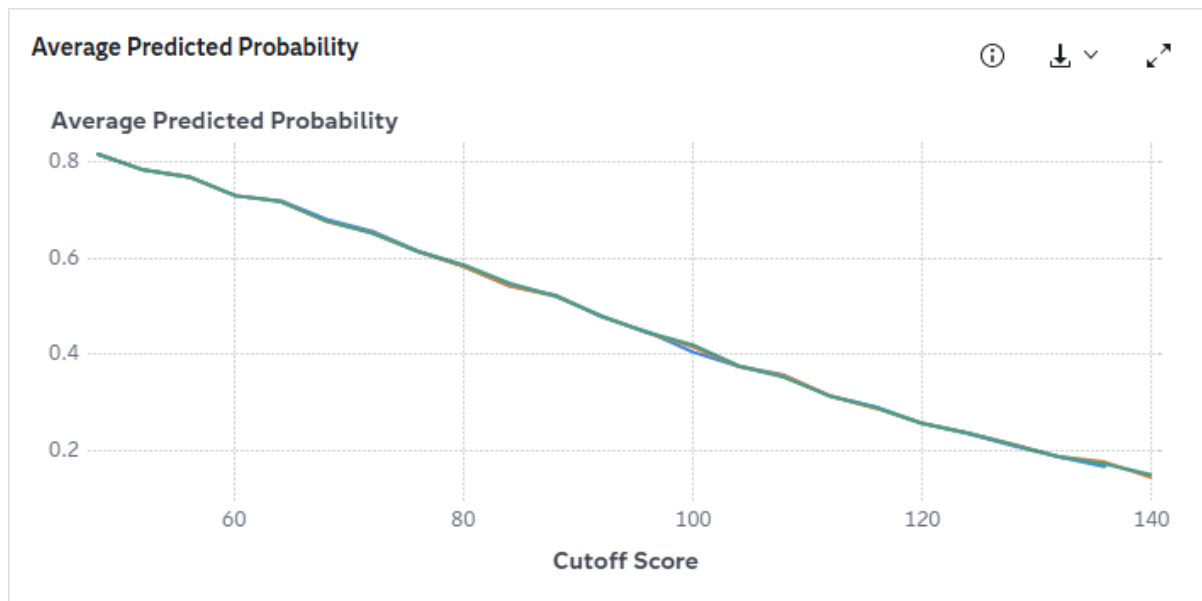


- **Gains Table** — Located on the **Assessment** tab. Displays several statistics for each score bucket.

Gains Table ⬇ ⬆

Data Role	Bucket	Score Bucket	Count	Event Count	N
TRAIN	25	Score >= 140	29	1	
TRAIN	24	136 <= Score < 140	13	2	
TRAIN	23	132 <= Score < 136	7	2	
TRAIN	22	128 <= Score < 132	49	7	
TRAIN	21	124 <= Score < 128	63	14	
TRAIN	20	120 <= Score < 124	52	19	
TRAIN	19	116 <= Score < 120	42	8	
TRAIN	18	112 <= Score < 116	77	25	
TRAIN	17	108 <= Score < 112	77	22	

- **Average Predicted Probability** — Located on the **Assessment** tab. As the score increases, the predicted probability of a bad risk applicant decreases.



5 Click **Close** to close the Results window.

Perform Reject Inference

The accepts data set that is used to develop a credit scoring model is structurally different from the through-the-door population to which the credit scoring model is applied. The number of events and nonevents in the target variable that is created for the credit scoring model is based on the records of applicants who were all accepted for credit. However, the population to which the credit scoring model is applied includes applicants who would have been rejected under the scoring rules that were used to generate the initial model.

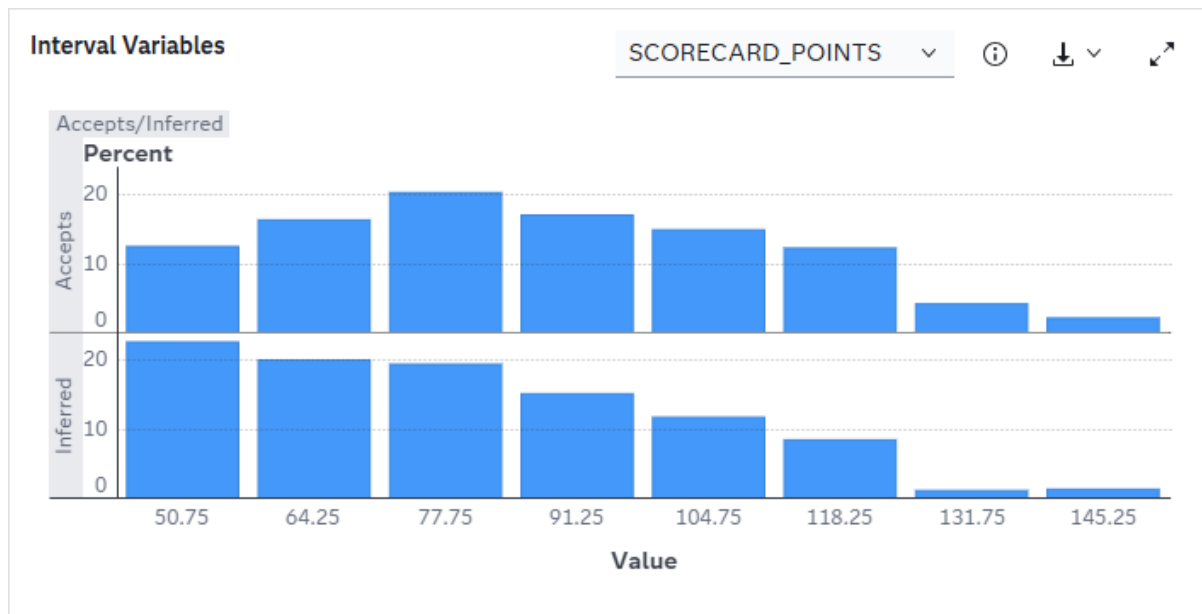
The reject inference approach uses the model that was trained using the accepted applications to score the rejected applications. The observations in the rejects data set are classified as inferred events and inferred nonevents. The inferred observations are then added to the accepts data set, which contains the actual event and nonevent records, to form an augmented data set. This augmented data set, which represents the through-the-door population, serves as the training data set for a second scorecard model.

To perform reject inference:

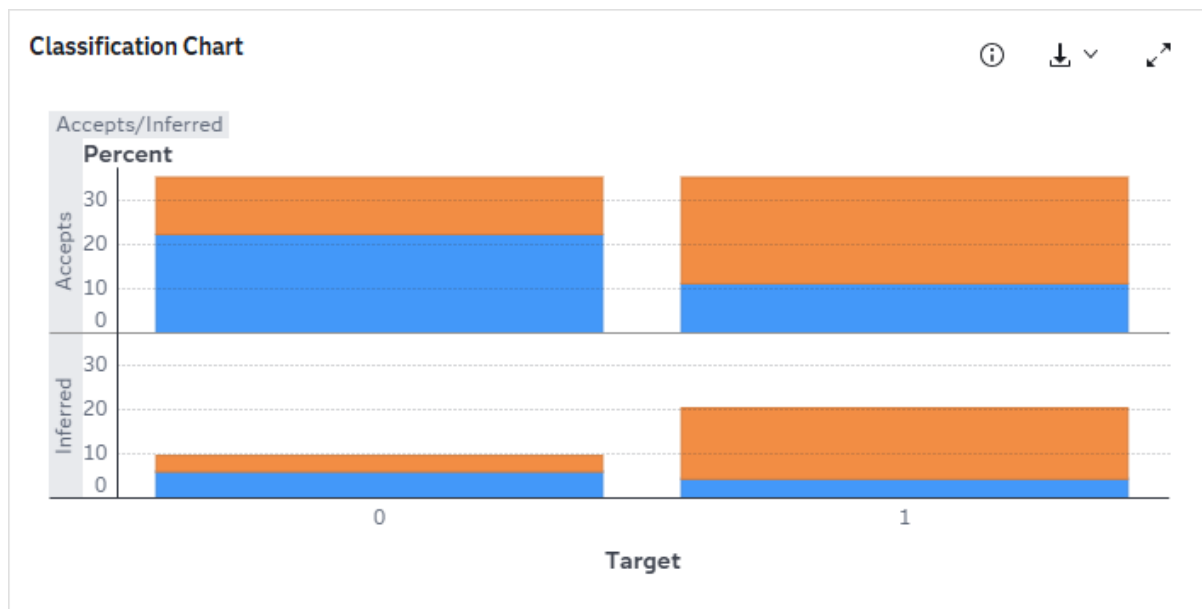
- 1 Right-click the **Scorecard** node and select **Move** ⇒ **Preprocessing**. Data preprocessing nodes, such as the Reject Inference node, cannot be added as a child node when the Scorecard node is specified as a supervised learning node. When moving the Scorecard node, notice that the Model Comparison node is removed from the pipeline. This is because the Scorecard node does not generate a model that can be used by subsequent nodes when specified as a preprocessing node.
- 2 Right-click the **Scorecard** node and click **Run**.

- 3 When the node run is complete, right-click the **Scorecard** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Reject Inference**.
- 4 Import the rejects data set:
 - a Select the Reject Inference node. In the options pane, under **Score data table name**, click **Browse**. The Choose Data window appears.
 - b In the upper right corner of the Choose Data window, select **Import data**. The Import Data window appears.
 - c In the left pane of the Import Data, click **Local files** and navigate to the folder where file `rm_rejects.sas7bdat` is stored. Select **rm_rejects.sas7bdat** and click **Open**.
 - d Click **Import item** in the upper right corner of the Import Data window. When the data is successfully imported, a note appears, saying that the data is ready for use.

 The table was successfully imported on Oct 7, 2019 10:33 AM and is ready for use.
 - e Once the data set is successfully imported, click **Add** in the lower right corner of the Import Data window. This returns you to the Choose Data window.
 - f In the Choose Data window, select **RM_REJECTS** in the list of available tables and click **OK**.
- 5 In the options pane, set **Inference method** to **Parceling**. The parceling method distributes the scored rejects into equal sized buckets that are defined by the **Score range method** that you specify. This example uses the default method to define the buckets based on the score range of the `rm_accepts` data set. The scored rejects are then randomly classified as an event or nonevent.
- 6 In the options pane, set **Event rate increase** to **1.2**. The proportion of event and nonevents in the `rm_rejects` data set is not expected to approximate the proportion of event and nonevents in the `rm_accepts` data set. You would expect that the event rate of the `rm_rejects` data set is higher than that of the `rm_accepts` data set. A value of 1.2 specifies that the event rate in the rejects data should be 20% greater than what is observed in the accepts data.
- 7 Right-click the **Reject Inference** node and select **Run**.
- 8 When the node has run successfully, right-click the node and select **Results** to view detailed information in each of the following sections of interest:
 - **Interval Variables** — Displays the distribution of the scorecard points, probability of the event, and probability of the nonevent for the accepts and inferred data sets.



- **Classification Chart** — Displays the distribution of the predicted event and nonevent for the accepts and inferred data sets.



- 9 Click **Close** to close the Results window.

Build the Final Scorecard

To build the final scorecard, you must repeat the grouping and scorecard creation steps on the augmented data set:

- 1 Right-click the **Reject Inference** node and select **Add child node** ⇒ **Data Mining Preprocessing** ⇒ **Interactive Grouping**.

- 2 Right-click the **Interactive Grouping** node and select **Add child node** ⇒ **Miscellaneous** ⇒ **Scorecard**.
- 3 Select the **Scorecard (1)** node.
- 4 In the options pane, set **Scorecard type** to **Detailed**.
- 5 Right-click the **Scorecard (1)** node and select **Run**.
- 6 When the node has run successfully, right-click the node and select **Results** to view detailed information about the final scorecard that is based on both accepted and rejected applicants.

Scorecard

		Group	Scorecard Points	Weight of Evidence	Event Rate GB = 1	Percent	Coefficient
Age	AGE < 23	1	-2	-1.22	81.69	14.73	-0.62
	23 <= AGE < 27	2	9	-0.66	69.91	18.58	-0.62
	27 <= AGE < 36 _MISSING_	3	21	0.04	54.02	31.11	-0.62
	36 <= AGE < 46	4	30	0.53	42.80	20.31	-0.62
	46 <= AGE	5	39	1.04	31.20	15.27	-0.62
Credit Cards	American Express VISA Citibank 						

- 7 Click **Close** to close the Results window.

Managing Projects

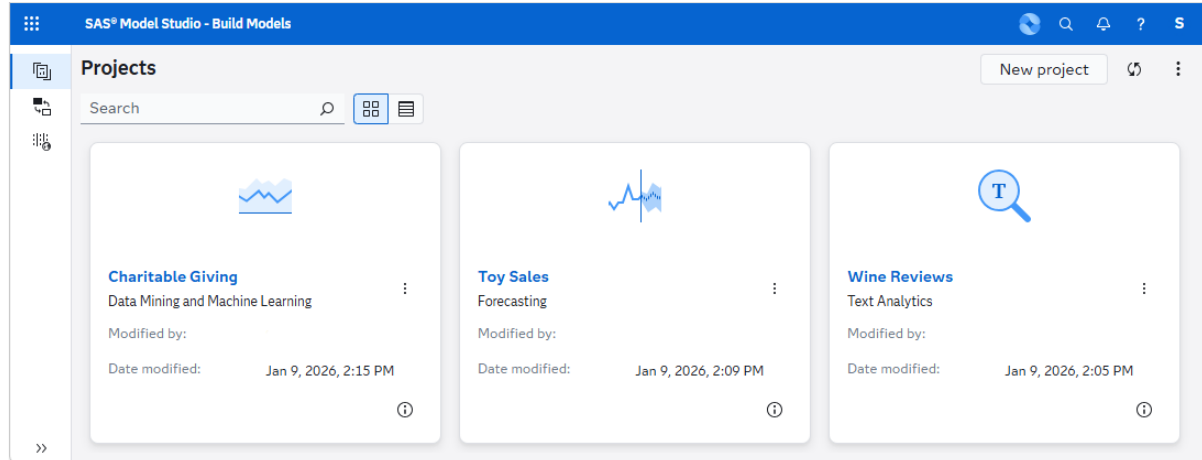
<i>Overview of SAS Model Studio Projects</i>	75
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Overview of SAS Model Studio Projects

A *project* is a top-level container for your analytic work in SAS Model Studio. You can view projects in the SAS Model Studio Projects page.

SAS Model Studio projects can be one of three types: Forecasting projects, Data Mining and Machine Learning projects, and Text Analytics projects. The project types that appear in your SAS Model Studio installation depend on the SAS licensing for your site.

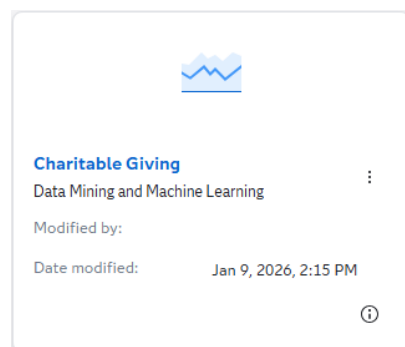
Depending on your project filter setting, existing projects in your environment appear either as graphic tiles or rows in a table of projects.



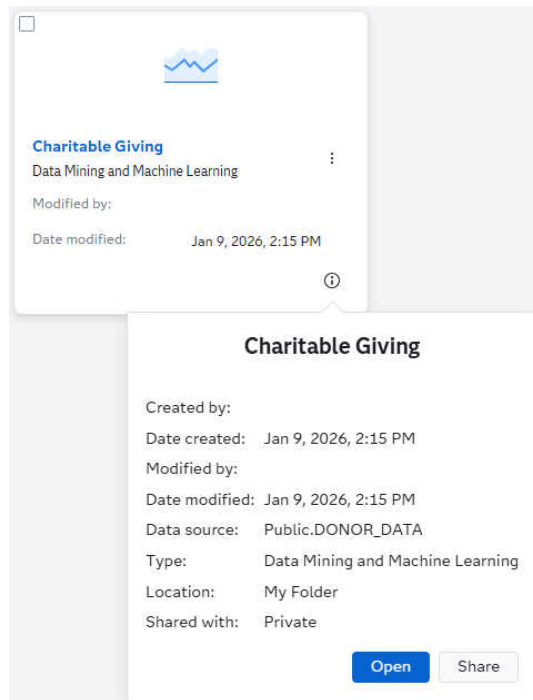
To alternate between table and tile project displays:

- Click  near the top of the page to show existing SAS Model Studio projects in a graphic tile matrix.
- Click  near the top of the page to show SAS Model Studio projects in a tabular list.

A SAS Model Studio project contains the data source, the pipelines that you create, and related project metadata (such as project type, project creator, share list, and last update history). If you create more than one pipeline in your project, analytic results that compare the performance of multiple pipelines are also stored in the project.



Click  in the lower right corner of the project tile to view additional information about the project.



Changing Ownership of a Project in SAS Model Studio

When you share a project, you can assign Read and Write permissions to other users. But there are certain tasks that have to be performed by the owner of the project. There are times when the owner is not available. Follow these steps to change ownership.

This process requires the SAS Viya command line interface (CLI). See [“Command-Line Interface: Overview” in SAS Viya Platform: Using the Command-Line Interface](#) for instructions for getting this CLI set up on your deployment before starting this task.

- 1 Locate the project on the Projects page of SAS Model Studio.
- 2 Click  on the project tile and select **Export** from the menu. If you are using the table view of the Projects page, select the project from the row and use the  at the top right corner of the page.

This opens a message stating that the project is being prepared for export. When it is ready, you are prompted to specify the location on your hard drive to download the project.

- 3 Transfer the downloaded ZIP file to the location where the SAS Viya CLI is installed.

- 4 Run this command to install the common-anaytics plug-in.

```
./sas-viya plugins install --repo SAS common-analytics
```

- 5 Run the SAS Viya command with the common-anaytics plug-in to change the ownership of the project.

```
./sas-viya common-analytics project change-user --source projectname.zip --user userid1
```

In this example, *projectname.zip* is the name of the project ZIP file and *userid1* is the ID for the user who you want to transfer ownership.

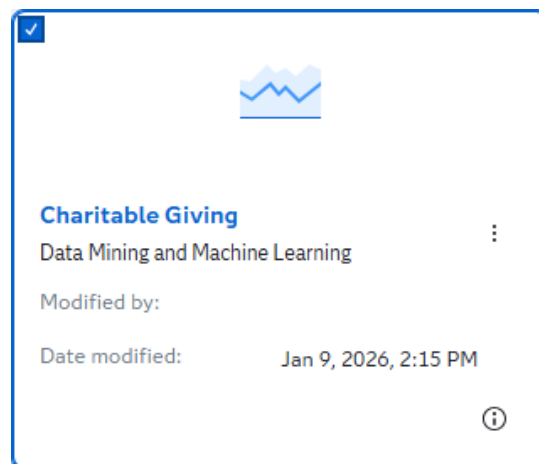
When the command completes, you have a new ZIP file with the name ***projectname_userid1.zip***.

- 6 Import the new ZIP file to SAS Model Studio.

The project is an exact copy of the original project, except that the ownership has changed.

Opening an Existing Project

You use the SAS Model Studio Projects page to access existing projects. If your SAS Model Studio Projects page displays project tiles, double-click the tile that you want to open, or click the link to the project name within the tile. SAS Model Studio opens the selected project.



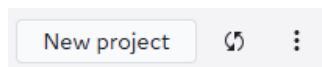
If your SAS Model Studio Projects page displays a project table, double-click the row that contains the desired project, or click the link of the project name to open it. SAS Model Studio opens the selected project to the last visited tab.

Projects								
Search			New project					
☐	Name ↑	Type	Modified By	Date Modified	Created By	Data Source	Permissions	Location
☒	Charitable Giving	Data Mining and Machine Learning		Jan 9, 2026, 2:15 PM		Public.DONOR_DATA		My Folder
☐	Toy Sales	Forecasting		Jan 9, 2026, 2:09 PM		Public.INSIGHTTOYDEMO		My Folder
☐	Wine Reviews	Text Analytics		Jan 9, 2026, 2:05 PM		Public.WINE_REVIEW		My Folder

Creating a New Project

You create new SAS Model Studio projects from the Projects page. To create a new project:

- 1 Click the **New Project** button in the upper right corner.



- 2 The New Project window appears.

New Project

Name: *
Spline Test

Type: *
Data Mining and Machine Learning

Template:
Blank template Browse

Data: *
Browse

Location:
My Folder Browse

Description:

Advanced

Save Cancel

Enter a name for your new project in the **Name** field.

- 3 Select a project type from the **Type** list. The choices are **Forecasting**, **Data Mining and Machine Learning**, and **Text Analytics**.
- 4 Select a template to use for your pipeline. *Templates* are pipelines that are pre-populated with configurations that can be used to create models quickly. The default template is the **Blank template**. The **Blank template** is the baseline pipeline that is pre-populated with only the **Data** node. For more information about the templates available in SAS Model Studio, see [Available Templates on page 126](#).
- 5 You must identify the data source that you want to use. .

Note:

The data source name cannot contain any of the following characters: / \ * ? " < > | : -. .

A project can have only one data source. However, some nodes can point to additional data sources. Select the **Browse** button to open the Choose Data window. Use the Choose Data window to select your data source and click **OK**.

- 6 (Optional) Select a folder in which to store the new project in the **Location** drop-down list.

Note: When creating a SAS Model Studio project, users should not select a repository folder, such as /DMRepository. This ensures that conflicts do not occur when registering models for that project into SAS Model Manager.

Location defaults to the folder that was most recently selected or **My Folder** if a folder has never been selected. The **Location** drop-down list includes **My Folder** and up to five of the most recently selected folders, if any. If the desired folder is not in the drop-down list, click **Browse** to select a different folder.

Note: When specifying a folder for the **Location** field, many of the folders listed can be selected but should be avoided. Selecting these folders makes the project visible to other users but does not grant access to all resources used within the project to the other users. This leads to errors encountered when opening and editing the project. For best results, use either **My Folder** or a folder that you or your system administrator has created for your use.

- 7 If you want to enter information about the project that might be useful to others, enter that content in the **Description** field.
- 8 Click **Advanced** to specify additional project creation options. In the New Project Settings window, you can specify the following settings:
 - **Advisor Options** — These settings enable you to specify settings such as the maximum number of class level values and the maximum percentage of missing values. Another option in this group enables you to control the interval cutoff value for numeric variables. If the number of unique levels exceeds the specified cutoff value, the variable is assigned the measurement level interval.
 - **Partition Data** — These settings enable you to partition the data set into subsets used for training, validation, and test. You can also specify the partition method. After the **Data** node is run, you can view the partition proportions, but you cannot edit them until you reload the data source. For more information, see [Replace the Data Source on page 108](#).
 - **Event-based Sampling** — These settings enable you to specify event-based sampling for the model. All of the observations that have the target event are included, and the observations with non-events are sampled to match the percentages specified for **Event** and **Non-event**. See [“Handling Rare Events” in SAS Viya: Machine Learning Advanced Topics](#).
 - **Node Configuration** — These settings enable you to control variable metadata propagation and to specify configuration code in Python:

- ❑ **Propagate variable metadata to the Transformations, Imputation, Replacement, and Filtering nodes** — enables the propagation of variable metadata to **Transformations, Imputation, Replacement, and Filtering** nodes, regardless of where they appear in the pipeline. For more information, see [“Assigning Variable Metadata” on page 109](#).
- ❑ **Prepend Python configuration code to all Open Source Code nodes** — specifies Python code that is automatically prepended to every **Open Source Code** that you include in your pipeline when the **Language** property is set to **Python**. To return all settings to their default values, click **Reset**.

Note: The Python code can be a maximum of 4000 characters.

- 9 Click the **Save** button to create your new project using the name, project type, and data source name that you specified.

After you create your new project, SAS Model Studio takes you to the **Data** tab of your new project page. Here, you can make adjustments to data source variable names, labels, type, role, and level assignments. For more information about the **Data** tab, see the [Data Management Overview](#) section.

Duplicating a Project

To duplicate a project in SAS Model Studio, complete the following steps:

- 1 Navigate to the **Projects** page, and perform one of the following:
 - Click  on the tile for a project, and then select **Duplicate** from the menu that appears.
 - Select the check box for a project. Click  in the project banner, and then select **Duplicate** from the menu that appears.

The Duplicate Project window appears.

- 2 Provide a name for the project in the **Name** field.
- 3 (Optional) Select a folder in which to store the new project in the **Location** drop-down list.

Note: This field is named **Location** starting in 2025.11. In prior releases, this field is named **Folder**.

Location defaults to the folder that was most recently selected or **My Folder** if a folder has never been selected. The **Location** drop-down list includes **My Folder** and up to five of the most recently selected folders, if any. If the desired folder is not in the drop-down list, click **Browse** to select a different folder.

Note: When specifying a folder for the **Location** field, many of the folders listed can be selected but should be avoided. Selecting these folders makes the project visible to other users but does not grant access to all resources used within the project to the other users. This leads to errors encountered when opening and editing the project. For best results, use either **My Folder** or a folder that you or your system administrator has created for your use.

4 (Optional) Provide a description for the project.

5 Click **Save**.

The duplicated project appears in the **Projects** page.

Sharing Projects

When you create a project, you become the project owner and the project is kept in private mode so that only you can view or edit it. You can share the project with other users and groups.

Note: If you intend to share a project with other users, you must store your data in a location that is accessible to all those other users.

Note:

If the project data source is an Azure Data Lake Storage (ADLS) caslib, when the other users open the project, a message that instructs them to authenticate the ADLS connection might appear. In that case, the other users must follow the instructions provided in the message.

When you share a project, you must specify how other users can access the project.

- If you give users only Read access to a project, they can view any part of the project but cannot make any changes.

Users with Read access to the project can also share the project with Read access to other users.

When you view the project list as a table, you can click  in the **Permissions** column to see who has Read access.

To view your project list as a table, click  in the upper left corner of the Projects page.

- If you give users Read and Edit access to a project, they can make changes to the project.

When a user edits the project, the project is not locked, and other users with Edit access can make changes at the same time. Simultaneous changes can cause conflicts. When a project is shared with multiple users with Edit access,

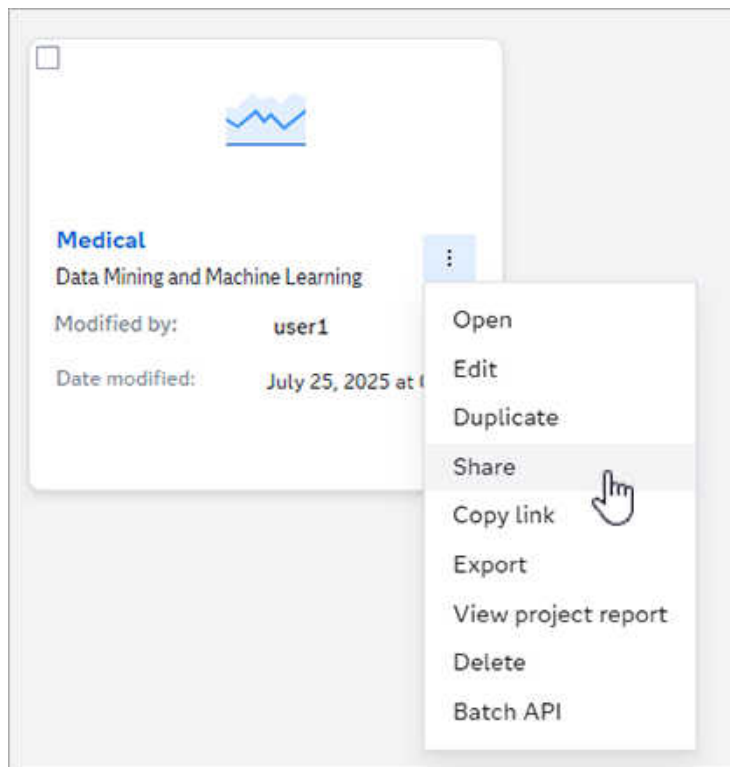
users should click **2** in the project banner to determine whether anyone else is editing the project at the same time. This action displays each user name along with the level of access and the name of the tab that is open.

If multiple users are working in the project, they should click  often in the project banner to make sure they have the latest changes.

Users with Edit access to the project can share the project with other users and give them Edit and Read permissions. Users with Edit access can also delete the project.

To share a project that you own:

- 1 On the Projects page, click  on the tile for the project and select **Share**.



The Share window is displayed.

Note: If you are viewing the project list as a table, select the project and click  on the upper right of the page and select **Share**.

You can select multiple projects to share.

- 2 Enter the name of one or more users or groups.

Share

Item name: Sales

Names:

[FSI] User 001 x

OpenStack Users x

[CE9] SAS Install x

Permission:

Can read

Can read

Can read and edit

> Shared with these users and groups (0):

Share

- 3 For each list of users, select the access level on the right.
 - **Can read** — the users can only view the project and cannot make any changes
 - **Can read and edit** — the users can view and make changes to the project

You can add a separate group of users by clicking **+**.

- 4 When you are finished, click **Share** to close the window and return to the projects page.

The project is now shared. You can open the Share window again if you need to change or remove access for any users. To change the level of access for any user, click .

To change users' access level, follow these steps.

- 1 Click  on the tile for the project and select **Share**.
- 2 Click . This opens the Edit Sharing window.
- 3 For each user, you can change their access to **Read**, **Read and edit**, or, to remove their access, select **Unshare**.

For users that have access to your project, you can send them links that take them directly to these specific locations in the project.

- **Project tiles** — From a project tile on the **Projects** page, click  and select **Copy link**.
- **Project tabs** — From any tab, click  and select **Copy link**.

- **Pipelines** — From a pipeline in the **Pipelines** tab, click  and select **Copy pipeline link**.
- **Nodes** — From any node in a pipeline, click  and select **Copy node link**.

For each action, the Copy Link window is opened. Click **Copy Link**. The link is copied to your clipboard and the window is dismissed. You can paste the link in any communications that you send to the other users. It will take them directly to the tab, pipeline, or node that you copied the link from.

Exporting and Importing a Project

Exporting a Project

To export a project, do the following:

- 1 On the Projects page, select the project that you want to export by clicking the check box in the project tile.
- 2 Click  in the upper right corner of the window, and select **Export**.

The project files are immediately downloaded. Project files are stored as JSON files.

Note: **Managed Variable** and **Decision Tree** nodes that have been interactively edited are not saved as configured when the pipeline is exported. This is because the interactive edits that were made cannot be persisted. If the exported project includes either or both of these nodes, the **Managed Variable** and **Decision Tree** nodes must be interactively edited after the pipeline is imported in order to restore the original changes.

Importing a Project

Note:

- If you are not an administrator, you can import projects only when you have access to update existing content that is present in the package or ZIP file and access to create new content that is present in the package or ZIP file. This content includes the exported resource and any component or pipeline templates that it references. It also includes transitive dependencies, so the content can consist of a chain of component or pipeline templates that were used to create the more immediate dependencies.

- If you are not an administrator, you cannot import another user's shared read-write or read-only project.

In order to import a project, the following must be true:

- The project source data must be available to the CAS server. If the data is currently not available, you can import the data when you import the project as described in this section.
- The owner of the project that is being imported must be signed in to the target environment. If you are a project owner, it is recommended that you promote your own individual projects.
- If the project is shared and the same users should access the project on the target environment, the project sharing must be re-created after it is imported. Project sharing permissions are not imported.

To import a project, do the following:

- 1 Click  in the upper right corner of the window, and select **Import**. If multiple products are installed, select **Import** ⇒ **Project type that you want to import**. Depending on the project type that you select, one of the following windows appear:
 - Import Forecasting Project
 - Import Data Mining Project
 - Import Text Analytics Project
- 2 In the Import Project window **File** field, specify the ZIP file that was saved when you exported the original project.
- 3 In the **Data** field, click **Browse** to open the Choose Data window. Select the data set that is associated with the project.

Note: If the project source data set is currently not available, click **Import data** to import the data set and load it into memory. For more information, see [“Importing Data” on page 106](#).

- 4 (Optional) Specify the destination of the imported project.

Note: The **Destination** setting is available starting with 2025.11.

Use the project's original location is selected by default. This setting stores the imported project in the location in which it was stored when it was exported. If the original location no longer exists, it is created when the project is imported. To specify a different location, select **Select the import location**, and then select the new location in the **Location** drop-down list. **Location** defaults to the folder that was most recently selected or **My Folder** if a folder has never been selected. The **Location** drop-down list includes **My Folder** and up to five of the most recently selected folders, if any. If the desired folder is not in the drop-down list, click **Browse** to select a different folder.

Note: When specifying a folder for the **Location** field, many of the folders listed can be selected but should be avoided. Selecting these folders makes the project visible to other users but does not grant access to all resources used within the project to the other users. This leads to errors encountered when opening and editing the project. For best results, use either **My Folder** or a folder that you or your system administrator has created for your use.

5 Click **Import**.

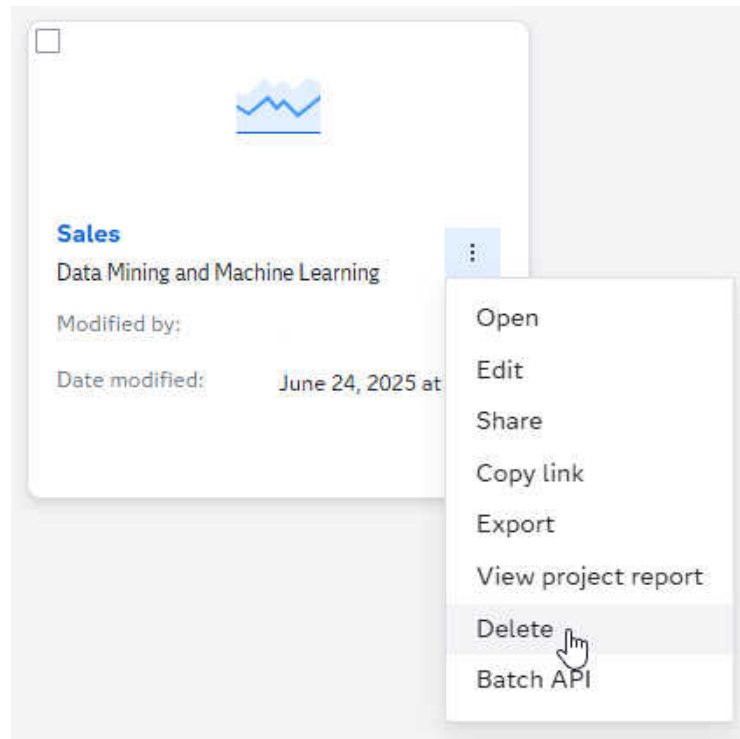
Note:

- The preceding steps apply only to importing projects that were created in the same version of SAS Model Studio. For information about how to import projects from previous versions of SAS Model Studio, see [“Promotions and Upgrades within the SAS Viya Platform” in SAS Viya: Machine Learning Advanced Topics](#).
 - If you import a project that contains a custom compute context that does not exist on the target system, the custom compute context is replaced with the default compute context.
 - If the **Comment** column on the Data tab in the imported project indicates that the initial value for a variable attribute was set by Global Metadata, be aware that the initial value might have been set by the imported project instead.
-

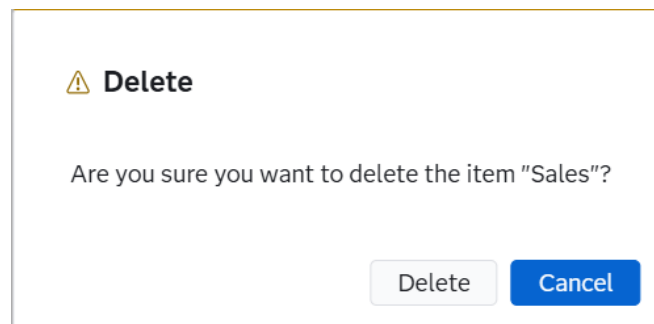
Deleting a Project

To delete a project:

- 1 Select the desired project by clicking the check box in the project tile, and then click  next to the Project page Toolbox.



- 2 Select **Delete**.
- 3 The Delete window appears, asking for confirmation of deletion.

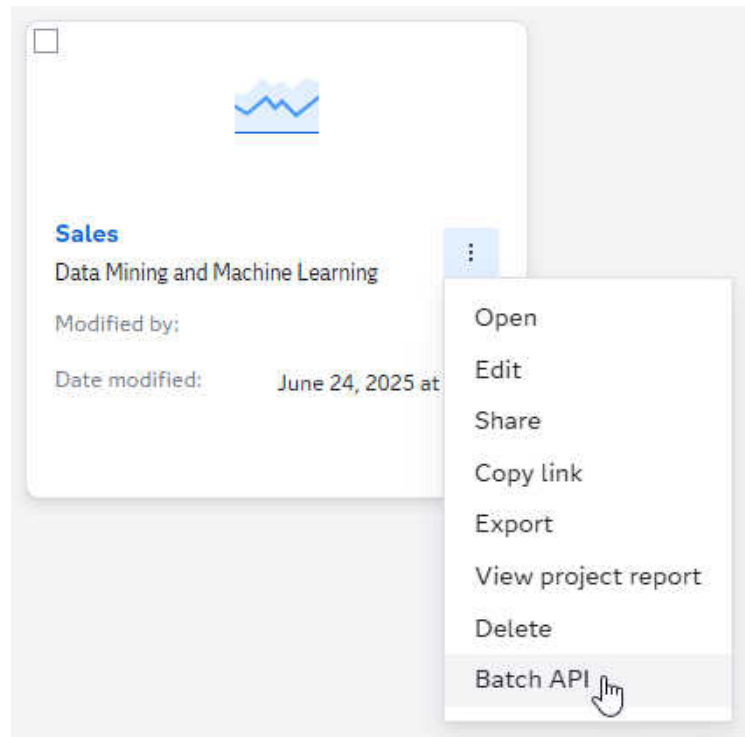


- 4 Click **Delete**.

Downloading Project Batch API Code

To download the batch API code for a project:

- 1 Select the desired project by clicking the check box in the project tile, and then click **:** next to the Project page Toolbox.



- 2 Select **Batch API**.
- 3 The Batch API window appears, which displays the batch API code for the project. Batch API code can be given in the following types:
 - **SAS** — Downloads code to run in a SAS programming environment.
 - **Python** — Downloads code to run in a Python programming environment.
 - **REST** — Downloads a text file with example REST calls that you can use in an application.

Close

- In the macro variable definition statements, replace the HOST and PORT placeholders with the host name and port of the SAS Viya Machine Learning server. Obtain a valid access token as described in [Obtain an Access Token](#) in *SAS Viya Platform: Authentication*. Replace the TOKEN placeholder with the token in the following format:

```
'bearer token'
```

Note: The token value must be enclosed in single quotation marks and it must be prefixed with `bearer` followed by a space.

Python

Update the `urlPrefix` and `oauthToken` variable assignments near the beginning of the code:

```
urlPrefix = "https://HOST:PORT"
projectId = "project-id"
dataUri = "data-source-uri"
oauthToken = "TOKEN"
```

In the `urlPrefix` variable assignment statement, replace the `HOST` and `PORT` placeholders with the host name and port of the SAS Viya Machine Learning server. Obtain a valid access token as described in [Obtain an Access Token in SAS Viya Platform: Authentication](#). In the `oauthToken` variable assignment statement, replace the `TOKEN` placeholder with the token in the following format:

```
bearer token
```

Note: The token value must be prefixed with `bearer` followed by a space.

If you have overrides in the project, they might generate conflicts when they are submitted by the batch code. You can update the batch code to automatically resolve these conflicts by adding `autoresolve="true"` to the code.

SAS

Find the PROC HTTP procedure that includes the following `IN` option with `firstTransaction` and `lastTransaction`.

```
in={"firstTransaction": "@first", "lastTransaction": "@last"}
```

Add the `autoResolve` setting, as follows:

```
in={"firstTransaction": "@first", "lastTransaction": "@last",
  "autoResolve": true}
```

Python

Find the `resubmit_overrides` function:

```
def resubmit_overrides(env):
```

Update the body statement with `autoResolve` assignment.

```
body = '{"firstTransaction\\": "@first\\", "lastTransaction\\": "@last\\",
  "autoResolve\\": true}'
```

Note:

If the project data source is an Azure Data Lake Storage (ADLS) caslib, when you run the batch code, an error message that instructs you to authenticate the connection might be returned. In that case, follow the instructions provided in the error message.

Project Banner

The banner for each project has the project name, tabs for navigating the project, and the following controls.



This icon runs all of the pipelines that have been created for a project. Also, ensure that your target variable is assigned before running any pipelines.



Refreshes the data on the page. Use this to make sure your changes do not conflict with changes made by other users in the project.



Indicates the number of users who are currently using the project. It shows only one when you are the only one in the project. Click the icon to see the other users in the project.

If the project is shared to permit other users to read and edit the project, multiple users can edit the project at the same time.



Controls settings that apply across the project as a whole.



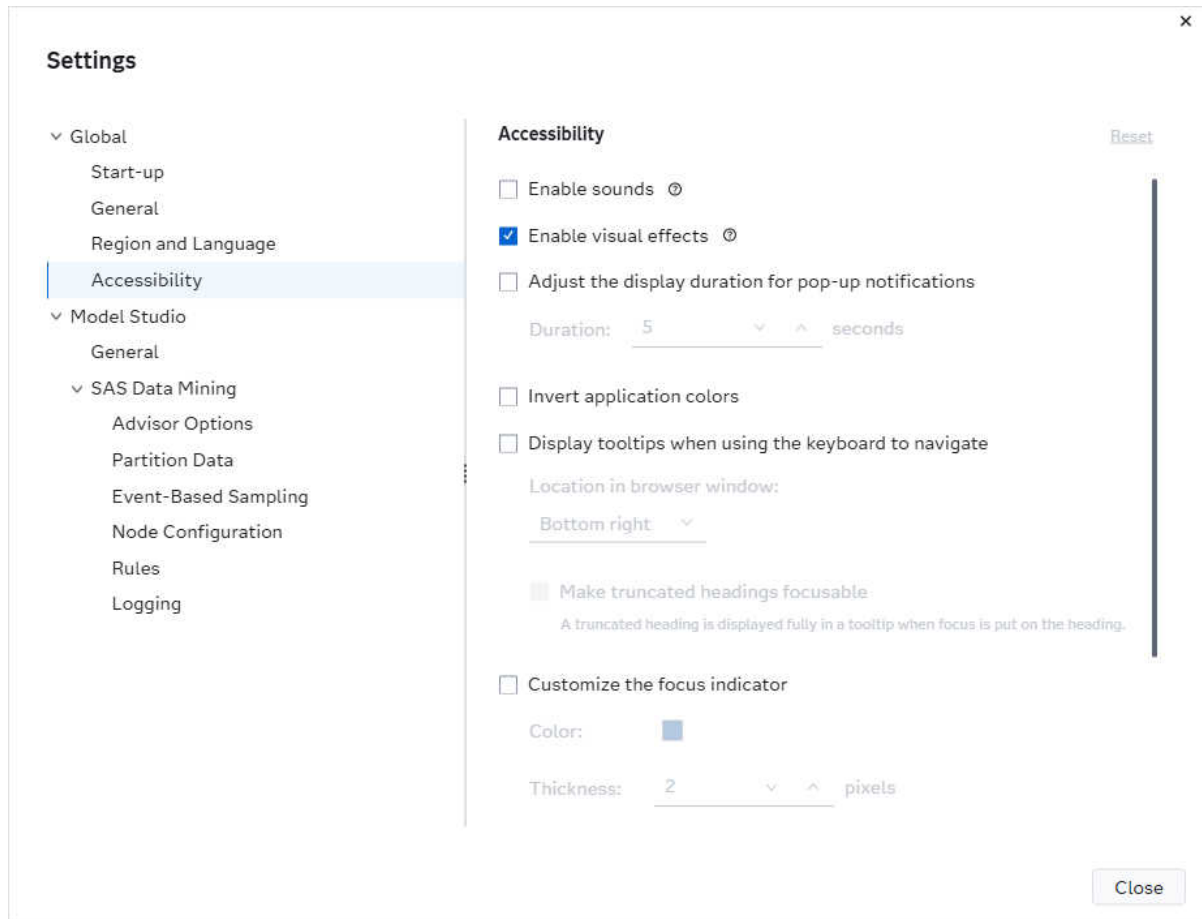
Opens the following menu of options.

- **Batch API** — Opens a window with the code used to run the project in batch. You can choose between SAS, Python, or REST API.
- **Copy link** — Opens a window that provide a link to the tab that you have open in the project. If you share this link with anyone who has access to the project, it opens the project to the tab that you are working in.
- **Force run all pipelines** — Runs all pipelines, including the pipelines that have already completed with success. This action was added in 2026.04.

Specifying Global Settings

When setting up your SAS Model Studio account, you might want to modify global settings for your account instance. To edit global settings:

- 1 In the upper right corner of the window, click the user name button, and select **Settings**.
- 2 The Settings window appears, enabling you to alter global settings.



- 3 For changing global settings, the following options are available:
 - **General** — These settings enable you to set your interface theme, reset messages, and choose a profile picture. To return all settings to their default values, click **Reset**.
 - **Region and Language** — These settings enable you to set locales for your browser and Java Runtime Environment. To return all settings to their default values, click **Reset**.
 - **Accessibility** — These settings enable you to specify sounds and visual effects for your interface. You can also adjust the display duration for pop-up notifications, invert colorings, display tooltips, and customize the focus indicator. To return all settings to their default values, click **Reset**.
- 4 For changing SAS Model Studio project settings at a global level, the following options are available for SAS Data Mining users:
 - **General** — Select **Reset Column Preferences** to reset table column preferences to their default values.
 - **Advisor options** — These settings enable you to specify settings such as the maximum number of class level values and the maximum percentage of missing values. Another option in this group enables you to control the interval cutoff value for numeric variables. If the number of unique levels exceeds the specified cutoff value, the variable is assigned the measurement level interval. To return all settings to their default values, click **Reset**.

- **Partition Data** — These settings enable you to partition the data set into subsets used for training, validation, and test. You can also specify the partition method. These proportions apply to new projects and can be changed at any time. To change the partition proportions in existing projects, see [Specifying Project Settings on page 97](#). By default, the data is partitioned as follows:

Training	60%
Validation	30%
Test	10%

.....

Note:

In 2026.05 and prior releases, you must specify integer values for the proportions. Starting with 2026.06, you can also specify decimal values.

.....

To return all settings to their default values, click **Reset**.

.....

Note: If the input table contains variable `_PartInd_`, then the **Partition Data** settings are ignored.

.....

- **Event-based Sampling** — These settings enable you to specify event-based sampling for the model.

.....

Note: In 2026.04 and prior releases, you must specify integer values for the percentages. Starting with 2026.05, you can also specify decimal values.

.....

All of the observations that have the target event are included, and the observations with non-events are sampled to match the proportions specified for **Event** and **Non-event**. By default, event-based sampling is disabled. If enabled, the **Event** and **Non-Event** percentages are both 50% by default. To return all settings to their default values, click **Reset**.

See “[Handling Rare Events](#)” in *SAS Viya: Machine Learning Advanced Topics*.

.....

Note: If the input table contains variable `_PartInd_`, then the **Event-Based Sampling** settings are ignored.

.....

- **Node Configuration** — These settings enable you to control variable metadata propagation and to specify configuration code in Python:
 - **Propagate variable metadata to the Transformations, Imputation, Replacement, and Filtering nodes** — enables the propagation of variable metadata to **Transformations**, **Imputation**, **Replacement**, and **Filtering** nodes, regardless of where they appear in the pipeline. For more information, see “[Assigning Variable Metadata](#)” on page 109.

- **Prepend Python configuration code to all Open Source Code nodes** — specifies Python code that is automatically prepended to every **Open Source Code** that you include in your pipeline when the **Language** property is set to **Python**. To return all settings to their default values, click **Reset**.

Note: The Python code can be a maximum of 4000 characters.

- **Rules** — These settings enable you to specify the rules used for comparing pipeline models:
 - **Model Comparison** — The model comparison settings enable you to specify the class selection statistic, the interval selection statistic, the selection partition, selection depth, and the ROC-based cutoff. For more information, see [Overview of Model Comparison](#) in the SAS Model Studio reference documentation.
 - **Model** — This setting enables you to override the default classification cutoff for a binary target. The default is 0.5. To override the default, select **Override the default classification cutoff**, and then enter the override value. The value must be between 0 and 1, exclusive.
 - **Assessment** — This setting enables you to specify the number of ROC cutoff values. This value affects the following statistics: Area under the curve, F0.5 score, F1 score, False discovery rate, False positive rate, Gamma, Gini coefficient, KS, ROC separation, and Tau.

When using an ROC-based statistic for the autotuning class target objective function, 100 cutoffs are always used so that the autotuning assessment statistics might not match the assessment for the final model.

Note:

Specifying a higher number of ROC cutoff values results in a longer run time, especially when the training table is large.

Note:

If the number of ROC cutoff values exceeds 1000, then the number of rows in the final table might be less than what is specified. This is because consecutive observations that have similar ROC statistics are removed.

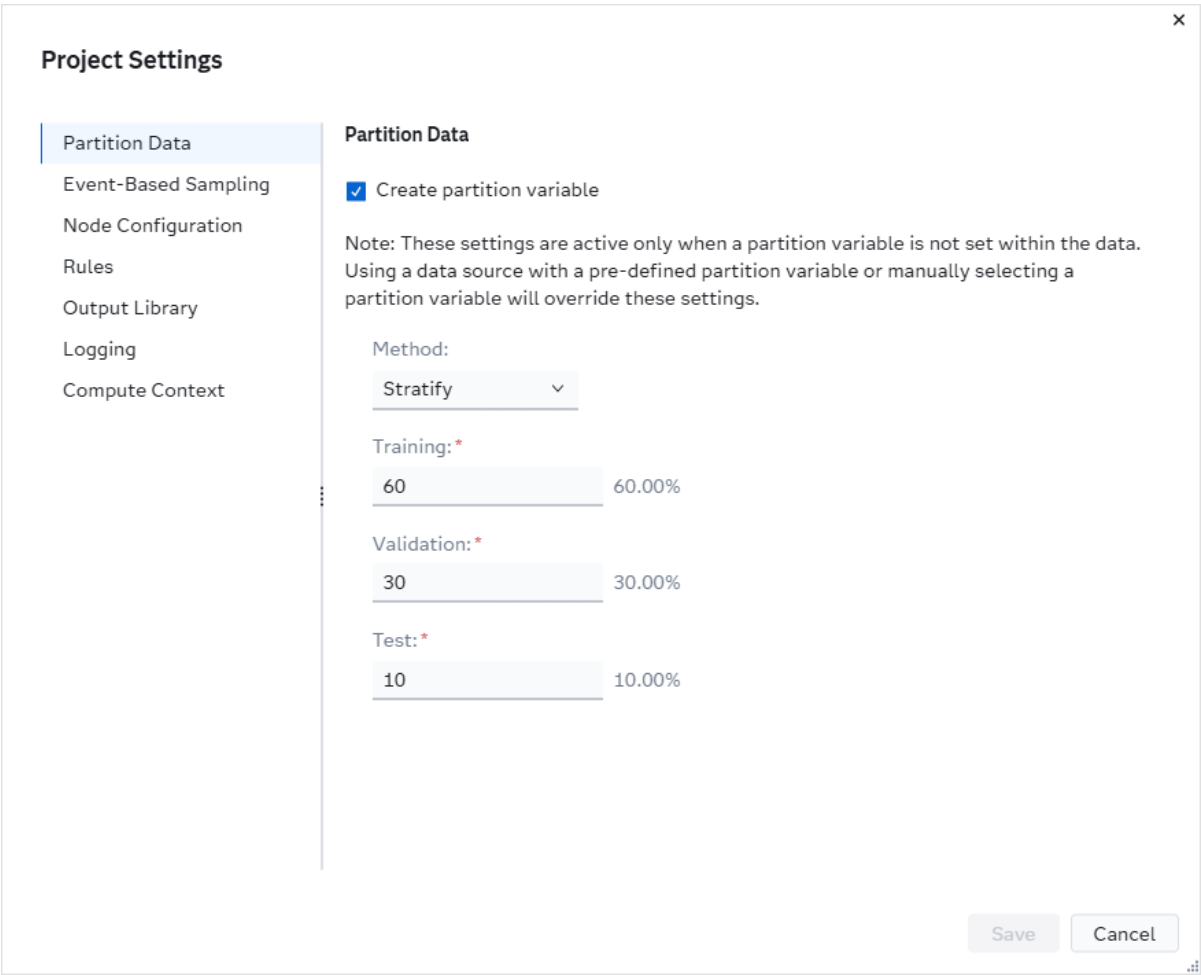
To return all settings to their default values, click **Reset**.

- **Logging** — These settings enable you to specify debug reporting, including options to resolve macro variables, add timings and headers, and retain temporary tables. To return all settings to their default values, click **Reset**.
- 5 Once the settings are appropriately configured, click **Close**. Settings are automatically saved.

Specifying Project Settings

For certain SAS Model Studio projects, you might need to modify project settings to set up models properly. Many of the project settings default to their corresponding global setting. To edit project settings:

- 1 Open the project, and then click  in the upper right corner of the window, under the user name button, and click **Project settings**.



- 2 The Edit Project Settings window contains several properties that might need to be altered for your projects. The following options are available:
 - **Partition Data** — These settings enable you to partition the data set into subsets used for training, validation, and test. You can specify the proportions for the subsets. You can also specify the partition method. By default, the data is partitioned as follows:

Training	60%
----------	-----

Validation	30%
Test	10%

Note:

In 2026.05 and prior releases, you must specify integer values for the proportions. Starting with 2026.06, you can also specify decimal values.

The **Partition Data** proportions can be edited after creating a project, but only before the **Data** node is run. After the **Data** node is run, you can view the partition proportions, but you cannot edit them until you reload the data source. For more information, see [Replace the Data Source on page 108](#).

Note: If the input table contains variable `_PartInd_`, then the **Partition Data** settings are ignored.

- **Event-Based Sampling** — These settings enable you to specify event-based sampling for the model.

Note: In 2026.04 and prior releases, you must specify integer values for the percentages. Starting with 2026.05, you can also specify decimal values.

All of the observations that have the target event are included, and the observations with non-events are sampled to match the proportions specified for **Event** and **Non-event**. By default, event-based sampling is disabled. If enabled, the **Event** and **Non-Event** proportions are both 50% by default.

See “[Handling Rare Events](#)” in *SAS Viya: Machine Learning Advanced Topics*.

Note: If the input table contains variable `_PartInd_`, then the **Event-Based Sampling** settings are ignored.

- **Node Configuration** — These settings enable you to specify variable metadata propagation and configuration code in Python:
 - ☐ **Propagate variable metadata to the Transformations, Imputation, Replacement, and Filtering nodes** — enables the propagation of variable metadata to **Transformations**, **Imputation**, **Replacement**, and **Filtering** nodes, regardless of where they appear in the pipeline. For more information, see “[Assigning Variable Metadata](#)” on page 109.
 - ☐ **Prepend Python configuration code to all Open Source Code nodes** — specifies Python code that is automatically prepended to every **Open Source Code** that you include in your pipeline when the **Language** property is set to **Python**. To return all settings to their default values, click **Reset**.

Note: The Python code can be a maximum of 4000 characters.

- **Rules** — These settings enable you to specify the rules used for comparing pipeline models:
 - **Model Comparison** — The model comparison settings enable you to specify the class selection statistic, the interval selection statistic, the selection partition, selection depth, and the ROC-based cutoff. For more information, see [Overview of Model Comparison](#) in the SAS Model Studio reference documentation.
 - **Model** — This setting enables you to override the default classification cutoff for a binary target. The default is 0.5. To override the default, select **Override the default classification cutoff**, and then enter the override value. The value must be between 0 and 1, exclusive.

TIP You can override this binary classification cutoff value at the node level by setting property **Binary Classification Cutoff** in the node's property panel. See [SAS Viya: Machine Learning Node Reference](#).

The default binary classification cutoff value is used during model training and for scoring holdout data. When you change the default classification cutoff in Project Settings, no action is triggered. You must rerun your project pipelines and rescore your holdout data to pick up the new binary classification cutoff value. Here is an example of user-specified binary classification cutoff value 0.333 in the Event Classification report in the node assessment results.

Figure 3.1 Event Classification Percentage Plot in Node Assessment Results



If you do not rerun your project pipelines after you change the binary classification cutoff value in Project Settings, then the new value is used only when scoring holdout data. In the following example, the binary classification cutoff value is changed to 0.333 in Project Settings, and then the holdout data is rescored before the pipeline is rerun.

Figure 3.2 Event Classification Percentage Plot in Score Holdout Data Results



In this case, the user-specified binary classification cutoff value is used for HOLDOUT only. For information about scoring holdout data, see [“Score Holdout Data” on page 157](#).

- **Assessment** – This setting enables you to specify the number of Receiver Operating Characteristic (ROC) cutoff values. This value affects the following statistics: Area under the curve, F0.5 score, F1 score, False discovery rate, False positive rate, Gamma, Gini coefficient, KS, ROC separation, and Tau.

When using an ROC-based statistic for the autotuning class target objective function, 100 cutoffs are always used so that the autotuning assessment statistics might not match the assessment for the final model.

Note:

Specifying a higher number of ROC cutoff values results in a longer run time, especially when the training table is large.

Note:

If the number of ROC cutoff values exceeds 1000, then the number of rows in the final table might be less than what is specified. This is

because consecutive observations that have similar ROC statistics are removed.

- **Output Library** — These settings enable you to specify the library for output from the model.

Note:

Saving node output data to an Azure Data Lake Storage (ADLS) caslib is not supported.

- **Logging** — These settings enable you to specify debug reporting, including options to resolve macro variables, add timings and headers, and retain temporary tables.

- 3 Once the settings are appropriately configured, click **Save**.

Note: The settings configured at the project level override any global settings that you configured for your SAS Model Studio instance.

Importing a Project from SAS Visual Analytics

SAS Model Studio users can use their SAS Visual Analytics reports to create projects. The following SAS Visual Analytics objects can be used to generate pipelines in SAS Model Studio:

- **Bayesian Network**
- **Decision Tree**
- **Factorization Machine**
- **Forest**
- **Generalized Additive Model**
- **Generalized Linear Model**
- **Gradient Boosting**
- **Linear Regression**
- **Logistic Regression**
- **Model Comparison**
- **Neural Network**
- **Nonparametric Logistic Regression**

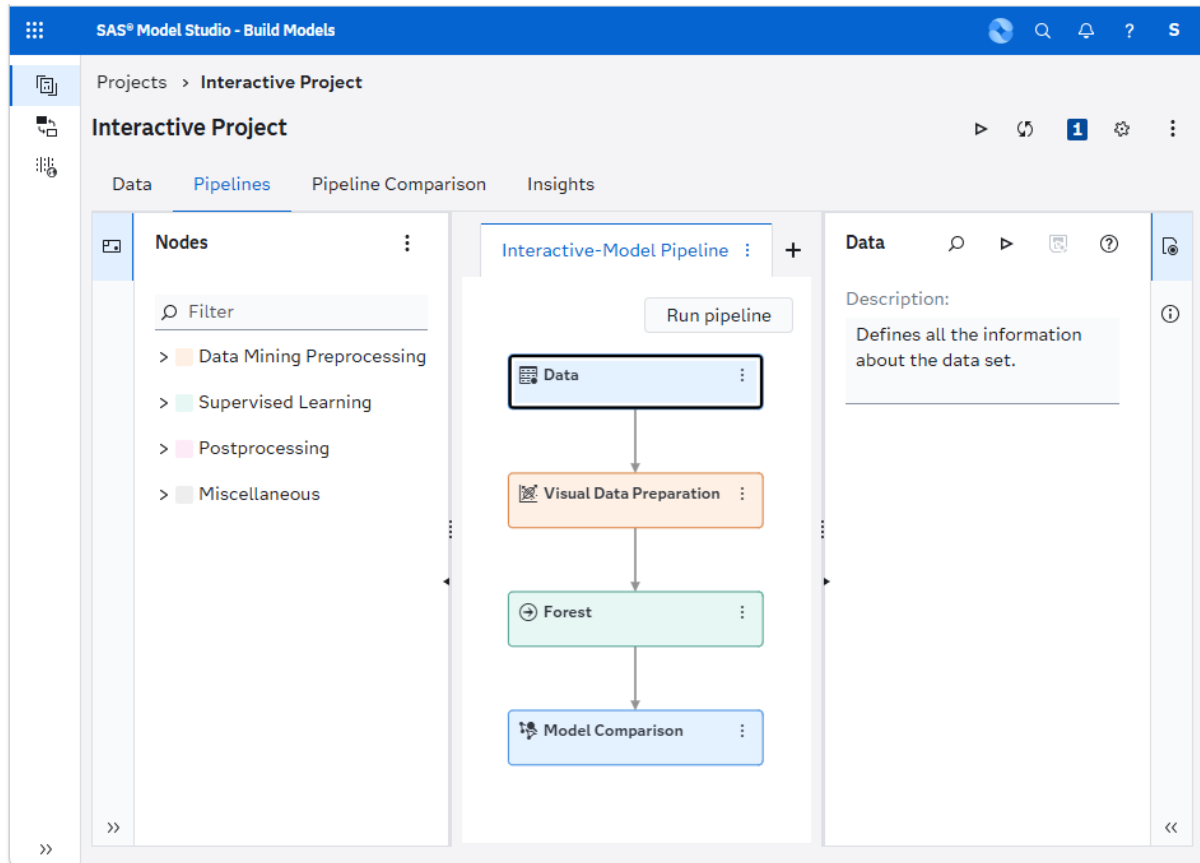
■ Support Vector Machine

Note: When importing a project with a Nonparametric Logistic Regression object, the pipeline in the project that you import into SAS Model Studio contains a GAM supervised learning node. If the Visual Analytics model uses a nominal target, the imported SAS Model Studio project uses a binary target that is converted from the nominal target.

The SAS Visual Analytics objects listed above correspond directly to nodes in SAS Model Studio. To export an object:

- 1 In SAS Visual Analytics, click the **Create Pipeline** button in the upper left corner of the object canvas, and select either **Add to new project** or **Add to existing project**.
- 2 If **Add to new project** is selected, the SAS Model Studio interface opens, and a project is created in SAS Model Studio. The name matches the name of the saved SAS Visual Analytics report. If the report in SAS Visual Analytics has no name, the project is named *Interactive Project* in SAS Model Studio. If **Add to existing project** is selected, the SAS Model Studio interface opens, and you are prompted to select an existing project. Only existing projects that meet the following conditions are listed for selection:
 - The target variable names match.
 - The target variable types match.
 - The response variable classification in SAS Visual Analytics is compatible with the SAS Model Studio target variable level. (See [“Integration with SAS Visual Analytics” on page 119.](#))
 - The data node in the SAS Model Studio project has previously been run.

When you select a project, a pipeline named *Interactive-Model Pipeline* is created in that project.



This pipeline looks identical to a normal SAS Model Studio pipeline, except for two key differences:

- SAS Model Studio generates a Data Mining Preprocessing node called **Visual Data Preparation**. The **Visual Data Preparation** node runs the score code necessary to perform all the data preparation steps that were performed in SAS Visual Analytics. The properties of this node are not available to edit.
- The SAS Visual Analytics objects are also represented. Here, **Forest** corresponds to the Forest object in SAS Visual Analytics. Supervised learning nodes can be edited in SAS Model Studio. Right-click the supervised learning node, and then select **Enable properties**. If you enable the properties of a node, the model is retrained, which might yield new results. Once properties have been enabled, the model cannot be switched back to use the original SAS Visual Analytics score code. **Model Interpretability** properties do not require retraining the model, so they are automatically enabled and do not affect the original SAS Visual Analytics score code. This pipeline can be run as-is, or nodes can be added to the pipeline to be run for comparison purposes.

When you import a project that contains a frequency variable from SAS Visual Analytics, consider the following implications:

- When you create a pipeline from a SAS Visual Analytics report that contains a frequency variable, that frequency variable is transferred with the report data only when you create a new project. Frequency variables are not transferred when you add a SAS Visual Analytics report as a pipeline to an existing project.

- Any negative frequency values that are in your SAS Visual Analytics data are replaced with zeros when you create a new project from a SAS Visual Analytics report.

See Also

[“Integration with SAS Visual Analytics” on page 119](#)

Upgrading a Project

If a project was created in an earlier version of SAS Model Studio, the project might need to be upgraded for the current version. For information about project upgrades, see [“Promotions and Upgrades within the SAS Viya Platform” in SAS Viya: Machine Learning Advanced Topics](#).

Working with Data

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Data Management Overview

SAS Model Studio provides several options for managing and modifying data. A project can have only one data source. However, some nodes can point to additional data sources. The Data tab enables you to modify variable assignments and manage global metadata. You can also retrain a model with new data, as long as the target variable in the new data set is the same as the original data set.

Note: Typically, the column names of the input data are not translated. However, certain system-generated UI labels and column names are automatically translated when you are using SAS Model Studio in a non-English language. If your input data contains a column name that matches one of these system-generated UI labels or column names, you might encounter unexpected translations of the column names of your input data.

Importing Data

When you create a new project or import a project, the project source data must be made available to the CAS server. To make the data available to CAS, you can import the data before you create or import the project by using the **Import** tab in SAS Data and AI Studio (known as SAS Data Explorer prior to SAS Viya 2026.06). For more information, see [“Making Data Available to CAS” in SAS Data Explorer: User’s Guide](#). Alternatively, you can import the data when you create or import the project by doing the following:

- 1 If you have not already done so, open the New Project or Import Data Mining Project window. See [“Creating a New Project” on page 79](#) or [“Importing a Project” on page 86](#).
- 2 In the **Data** field, click **Browse**. The Choose Data window appears.
- 3 In the top right corner, click **Import data** to open the Import Data window.
- 4 Drag the desired local data set directly to the window. SAS Model Studio parses the data set and pre-populates the window with data set configurations.

Import Data

Imports + Add files

- insightToyDemo.sas7bdat (Local file)

insightToyDemo.sas7bdat [Import item] [Import all]

Target table name: *

insightToyDemo

Target location: *

cas-shared-default/Public

If target table name exists:

☒ Cancel import

☐ Replace file

Password:

Enter password

Encryption key:

Enter key

Character multiplier:

1

☐ Create unique ID column

[Add] [Cancel]

Note:

Data set names and variable names can have a maximum length of 32 bytes.

You cannot create a project if the data set name contains any of the following characters: / \ * ? " < > | : -

Note:

Azure Data Lake Storage (ADLS) does not support custom SAS file formats such as SASHDAT and SAS7BDAT.

- 5 After configuring the data set import properties, click **Import item**. After successful import, the following message appears:

✓ The table was successfully imported on Mar 19, 2019 03:17 PM and is ready for use.

Click **Add**.

- 6 In the Choose Data window, click **OK**.

Follow the remaining steps outlined in [“Creating a New Project” on page 79](#) or [“Importing a Project” on page 86](#).

User-Defined Formats

Your data sets might include formats that are not natively supported by SAS Viya. To enable SAS Viya to recognize these formats:

- 1 Upload your format to a CAS format library.
- 2 Move your format to the compute server machine. If the formats are Windows formats and the machine is UNIX, then you need to use PROC CPORT and PROC CIMPORT to move the format. For more information, see the [PROC CPORT](#) and [PROC CIMPORT](#) documentation in *Base SAS 9.4 Procedures Guide*.
- 3 Add the format to the compute server session search path. For example, you can add the following code to the autoexec_usermods.sas autoexecutable SAS file in `/opt/sas/viya/config/etc/compsrv/default`:

```
libname format '</home/filepath/casuser/>';
options fmtsearch=(format.emfmt);
```

See [Managing User-Defined Formats in SAS Viya](#) in the SAS Viya Administration documentation for more information.

Replace the Data Source

A project can have only one data source. You might want to replace the data source in your project to do one of the following:

- Retrain a model with new data.
- Change the way that the data is partitioned after a pipeline is run.
- Reload the input table with additional records.
- Reload the input table with additional variables.
- Load the input table with a different name.

To replace the data source of a project:

- 1 Open a project. On the Data tab, click  to open the Data Sources pane.
- 2 In the upper left corner of the Data Source pane, click .
- 3 The Choose Data window appears. If the desired data set has already been added to the SAS Model Studio repository, select it from the list on the **Available** tab. If the desired data set is local to your environment, select the **Import** tab, and follow the instructions in [Importing Data on page 106](#).
- 4 Click **OK**. The Data tab now displays details about the new data set.

Note: To retrain a model with a new data set, the new data set must use the same target variable as the original data set. For more information, see [Managing Global Metadata on page 118](#).

Refresh the Project Data

If your project's source data has changed, you can refresh the project data sources on the **Data** tab.

Note: If you delete the table used to create a project, an error appears on the **Data** tab. However, you can continue to work on any projects that use that table, including adding nodes and running pipelines.

To refresh your projects data, do the following:

- 1 Navigate to the project's **Data** tab.

- 2 In the left pane, click .

All data sources are refreshed if their content has changed.

Note: You can also use a data refresh to remove any variables that are unassigned. For example, you can remove the unassigned variables from the original data source and perform a data refresh with the original data source. Or you can perform a data refresh using a copy of the original data source with the unassigned variables removed.

The data source for a project is kept in a CAS table in memory. In general, the source for this CAS table comes from a table stored on disk. If the source for the project's CAS table is not stored on disk, and CAS is restarted, you need to reload the table to memory in CAS. This triggers a refresh notification, even if the data has not actually been changed.

After the project data is refreshed, all of the nodes in the pipelines are placed in the out-of-date state. In that case, rerun the pipelines to update the nodes.

Managing Variable Assignments

Assigning Variable Metadata

To specify variable properties:

- 1 On the **Data** tab, select the desired variables.
- 2 The right pane enables you to specify several properties of the variables, including the following:

- **Role**
- **Level**
- **Order**

The **Order** attribute affects how the target event is selected for a class target variable or a class input variable, and how partition values are assigned for a partition variable. When assigning a role of **Target** to an existing class variable in the project table, the **Order** attribute affects the selection of the target event for the target variable as follows:

- 1 The variable levels are sorted according to the specified **Order**, which defaults to **Descending**.
- 2 The first level in the sorted levels is selected as the target event.

When assigning a role of **Partition** to an existing variable in the project table, the **Order** attribute is used to assign the partition values. By default, the

Order attribute assigns partitions using the descending order of the variable. That is, the first value of the sorted variable **2** is assigned to test. The second value of the sorted variable **1** is assigned to train. The third value of the sorted variable **0** is assigned to validation.

Specifying a value other than **Default** for **Order** can affect the assignment of these values. For example, specifying **Ascending** for **Order** changes the assignment of partitions to the following values:

- A value of 0 is assigned to observations used for test.
- A value of 1 is assigned to observations used for training.
- A value of 2 is assigned to observations used for validation.

Input variables are impacted by the **Order** attribute as follows:

- If the **Order** attribute is set to **Default** for an input variable, then the value specified for the **Class input order** node property is used by applicable downstream nodes:
 - GAM (Generalized Additive Model)
 - GLM (Generalized Linear Model)
 - Linear Regression
 - Logistic Regression
 - Quantile Regression

The **Order** attribute is not applied to any other nodes.

- If you specify an **Order** attribute other than **Default** on the **Data** tab, it overrides a node's **Class input order** property.

- **Transform**
- **Impute**
- **Lower Limit**
- **Upper Limit**
- **Assess this variable for bias**

Note: This feature can be enabled only for nominal and binary variables. A **Fairness and Bias** tab is added to the **Results** view of any supervised learning nodes present in your pipeline when you select a variable to assess for bias.

Note: While a pipeline is running, variable metadata assignment is disabled.

For the **Transform**, **Impute**, **Lower limit**, and **Upper limit** properties, altering these values in the **Data** tab does not directly modify the variable. Instead, it sets metadata values for these properties. In 2025.12 and prior releases, this variable metadata is available to the **Transformations**, **Imputation**, **Replacement**, and **Filtering** nodes only when they are a direct child of the **Data** node. Starting with 2026.01, project setting **Propagate variable metadata to the Transformations, Imputation, Replacement, and Filtering nodes** under **Node Configuration** can be

used to enable the propagation of the variable metadata to these nodes regardless of where they appear in the pipeline. (See “[Specifying Project Settings](#)” on page 97). By default, this option is not selected, in which case the prior behavior is used. When selected, variable metadata propagation is enabled for **Transformations**, **Imputation**, **Replacement**, and **Filtering** nodes, except in the following cases:

- The node is preceded by a **Feature Extraction** or **Feature Machine** node.
- The node is preceded by a **Variable Clustering** node that uses the **Cluster component** method for the cluster representation.
- The node is preceded by a **Manage Variables** node that overrides the **Transform** and **Impute** methods, and the **Lower limit** and **Upper limit** values.

In these cases, the methods that are specified on the **Data** tab are not available to the **Transformations**, **Imputation**, **Replacement**, and **Filtering** nodes.

For a **Transformations** node, the **Lower limit**, **Upper limit**, and **Impute** metadata is not propagated to a subsequent **Imputation**, **Replacement**, or **Filtering** node in a pipeline when binning and encoding transformation methods such as **Bucket binning**, **Quantile binning**, **Level encoding**, and **Target encoding** are specified. This is because these methods change the variable type. When transformation method **Inverse**, **Inverse square**, or **Inverse square root** is specified, the **Lower limit** and **Upper limit** metadata is not propagated. This is because the inverse of the lower limit is greater than the inverse of the upper limit. When any of the remaining transformation methods are specified, before the **Lower limit**, **Upper limit**, and **Impute** metadata is propagated, the **Lower limit** and **Upper limit** values are first transformed, and the **Impute Custom constant value**, if specified, is first transformed.

Assigning Variable Metadata Details

- The following options are available when specifying variable roles:
 - **Assessment** — Specifies that the variable be used for decision processing. The Assessment role is currently not used in SAS Model Studio.
 - **Classification** — Specifies that the variable be used for model classification for a class target. For example, if the variable BAD is set as the target variable, the I_BAD variable has the 0 or 1 prediction based on the predicted probabilities and the cutoff used. The classification cutoff is applied only to binary targets.
 - **Exposure** — Specifies that the variable is a numeric variable that is used as a scaling measure of risk, which is often referred to as an exposure. Only one variable with the role of **Exposure** is allowed. This role is available beginning with SAS Viya 2025.05.
 - **Filter** — Specifies that the variable be used for filtering. For variables with the role of Filter, observations are filtered out when the value = 1 and kept when the value = 0. The **Filtering** node and the **Anomaly Detection** node create a variable with the Filter role that is used in subsequent nodes.

- **Frequency** — Specifies that the variable be used as a frequency variable. Variables that are assigned this role must be numeric and nonnegative. If a value for that variable is not an integer, that value is truncated. When a frequency variable is both specified and valid, it is used by most nodes with the following exceptions:
 - The Text Mining node
 - The Anomaly Detection node
 - The Gaussian Process Classification node
 - The Gaussian Process Regression node
 - The Quantile Regression node
- **Geolocation** -- Specifies that the variable is a geographically encoded variable.

Note: This role is available beginning with SAS Viya 2025.03.

- **ID** — Specifies that the variable is an ID variable. A variable becomes an ID variable when the variable name begins with "id", ends with "id", or when you specify **ID** as the variable role on the **Data** tab. In addition, the Cardinality action set might be used to determine whether a variable is set as an ID variable. For more information, see [Cardinality Action Set](#) in *SAS Visual Statistics: Programming Guide*.
- **Geolocation** -- Specifies that a variable is used for geographically encoded data. This role is available starting with SAS Viya 2025.03.
- **Input** — Specifies that the variable be used as an input variable in your pipeline.
- **Key** — Specifies that the variable is a unique identifier for all observations. The Key role is used by the **Text Mining** node and is used in the generation of the observation-based Model Interpretability reports.
- **Offset** — Specifies that the variable is a numeric variable that is used by the **GLM**, **Gradient Boosting**, **Logistic Regression**, and **Ratemaking** nodes. An offset variable is typically used for a covariate with a known slope. The variable that is specified is not estimated, but instead is added directly to the model. Only one variable with the role of **Offset** is allowed.
- **Partition** — Specifies that the variable be used for partitioning your data set. The variable role **Partition** can be assigned only to variables with three or fewer unique levels. When partitioning the project, SAS Model Studio creates the **PartInd** variable that is used to identify each set of partitions. A value of 0 is assigned to observations used for validation. A value of 1 is assigned to observations used for training. A value of 2 is assigned to observations used for test.

When assigning a role of **Partition** to an existing variable in the project table, SAS Model Studio uses the **Order** attribute to assign the partition values. By default, the **Order** attribute assigns partitions using the descending order of the variable. That is, the first value of the sorted variable **2** is assigned to test. The second value of the sorted variable **1** is assigned to train. The third value of the sorted variable **0** is assigned to validation.

Specifying a value other than **Default** for **Order** can affect the assignment of these values. For example, specifying **Ascending** for **Order** changes the assignment of partitions to the following values:

- A value of 0 is assigned to observations used for test.
- A value of 1 is assigned to observations used for training.
- A value of 2 is assigned to observations used for validation.

TIP You can manually map the partition variable values to each partition by selecting **Map Partition Levels** after specifying a partition variable.

- ☐ **Prediction** — Specifies that the variable is used during model assessment. This variable is the prediction for an interval target or the posterior probabilities for a class target.
- ☐ **Rejected** — Specifies that the variable be excluded from all analysis in your pipeline.
- ☐ **Residual** — Specifies that the variable is an error residual. This role is used only for informational purposes.
- ☐ **Segment** — Specifies that the variable is a segment variable. Segment variables are created by the **Clustering** node for cluster IDs created. The segment variable is also used in the **Segment Profile** node.
- ☐ **Target** — Specifies that the variable is the target variable. A target variable is required for a project that includes supervised models. Only one variable with the role of **Target** is allowed. When a nominal variable is assigned the role **Target**, the **Specify the Target Event Level** option is available, which enables you to specify the target event level. The specified target event level corresponds to the target event. If a target event level is not specified, the target event is determined as follows:
 - 1 The variable levels are sorted according to **Order**, which defaults to **Descending**.
 - 2 The first level in the sorted levels is selected as the target event.

For example, assume that you have a nominal target variable that is formatted as shown in the following table.

Raw Value	Formatted Value
0	Very Low
1	Low
2	Medium
3	High

The target event is selected as shown in the following table.

Specified Order	Sorted Level Order	Selected Target Event
Ascending	0, 1, 2, 3	0 (Very Low)
Descending (default)	3, 2, 1, 0	3 (High)
Formatted ascending	High, Low, Medium, Very Low	High (3)
Formatted descending	Very Low, Medium, Low, High	Very Low (0)

Note: For a Nominal target, the formatted target values collectively must not include all of the following special characters: blank space, semicolon (;), comma (,), period (.) and asterisk (*). If all 5 of the special characters appear in the target values, assessment results cannot be generated. At least one of the special characters must not appear in the target values.

- ❑ **Text** — Specifies that the variable is a text variable. The Text role is used by the **Text Mining** node.
- ❑ **Time ID** — Specifies that the variable is a time variable. This role is used only for informational purposes.
- ❑ **Weight** — Specifies that the variable be used to incorporate weighted observations into the model. A variable must be numeric and cannot contain negative values. The weight variable is not used for model assessment, model interpretability, and model bias assessment because every observation is considered to be of equal value for those computations. When a weight variable is both specified and valid, it is used by most nodes with the following exceptions:
 - When either **Robust PCA** or **Singular value decomposition** are specified for the **Feature extraction method** on the **Feature Extraction** node
 - The Text Mining node
 - The Clustering node when the data contains class inputs
 - The Bayesian Network node
 - The Gaussian Process Classification node
 - The Gaussian Process Regression node
- The following options are available when selecting variable levels:
 - ❑ **Binary**
 - ❑ **Interval**
 - ❑ **Nominal**
 - ❑ **Ordinal**

Note: All nodes, except for the Interactive Grouping node, analyze **Ordinal** variables as **Nominal** variables. For more information about the Interactive Grouping node, see [“Interactive Grouping Properties” in SAS Viya: Machine Learning Node Reference](#).

- The following options are available when selecting variable order:

- ☐ **Ascending**
- ☐ **Default**
- ☐ **Descending**
- ☐ **Formatted Ascending**
- ☐ **Formatted Descending**

SAS Model Studio uses the **Order** attribute to assign the partition values when a role of **Partition** is assigned to an existing variable in the project table. By default, the **Order** attribute assigns partitions using the descending order of the variable. That is, the first value of the sorted variable **2** is assigned to test. The second value of the sorted variable **1** is assigned to train. The third value of the sorted variable **0** is assigned to validation.

Specifying a value other than **Default** for **Order** can affect the assignment of these values. For example, specifying **Ascending** for **Order** changes the assignment of partitions to the following values:

- ☐ A value of 0 is assigned to observations used for test.
- ☐ A value of 1 is assigned to observations used for training.
- ☐ A value of 2 is assigned to observations used for validation.

Note: You can manually map the partition variable values to each partition by selecting **Map Partition Levels** after specifying a partition variable.

- The following options are available when selecting variable transformations:

- ☐ For class variables, the options are as follows:

- **(none)**
- **Bin rare nominal levels**
- **Default**
- **Level count encoding**
- **Level encoding**
- **Level proportion encoding**
- **Target encoding**
- **WOE encoding**

The **Bin rare nominal levels** option is available only for nominal level variables.

- ☐ For interval variables, the options are as follows:

- **(none)**

- **Best**
- **Bucket binning**
- **Centering**
- **Default**
- **Exponential**
- **Inverse**
- **Inverse square**
- **Inverse square root**
- **Log**
- **Log10**
- **Quantile binning**
- **Range standardization**
- **Square**
- **Square root**
- **Standardization**
- **Tree-based binning**

The **Best** transform option performs several transformations and uses the transformation that by default, minimizes the skewness of the variable distribution.

Note: **Log**, **Log 10**, **Square root**, and **Inverse square root** add an offset to variables to ensure positive values. **Inverse** and **Inverse square** add an offset to variables to ensure nonzero values. This prevents creating missing values during the transformation when input variable values are zero.

- The following options are available when you are specifying how to impute missing variable values:
 - For class variables, the options are as follows:
 - **(none)**
 - **Cluster count**
 - **Count**
 - **Custom constant value**
 - **Default**
 - **Default constant value**
 - **Distribution**

Cluster count, **Count**, and **Distribution** are not available for unary variables.
 - For interval variables, the options are as follows:
 - **(none)**
 - **Cluster mean**

- Custom constant value
- Default
- Default constant value
- Distribution
- Maximum
- Mean
- Median
- Midrange
- Minimum
- Trimmed maximum
- Trimmed mean
- Trimmed midrange
- Trimmed minimum
- Winsorized maximum
- Winsorized mean
- Winsorized midrange
- Winsorized minimum

You can specify a **Lower limit** and an **Upper limit** to use for the **Metadata limits** method in the **Filtering** node or the **Replacement** node.

Changing Variable Metadata for an Existing Project

For an existing project, except for the **Target** variable, you can change the role and level for variables in the input data. However, making these changes can have the following impacts:

- A pipeline that has already been run is reset and must be rerun.
- A pipeline that was automatically generated by SAS is unlocked and reset and must be rerun.

In these cases, a window appears when you change the variable metadata. The window indicates that making the change unlocks any automatically generated pipelines and resets any pipelines that have already been run. You can cancel the change or proceed with it and rerun the pipeline or pipelines afterward.

For a pipeline that includes a **Decision Tree** node that is in interactive mode, an error might occur when you attempt to open the **Decision Tree** node. In that case, you can close the interactive editor and leave the existing edits in place, or you can clear the existing edits and edit the node again as follows:

- 1 If you have not already done so, close the interactive editor window.

- 2 Right-click the **Decision Tree** node and select **Enable Properties** to clear the existing edits.
- 3 Right-click the node and select **Run**.
- 4 Right-click the node, select **Open**, and then edit the node as needed. For more information, see [“Interactively Edit a Decision Tree” in SAS Viya: Machine Learning Node Reference](#).

Managing Global Metadata

In SAS Model Studio, *metadata* is defined as the set of variable roles, measurement levels, and other configurations that apply to your data set.

When creating multiple projects using similar data sets (or when using a single data set), you might find it useful to store the metadata configurations for usage across projects. SAS Model Studio enables you to do this by collecting the variables in a repository called Global Metadata. By storing your metadata configurations as global metadata, the configurations apply to new data sets that contain variables with the same name.

To save a variable as global metadata:

- 1 On the **Data** tab, select the desired variables.
- 2 In the right pane, set the desired variable metadata assignments.
- 3 Click  in the upper right corner of the right pane.
- 4 A window appears, confirming that the operation was successful.

To examine and manage the variables designated as global metadata:

- 1 Click  in the upper left corner of the window.
- 2 The Global Metadata window appears.

Global Metadata								
		Filter						
☐	Variable Name	Label	Type	Role	Assess for Bias	Level	Order	Comment
☐	CONTROL_NUMBER		Character	ID		Nominal	Default	The variable exceeds the
☐	DONOR_AGE		Numeric	Input		Interval	Default	
☐	URBANICITY		Character	Input		Nominal	Default	
☐	WEALTH_RATING		Numeric	Input		Nominal	Default	

This window displays a table that contains all variables specified as global metadata, as well as their metadata assignments.

- 3 To remove a variable from the global metadata repository, select the desired variable and click  in the upper right corner of the window.

Note: If the advisor determines that a variable should be assigned the role of **Rejected** and you change the role of that variable, the new role assignment in the global metadata does not carry over to other projects. You must manually reassign the role of those variables on a project-by-project basis.

Integration with SAS Visual Analytics

SAS Visual Analytics enables you to transfer certain analytical models from SAS Visual Analytics to SAS Model Studio. You can transfer the model to a new project or an existing project in SAS Model Studio. To move a model from SAS Visual Analytics to SAS Model Studio, click **Create pipeline**, and then select **Add to new project** or **Add to existing project**.

If you select **Add to new project**, the pipeline is added to a new project in SAS Model Studio. The new project contains the following elements:

- the active data set
- score code to apply all data processing, filtering, and transformations
- score code to run the model that was exported

Some of the metadata for the new project is set based on what is defined in the SAS Visual Analytics project. This metadata includes the partition variable, if a partition variable was used, and the target variable that corresponds to the response variable in the SAS Visual Analytics project. If Frequency and Weight variables are used in the SAS Visual Analytics project, they are transferred to SAS Model Studio as well.

If you select **Add to existing project** and then select an existing SAS Model Studio project, the pipeline is added to the selected project.

In either case, by default, when you run a SAS Visual Analytics pipeline in SAS Model Studio, the project table is scored, and the assessment reports are generated for the SAS Visual Analytics modeling node. In SAS Model Studio, if you want to retrain the model or modify the properties, right-click the node, and then select **Enable properties**. This enables you to modify the model's properties, if desired, and it causes the model to be retrained. Model interpretability properties are always automatically enabled. You can add and delete nodes in this pipeline as in any other SAS Model Studio pipeline. You can use the SAS Model Studio model comparison and pipeline comparison tools to evaluate your transferred models against any new models.

At this time, the supported models are Bayesian Network, Decision Tree, Factorization Machine, Forest, Generalized Linear Model, Gradient Boosting, Linear Regression, Logistic Regression, Neural Network, and Support Vector Machine. However, within these models, if the SAS Visual Analytics model contains a Generalized Linear Model, Linear Regression, or Logistic Regression that uses an offset or group by variable, the pipeline cannot be created in SAS Model Studio.

There are a few caveats to note when transferring a model from SAS Visual Analytics to SAS Model Studio:

- In order to add a SAS Visual Analytics model to an existing SAS Model Studio project, the following requirements must be met:
 - The target variable name and type in the SAS Model Studio and SAS Visual Analytics projects must match.
 - If the SAS Visual Analytics response variable is category, then the SAS Model Studio target variable must be binary or nominal, and the number of levels for both variables must match.
 - If the SAS Visual Analytics response variable is measure, then the target variable level in SAS Model Studio must be interval.
 - If the SAS Model Studio project uses a weight variable, frequency variable, or both, then in the SAS Visual Analytics model, the weight and frequency variables must not be used as input or response variables, or to derive other variables.

Only the existing SAS Model Studio projects that meet all of the applicable requirements are listed when you select **Add to existing project** in SAS Visual Analytics.

If the existing SAS Model Studio project uses a partition variable, then that same variable, if used in a SAS Visual Analytics report, must also be a partition variable. If the variable used in the report is not a partition variable, the mismatch does not prevent the SAS Model Studio project from being listed as an available project. However, if you select a SAS Model Studio project that has that variable with the partition role, then the transfer process fails, and the transferred model is automatically removed from the SAS Model Studio project.

- If the SAS Visual Analytics model contains a Logistic Regression with a non-binary response variable, when the Logistic Regression model is copied to a new SAS Model Studio project, the multinomial (non-binary) response variable is mapped to a binary response variable, and the transformed target variable level is set to Binary in SAS Model Studio.
- Instead of using the variable name that exists in the original data set, SAS Visual Analytics prefers to use the variable label. However, SAS Model Studio prefers to use the variable name as it exists in the original data. Therefore, if the variable names and variable labels in your input data are different, you might experience some unexpected naming issues when a model is transferred. SAS Model Studio displays both the variable name and the variable label in the **Variables table** layout of the Data pane.
- SAS Visual Analytics creates a custom name for target variables. This new variable is indicated with a label in SAS Model Studio.
- When you are exporting from the Model Comparison object, only the champion model is exported.
- Certain actions that create a data item in SAS Visual Analytics are performed in the **Visual Data Preparation** node in SAS Model Studio. For example, when you derive a cluster ID in SAS Visual Analytics, a data item is created in the **Data** pane. If you specify this created data item in a SAS Visual Analytics model that is transferred to SAS Model Studio, it does not appear in the **Data** pane of your project. Instead, it is re-created when the **Visual Data Preparation** node runs.

- When you run a transferred SAS Visual Analytics modeling node in the SAS Model Studio project, the assessment results might differ from the results in SAS Visual Analytics. Possible reasons for differences are as follows:
 - The input table data is different.
 - If the project data is partitioned, the partitions might be different.
 - Some of the predictor variables that are used in the SAS Visual Analytics models are not present in the SAS Model Studio project table.
- If you enable the properties of a transferred SAS Visual Analytics modeling node in SAS Model Studio, the model is retrained, which might yield new results. Once properties have been enabled, the model cannot be switched back to use the original SAS Visual Analytics score code. **Model Interpretability** properties do not require retraining the model. Therefore, they are automatically enabled and do not affect the original SAS Visual Analytics score code.
- You cannot save your pipeline as a template to the exchange.
- If you transfer a Gradient Boosting, Decision Tree, or Forest model, the value for the **Maximum levels** property that is mapped to SAS Model Studio is one fewer than the value that you specified in SAS Visual Analytics.

Finally, you cannot transfer a model from SAS Model Studio to SAS Visual Analytics. However, you can copy the input data to SAS Visual Analytics for exploration and visualization. See [“Explore and Visualize Your Data” on page 121](#).

Explore and Visualize Your Data

The **Output Data** tab is located in the upper left corner of the **Results** window. It enables you to generate output data. A row is created for each observation, and all original and created columns are displayed. The created columns can contain imputation results, model prediction results, or any other information created by the node.

In order to generate output data, click the **Output Data** tab, and then click **View output data** in the middle of the screen. When the output data is successfully generated, the table containing that data appears. You can save the output data for later use by clicking  in the upper left corner of the **Data sources** tab.

Note: Table names cannot exceed 247 bytes.

Note:

In 2024.12, saving component output tables to an Azure Data Lake Storage (ADLS) caslib is not supported. Starting with 2025.01, saving component output tables to an ADLS caslib is supported.

You can also visualize your output data by clicking  in the upper left corner of the **Data sources** tab. The Explore and Visualize Output Data window appears, and you are prompted to select a CAS library that you want to save your output table in.

Note: To use the **Explore and visualize** functionality in SAS Model Studio, you must be able to create and edit reports in SAS Visual Analytics.

When you have selected a CAS library, click **Explore and visualize** in the lower right corner of the window. This redirects you to SAS Visual Analytics, where you can use a variety of tools to model your data. For information about using SAS Visual Analytics, see [SAS Visual Analytics: Getting Started with Reports](#).

Working with Templates

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Overview of Templates

SAS Model Studio supports templates as a method for creating statistical models quickly. A *template* is a special type of pipeline that is pre-populated with configurations that can be used to create a model. A template might consist of multiple nodes or a single node. SAS Model Studio includes a set of templates that represent frequent use cases, but you can also create models themselves and save them as templates in the toolkit.

Creating a New Template from a Pipeline

To create a template from a pipeline:

- 1 Click  next to the pipeline tab in the upper left corner of the canvas.
- 2 Select **Save to The Exchange**.
- 3 In the Save Pipeline to The Exchange window, enter a **Name** and **Description** for the new template.
- 4 Click **Save**.

Note: **Managed Variable** and **Decision Tree** nodes that have been interactively edited are not duplicated as configured when the pipeline is saved to the Exchange. This is because the interactive edits that were made cannot be persisted. If the saved pipeline includes either or both of these nodes, after the pipeline is added to a project from the Exchange, the **Managed Variable** and **Decision Tree** nodes must be interactively edited in order to restore the original changes.

You can also create templates from singular nodes. To create a template from a node:

- 1 Right-click on the desired node. Select **Save As**. The Save Node to The Exchange window appears.
- 2 In the Save Node to The Exchange window, enter a **Name** and **Description** for the new template.
- 3 Click **Save**.

Note: A **Manage Variables** node cannot be saved to the Exchange because any variable metadata changes that are made in the node cannot be persisted. The **Save as** option is disabled.

Note: Starting in 2025.05, when a **Decision Tree** node is in interactive mode, it cannot be saved to the Exchange because the interactive changes cannot be persisted. In that case, the **Save as** menu option is disabled. For more information, see [“Interactively Edit a Decision Tree” in SAS Viya: Machine Learning Node Reference](#).

Creating a New Template in The Exchange

- 1 Click  in the upper left corner of the screen. The Exchange page opens. This page enables you to examine all available templates. The Exchange stores node and pipeline templates for SAS Visual Machine Learning, SAS Visual Text Analytics, and SAS Visual Forecasting applications.
- 2 To create a new template, select the existing template most similar to your desired template. You will duplicate and modify this template.
- 3 Click  in the upper right corner of the screen and select **Duplicate**
- 4 For node templates, the Duplicate Node window appears. For pipeline templates, the Duplicate Pipeline window appears. Enter a name and a description for the new template.

- 5 Click **Save**. Your new template appears in the list of templates.

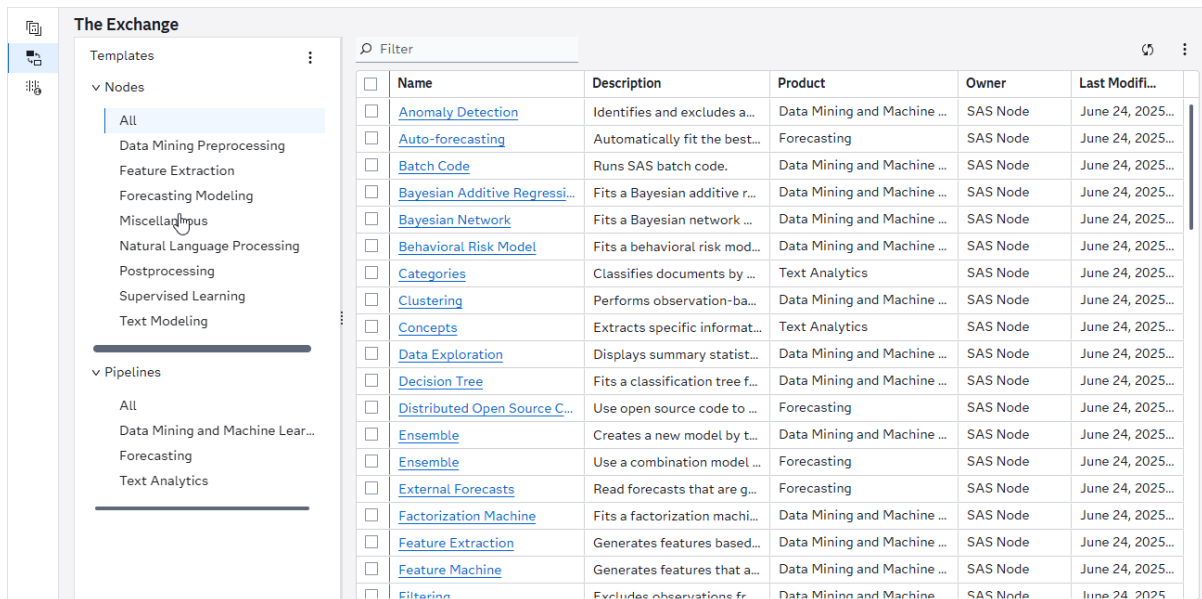
Nodes and pipelines that are created using these custom templates are copies of the templates. The nodes and pipelines can be further modified when they are used in a project. The templates can be removed without affecting any of the nodes and pipelines that were created from them.

You can also save node and pipeline templates from the project pipeline. For more information, see [“Creating a Template from a Pipeline” on page 145](#).

Modifying an Existing Template

If you have sufficient permissions, you can modify existing templates. To modify a template:

- 1 Click  in the upper left corner of the screen.
- 2 The Exchange page opens. This page enables you to examine all available templates. The Exchange stores node and pipeline templates for SAS Visual Data Mining and Machine Learning, SAS Visual Text Analytics, and SAS Visual Forecasting applications.



Name	Description	Product	Owner	Last Modified
Anomaly Detection	Identifies and excludes a...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Auto-forecasting	Automatically fit the best...	Forecasting	SAS Node	June 24, 2025...
Batch Code	Runs SAS batch code.	Data Mining and Machine ...	SAS Node	June 24, 2025...
Bayesian Additive Regressi...	Fits a Bayesian additive r...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Bayesian Network	Fits a Bayesian network ...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Behavioral Risk Model	Fits a behavioral risk mod...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Categories	Classifies documents by ...	Text Analytics	SAS Node	June 24, 2025...
Clustering	Performs observation-ba...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Concepts	Extracts specific informat...	Text Analytics	SAS Node	June 24, 2025...
Data Exploration	Displays summary statist...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Decision Tree	Fits a classification tree f...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Distributed Open Source C...	Use open source code to ...	Forecasting	SAS Node	June 24, 2025...
Ensemble	Creates a new model by t...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Ensemble	Use a combination model ...	Forecasting	SAS Node	June 24, 2025...
External Forecasts	Read forecasts that are g...	Forecasting	SAS Node	June 24, 2025...
Factorization Machine	Fits a factorization machi...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Feature Extraction	Generates features based...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Feature Machine	Generates features that a...	Data Mining and Machine ...	SAS Node	June 24, 2025...
Filtering	Excludes observations fr...	Data Mining and Machine ...	SAS Node	June 24, 2025...

- 3 To access a particular template, click the template name. This opens the Node Template or Pipeline Template window. If you do not have Edit privileges for a given template, you will see **(Read-Only)** displayed in the window.

In the Node Template or Pipeline Template window, you can make changes and configure the nodes in the pipeline. Changes are saved automatically to the template.

Note: While you are editing a template, nodes can be re-configured, but no nodes can be added or deleted.

Available Templates

The following Node templates are included with SAS Model Studio:

Node Name	Node Description	Product
Anomaly Detection	Identifies and excludes anomalies (observations) using the support vector data description.	Data Mining and Machine Learning
Auto-forecasting	Use an ESM, ARIMAX, IDM, or UCM model to generate forecasts.	Forecasting
Batch Code	Runs SAS batch code.	Data Mining and Machine Learning
Bayesian Network	Fits a Bayesian network model for a class target.	Data Mining and Machine Learning
Categories	Classifies documents by subject.	Text Analytics
Clustering	Performs observation-based clustering for segmenting data.	Data Mining and Machine Learning
Concepts	Extracts specific information from text.	Text Analytics
Data Exploration	Displays summary statistics and plots for variables in your data table.	Data Mining and Machine Learning
Decision Tree	Fits a classification tree for a class target or a regression tree for an interval target.	Data Mining and Machine Learning
Ensemble	Creates a new model by taking a function of posterior probabilities (for class targets) or the predicted values (for interval targets) from multiple models.	Data Mining and Machine Learning

Node Name	Node Description	Product
External Forecasts	Reads forecasts that are generated by an external source.	Forecasting
Feature Extraction	Generates features based on PCA, robust PCA, SVD, or autoencoders to use as inputs. Note that PCA, SVD, and RPCA use interval inputs only.	Data Mining and Machine Learning
Feature Machine	Generates features that address one or more identified transformation policies.	Data Mining and Machine Learning
Filtering	Excludes observations from analysis based on specified criteria.	Data Mining and Machine Learning
Forest	Fits a forest model, which consists of multiple decision trees based on different samples of the data and different subsets of inputs.	Data Mining and Machine Learning
GLM	Fits a generalized linear model for an interval target with a specified target distribution and link function.	Data Mining and Machine Learning
Gradient Boosting	Fits a gradient boosting model, which builds a sequential series of decision trees.	Data Mining and Machine Learning
Hierarchical Forecasting	Generates forecasts for each level of the specified hierarchy.	Forecasting
Hierarchical Forecasting (Pluggable)	Generates forecasts using hierarchical forecasting model.	Forecasting
Imputation	Imputes missing values for class and interval inputs using the specified methods.	Data Mining and Machine Learning
Linear Regression	Fits an ordinary least squares regression model for an interval target.	Data Mining and Machine Learning
Logistic Regression	Fits a logistic regression model for a binary or nominal target.	Data Mining and Machine Learning

Node Name	Node Description	Product
Manage Variables	Modifies the metadata of variables.	Data Mining and Machine Learning
Model Composer	Automatically tunes hyperparameters for multiple model types concurrently with optimal allocations, and then selects the top model.	Data Mining and Machine Learning
Multistage Model	Generates forecasts using a multistage forecasting model.	Forecasting
Naive Model	Generates forecasts using naive model.	Forecasting
Neural Network	Fits a fully connected neural network model.	Data Mining and Machine Learning
Non-seasonal Model	Generates forecasts using a non-seasonal ESM, ARIMAX, or UCM model.	Forecasting
Open Source Code	Runs Python or R code.	Data Mining and Machine Learning
Panel Series Neural Network	Generates forecast using fully connected neural network model.	Forecasting
Quantile Regression	Fits a quantile regression model for an interval target.	Data Mining and Machine Learning
Regression for Time Series	Uses a regression model to generate forecasts	Forecasting
Replacement	Replaces data values such as outliers and unknown class levels with specified values.	Data Mining and Machine Learning
Retired Series	Generates forecasts for retired series using a specified value.	Forecasting
SAS Code	Runs SAS code.	Data Mining and Machine Learning
Save Data	Saves data exported by a node in a pipeline to a CAS library.	Data Mining and Machine Learning

Node Name	Node Description	Product
Score Code Import	Imports SAS score code.	Data Mining and Machine Learning
Score Data	Scores a table using the score code generated by predecessor nodes and saves the scored table to a CAS library.	Data Mining and Machine Learning
Seasonal Model	Generates forecasts using a seasonal ESM, ARIMAX, or UCM model.	Forecasting
Segment Profile	Examines segmented data and enables identification of factors that differentiate the segments from the population.	Data Mining and Machine Learning
Sentiment	Analyzes attitudes expressed in documents.	Text Analytics
Stacked Model (NN + TS) Forecasting	Generates forecasts using stacked model (Neural Network + Time Series).	Forecasting
SVM	Fits a support vector machine via interior-point optimization for a binary target.	Data Mining and Machine Learning
Temporal Aggregation Model	Generates forecasts using a temporal aggregation model.	Forecasting
Text Mining	Parses and performs topic discovery to prepare text data for modeling.	Data Mining and Machine Learning
Text Parsing	Prepares text for terms analysis.	Text Analytics
Topics	Assigns documents to topics.	Text Analytics
Transformations	Applies numerical or binning transformations to input variables.	Data Mining and Machine Learning
Variable Clustering	Performs variable clustering to reduce the number of inputs.	Data Mining and Machine Learning
Variable Selection	Performs unsupervised and several supervised methods of variable	Data Mining and Machine Learning

Node Name	Node Description	Product
	selection to reduce the number of inputs.	

The following Pipeline templates are included with SAS Model Studio:

Pipeline Name	Pipeline Description	Product
Advanced template for class target	Extends the intermediate template for class target with neural network, forest, and gradient boosting models, as well as an ensemble.	Data Mining and Machine Learning
Advanced template for class target with autotuning	Advanced template for class target with autotuned tree, forest, neural network, and gradient boosting models.	Data Mining and Machine Learning
Advanced template for interval target	Extends the intermediate template for interval target with neural network, forest, and gradient boosting models, as well as an ensemble.	Data Mining and Machine Learning
Advanced template for interval target with autotuning	Advanced template for interval target with autotuned tree, forest, neural network, and gradient boosting models.	Data Mining and Machine Learning
Advanced template for risk modeling	Risk modeling pipeline that extends the basic risk modeling template with a Reject Inference node and subsequent Interactive Grouping and Scorecard nodes.	Data Mining and Machine Learning
Auto-forecasting	Forecasting pipeline with automatic modeling.	Forecasting
Auto-forecasting (Intermittent)	Forecasting pipeline with automatic, intermittent modeling.	Forecasting
Base Forecasting	Forecasting pipeline with no modeling components added by default.	Forecasting
Basic template for class target	A simple linear flow: Data, Imputation, Logistic Regression, Model Comparison.	Data Mining and Machine Learning

Pipeline Name	Pipeline Description	Product
Basic template for interval target	A simple linear flow: Data, Imputation, Linear Regression, Model Comparison.	Data Mining and Machine Learning
Basic template for risk modeling	Risk modeling pipeline that contains an Interactive Grouping node followed by a Scorecard node.	Data Mining and Machine Learning
Blank Template	A Data Mining pipeline that contains only a data node.	Data Mining and Machine Learning
Demand Classification	Forecasting pipeline with demand classification segmentation.	Forecasting
External Forecasts	Forecasting pipeline with external forecasts.	Forecasting
External Segmentation	Forecasting pipeline with external segmentation.	Forecasting
Feature engineering template	Data mining pipeline that performs feature engineering.	Data Mining and Machine Learning
Hierarchical Forecasting	Forecasting pipeline with hierarchical modeling.	Forecasting
Intermediate template for class target	Extends the basic template with a stepwise logistic regression model and a decision tree.	Data Mining and Machine Learning
Intermediate template for interval target	Extends the basic template with a stepwise linear regression model and a decision tree.	Data Mining and Machine Learning
Naive (Moving Average) Forecasting	Forecasting pipeline with naive, moving average modeling.	Forecasting
Naive Forecasting	Forecasting pipeline with naive modeling.	Forecasting
Non-seasonal Forecasting	Forecasting pipeline with non-seasonal modeling.	Forecasting
Regression Forecasting	Forecasting pipeline with regression modeling.	Forecasting

Pipeline Name	Pipeline Description	Product
Retired Forecasting	Forecasting pipeline with retired modeling.	Forecasting
Seasonal Forecasting	Forecasting pipeline with seasonal modeling.	Forecasting
Text Analytics: Assisted Concept Rule Creation	Use Textual Elements to quickly generate custom concept rules.	Text Analytics
Text Analytics: Data Access	Text Analytics pipeline that contains a single Data node.	Text Analytics
Text Analytics: Generate Concepts, Topics, and Categories	Text Analytics pipeline for model generation with Concepts, Text Parsing, Sentiment, Topics, Categories.	Text Analytics
Text Analytics: Topic Discovery	Text Analytics pipeline that uses text parsing and machine learning to discover topics.	Text Analytics

Working with Pipelines

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Overview of Pipelines

SAS Model Studio projects are built around one or more pipelines. A *pipeline* is a process flow diagram that can be used to represent a sequence of analytical tasks. These analytical tasks are represented as individual nodes in a pipeline.

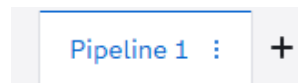
By default, the initial pipeline for a project uses the template that was specified when the project was created. You can create new pipelines using different templates, and you can make changes to the initial pipeline.

Creating a New Pipeline

In SAS Model Studio, pipelines contain the nodes that process data and create models. A project can contain multiple pipelines.

To create a new pipeline:

- 1 Navigate to the **Pipelines** tab.
- 2 Click **+** next to the current pipeline tab in the upper left corner of the canvas.



The New Pipeline window appears.

- 3 Give the pipeline a name and an optional description.
- 4 Click **OK**.

In the **Template** field, your recently used templates are available. To use a template that you have not used recently, select **Browse templates** and select a template in the Browse Templates window.

New Pipeline

Name: *
Pipeline 3

Description:

☒ Select a pipeline template

Template:
Blank template ▼ Browse

☐ Automatically generate the pipeline ⓘ

☒ Set automation time limit ⓘ

15 minutes

Advanced Settings

Save Cancel

Note: Automatic pipeline generation is available only in SAS Visual Data Mining and Machine Learning projects.

You can also duplicate a pipeline. Click  next the current pipeline tab in the upper left corner of the canvas and click **Duplicate**. The duplicate functionality is not available in SAS Visual Text Analytics.

Actions on the Pipeline

Click  on the current pipeline tab to perform the following actions:

- **Run** — Runs the entire pipeline.
- **Force run** — Runs a pipeline that has already completed a successful run. This action might be needed if there was an update applied outside of the SAS Viya environment (for example, a Python or R program). This action was added in release 2026.04.

- **Stop** — Stops a pipeline that is running.
- **Duplicate** — Creates a duplicate pipeline. The name is appended with a number. You can rename the duplicate after it is created. The duplicate functionality is not available in SAS Visual Text Analytics.
- **Rename** — Enables you to rename the pipeline.
- **Save to The Exchange** — Saves the pipeline with the nodes and any settings applied to those nodes as a template to The Exchange. The new template can be used in other projects.
- **Delete** — Deletes the pipeline. This option is available when you have more than one pipeline in your project.
- **Show overview map** — Places a map of the pipeline in the upper left corner of the canvas.
- **Expand header** — Provides a space at the top of the tab to add a description or other text that might be useful. When you select **Expand header** and click inside the header area, a rich-text editor enables you to add headers and text, change text attributes, and add links.

Note: The content of the header cannot exceed 4000 characters.

Note: To open a link in the editor content, press and hold Ctrl, and then click the link.

- **Copy graph to clipboard** — Enables you to copy the pipeline and paste it into rich text applications such as a word processor or email. The image cannot be pasted directly into image editors. However, you can select it from the rich text application and then paste it into the image editor.

Note: You can also copy any plot or graph image in SAS Model Studio to your clipboard. Right-click the image to access the **Copy graph to clipboard** selection.

- **Copy pipeline link** — Copy the link to a pipeline in your project to your clipboard. Only users with Read permission for the project that contains the pipeline can open the link.
- **Unlock** — For automatically generated pipelines, select this option to edit the pipeline.
- **Logs** — For automatically generated pipelines, select this option to view the pipeline creation log.

Before you add a node to a pipeline, you can hover your cursor over the node to see a quick description of that node. Follow these steps:

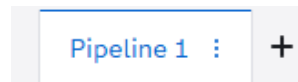
- 1 Navigate to the **Pipelines** tab, and click .
- 2 Select one of the node categories to see the list of nodes that it contains.
- 3 Hover your cursor over a node name to see the description for that node.

Automated Pipeline Creation

Initiating the Automated Pipeline Creation Process

SAS Viya can use automated machine learning to dynamically build a pipeline that is based on your data. To initiate the automated pipeline creation process, do the following:

- 1 Navigate to the **Pipelines** tab.
- 2 Click **+** next to the current pipeline tab in the upper left corner of the canvas.



The New Pipeline window appears.

- 3 In the New Pipeline window, select **Automatically generate the pipeline**.

.....
Note: The data advisor must finish running, and a variable must be assigned the role **Target** before the **Automatically generate the pipeline** option is available.
.....

×

New Pipeline

Name:

Pipeline 2

Description:

☐ Select a pipeline template

Template:

Blank template ▼ Browse

☒ Automatically generate the pipeline ?

☒ Set automation time limit ?

10 minutes

Advanced Settings

Save Cancel

- 4 To specify a maximum time limit for pipeline generation, leave **Automation time limit** checked, and change the maximum time in the **minutes** field, if needed.
- 5 To specify sampling preferences and which algorithms to use for candidate models when automatically generating the pipeline, click **Advanced Settings** to open the Advanced Settings window.
- 6 On the **Algorithms** tab in the Advanced Settings window, select the algorithms that you want to be either considered or forcibly included in the automatically generated pipeline. Algorithms that are selected for forcible inclusion are automatically added to the final pipeline.

Note: A LightGBM model is also considered in the final pipeline when a Gradient Boosting model is considered.

By default, either a linear regression model or a logistic regression model is selected for forcible inclusion. This ensures compatibility with previous versions of SAS Viya, and provides an interpretable model for regulatory purposes.

Note: Hover over the ⓘ icon for the algorithms where it exists for information about the scenarios in which those algorithms are supported.

- 7 Navigate to the **Sampling** tab. Sampling preferences enable you to reduce the amount of data and therefore the amount of time used to generate the pipeline. If you select **Enable sampling**, specify one of the following:
 - **Number of training rows**
 - **Percentage of training rows**
- 8 Click **OK** to close the Advanced Settings window.
- 9 In the New Pipeline window, click **Save** to close the New Pipeline window and initiate the automated pipeline creation process.

A new **Pipelines** tab is created and the pipeline generation process begins. During pipeline generation, shared projects are locked to your user ID, even if you exit the project. The process terminates when it is complete, the time limit passes, or you click **Stop**. The generated pipeline then appears in the new **Pipelines** tab.

About the Automated Pipeline Creation Process

When you initiate the automated pipeline creation process, SAS Viya uses automated machine learning to dynamically build a pipeline that is based on your data. This process automatically performs data preparation, model building, model comparison, and model selection on your data to create the pipeline. During pipeline construction, shared projects are locked to the user who started the process, even if that user exits the project.

SAS Viya builds the final pipeline within the time limit that is specified in **Set automation time limit**. The process for building the pipeline involves implementing multiple candidate pipelines, each with different combinations of models with different parameter settings. The models are chosen from the following pool:

- Decision Tree
- Ensemble
- Forest
- GAM
- Gradient Boosting
- Light Gradient Boosting (LightGBM)

Note: LightGBM does not currently support variable names with special characters.

- Neural Network
 - Regression
 - SVM
-

Note: SVM applies only when the target variable is binary.

The candidate pipelines are implemented using the same model parameter values that are used in the SAS Model Studio advanced template. For Gradient Boosting and Forest models, meta-learning is used to set smart default values for the hyperparameters.

The automated pipeline creation process attempts to improve model accuracy by using the following techniques:

- Applying the **Best** transformation method to interval inputs
- Binning or encoding variables to create new features
- Imputing or creating missing value indicators
- Generating a set of new features for each input based on its characteristics
- Extracting new features from a variable with a date component when the role of that variable is set to **ID** or **Input**
- Processing textual data to use as input in subsequent nodes
- Selecting important inputs
- Creating an ensemble model of two or more of the top branches of a pipeline

Subsequent tasks for the candidate pipelines that are built might include the following:

- Tune the models — If sufficient time remains within the automation time limit, perform hyperparameter tuning on the models in the candidate pipelines. Hyperparameter tuning helps find the hyperparameter values that maximize model performance. The hyperparameter tuning process uses a genetic algorithm that searches over a range of hyperparameter values and selects the best performing hyperparameter value combination.
- Compare pipelines — for more information about comparing pipelines, see [“Comparing Pipelines” on page 149](#).
- Review model interpretability results — enable model interpretability properties in the Supervised Learning nodes. These properties include variable importance and partial dependency (PD) plots, as well as ICE, LIME, and HyperSHAP techniques.

The final pipeline selection is based on the model comparison selection statistic and data partition that are specified in the project settings. The selected pipeline contains the top five models.

Note: Assessment is performed on the validation data set.

The selected pipeline is then constructed and displayed on the **Pipelines** tab.

About Automatically Generated Pipelines

Pipelines that are created by SAS automation are indicated by the  icon on the **Pipelines** tab.

After your pipeline is created, it is in a locked state and has not been run. This means that the pipeline cannot be edited and results are not immediately available. The pipeline also includes a note that indicates it was initially generated by SAS automation. If you set a time limit or manually stop the pipeline generation, the best pipeline that is generated in that time frame is displayed, but it might not be the optimal pipeline.

To run the pipeline, click the **Run pipeline** button in the project workspace. You do not need to unlock the pipeline to run it.

To edit the pipeline, including the note, you must unlock it by clicking the **Unlock pipeline** button at the top of the project workspace. Unlocking a pipeline enables you to edit the pipeline as if it were a user-created pipeline. The original note remains until you edit or remove it.

Automated Pipeline Creation Using REST API

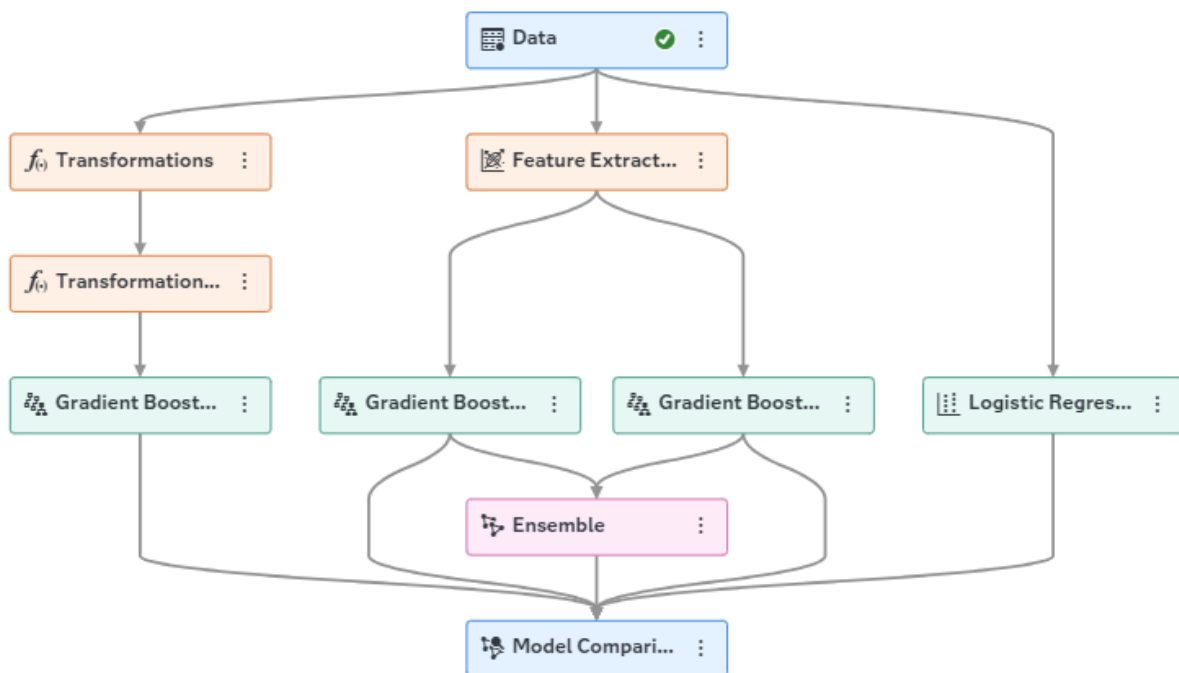
Another way to automatically generate pipelines in SAS Model Studio is through RESTful interfaces. The Machine Learning Pipeline Automation REST API is a set of endpoints that enables you to control additional parameters that are not available in the SAS Model Studio user interface.

An end-to-end example in Python to automatically generate pipelines using this REST API is provided in the `example1.py` file in the [sas-viya-dmml-pipelines GitHub repository](#). Other languages or REST clients can also be used in place of Python.

For more information about the REST API and the authentication tokens needed for access, see category [Auto Machine Learning](#) under Explore REST APIs by Category on [developer.sas.com](#). More details about automatically generate pipelines using the REST API can be found in [Automation in SAS® Visual Data Mining and Machine Learning](#) by Czika, Medins, and Myneni.

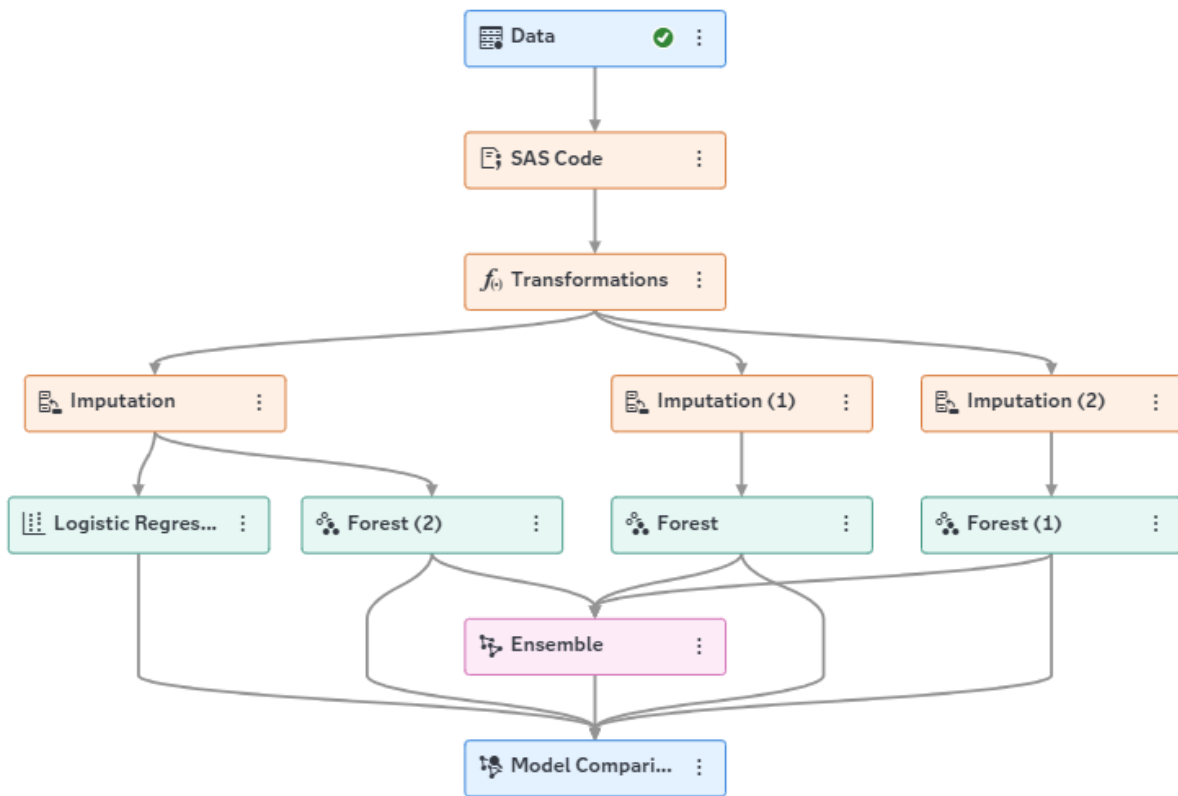
MLPA Example Using the SAS Model Studio User Interface

A data set that contains the information about credit card clients is used in this example to predict the probability that an individual defaults on credit card payments. An example of the automatically generated pipeline using the default Project Setting values is shown below.



The best models that were found include four forest models as well as an ensemble model. Each model uses different data preprocessing and transformation methods.

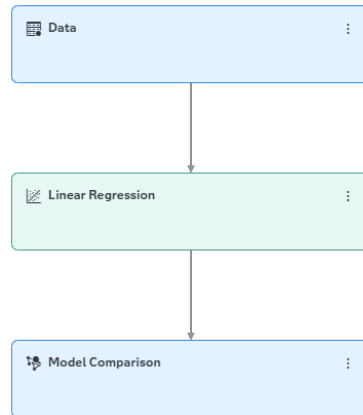
By changing the Class selection statistic option in Project Settings from the default value Kolmogorov-Smirnov statistic (KS) to Multiclass log loss, generating a new pipeline results in a different set of data preprocessing steps. For example, weight-of-evidence (WOE) encoding is applied to high-cardinality inputs, alternative imputation techniques are performed, and the best ensemble model includes only the forest models.



Modifying a Pipeline

After creating a new pipeline, you are ready to create functionality by adding nodes to the pipeline in either of the following ways:

- 1 Expand the **Nodes** pane on the left side of the canvas. By default, the **Blank Template** still contains a data node. These steps assume that you are starting with the **Blank Template**.
- 2 Select a node from the **Data Mining Preprocessing**, **Supervised Learning**, or **Miscellaneous** sections, click and drag it so that the node icon is positioned over the **Data** node, and release the cursor. The new node is added to the canvas, automatically connected to the **Data** node.



Similarly, you can add more nodes to the pipeline, either connected to the **Data** node, or to the other nodes. Click and drag the new node so that the icon is positioned over the existing node. The new node is added to the pipeline, connected to the node that it was positioned over. See the restrictions (described below) that govern how nodes can be connected to each other.

- 3 Alternatively, right-click on the **Data** node, and select **Add child node**. Select a node from the **Data Mining Preprocessing**, **Supervised Learning**, or **Miscellaneous** options.

Similarly, you can connect more nodes to existing nodes by either selecting **Add child node** (creating more successor nodes) or **Add parent node** (creating predecessor nodes). As with the other method for adding nodes, there are some restrictions as to how nodes can be connected to each other.

- 4 You can copy a node from a pipeline and paste it into the same or another pipeline in the same project. Any changes you have made to the node are copied along with the node. This includes any changes that you made in the Options pane or to the code.
 - a Right-click the node that you want to copy and select **Copy**. You can also select the node and copy using Ctrl+C. The node is copied to your clipboard.
 - b Select the target node in a pipeline where you want to paste the copied node. The copied node will be pasted after the target node.
 - c Right-click the target node and select **Paste**. You can also paste the node using Ctrl+V.

Note: The **Copy** and **Paste** options are disabled for the following nodes:

- **Manage Variables**
 - **Interactive Grouping**
 - **Visual Data Preparation**
 - **Decision Tree** (in interactive mode)
-

You can also add nodes to your pipeline. This can be useful if multiple users are working in the same project. Click **⋮** next to the current pipeline tab in the upper

left corner of the canvas and click **Expand header**. Click **Collapse header** to hide the notes.

SAS Model Studio has a series of rules that govern where nodes can be placed:

- 1 **Data Mining Preprocessing** nodes can follow the **Data** node or other **Data Mining Preprocessing** nodes. They cannot follow **Supervised Learning** or **Postprocessing** nodes.
- 2 **Supervised Learning** nodes can follow the **Data** node or **Data Mining Preprocessing** nodes. They cannot follow **Postprocessing** nodes or other **Supervised Learning** nodes.
- 3 **Postprocessing** nodes can follow only **Supervised Learning** nodes. They are invalid elsewhere.
- 4 **Miscellaneous** nodes can follow any SAS Model Studio nodes except for the **Model Comparison** node. The **Model Comparison** node is generated when any **Supervised Learning** node is added to the pipeline, and is automatically connected to follow the added **Supervised Learning** node.
- 5 The **Data Exploration**, **Model Comparison**, **Save Data**, and **Score Data** nodes are terminal nodes—that is, no nodes can follow these nodes in your pipeline.

You can delete nodes from your pipeline by right-clicking the node, selecting **Delete**, and clicking **Delete** on the Delete window.

Note:

The **Model Comparison** node in a pipeline cannot be deleted directly. The **Delete** option is disabled in the **Model Comparison** node pop-up menu. The **Model Comparison** node is deleted automatically when the last remaining Supervised Learning node in the pipeline is deleted.

Creating a Template from a Pipeline

To create a template from a pipeline:

- 1 Click  next to the pipeline tab in the upper left corner of the canvas.
- 2 Select **Save to The Exchange**.
- 3 In the Save Pipeline to The Exchange window, give the template a name and an optional description.
- 4 Click **Save**.

Note: If the pipeline includes one or more **Manage Variables** nodes, variable metadata changes that were made in these nodes are not persisted when the pipeline is saved to the Exchange.

Running a Pipeline

There are two ways to run a pipeline:

- Run all the nodes of the pipeline sequentially, starting with the **Data** node. This is done by clicking **Run pipeline** in the upper right corner of the canvas.
- Run one branch of the pipeline, running only the selected node, and all nodes preceding that node by arrows. This is done by right-clicking a node, and selecting **Run**.

To interrupt a running pipeline, click **Stop pipeline** in the upper right corner of the canvas.

If you have multiple pipelines in your project, you can run all of them at the same time. Here is how to run all of your pipelines simultaneously:

- 1 Navigate to either the **Data** tab or the **Pipelines** tab.
- 2 Click  in the upper right corner of the window.

Node Status

As you look at a pipeline, hold your pointer over each node to see its status.

- When you first create the pipeline, all nodes have a status of **Initialized**.
- After a node in the pipeline has successfully finished execution, the node shows a status of **Successful**.
- A node with a status of **Pending** is waiting for other nodes in the pipeline to complete before starting.
- When a node starts executing its functions, its status is **Running**. You cannot make any changes to the node until it finishes running.
- If a pipeline run fails, check to see which node has a **Failed** status.
- If you make any changes to the project settings, variable assignments, or the project training table the nodes change to **Out-of-date** status, and the pipeline must be run again.

- If a pipeline is stopped during processing, the status of the subsequent nodes in the pipeline is **Canceled**. This can occur for one of the following reasons:
 - You stop a pipeline before it completes.
 - There is a failure in one of the nodes in the pipeline. For example, if the pipeline has only one modeling node and it fails, the subsequent Model Comparison and Output nodes show a status of Canceled.

Actions on Nodes in the Pipeline

Right-click any node in a pipeline or click  on a node to open a menu of actions. The actions that appear in the menu depend on the node type and the node state. You can select one of the following actions, when applicable:

- **Add challenger model** — Adds a model as a challenger on the **Pipeline Comparison** tab. See [“Comparing Pipelines” on page 149](#). This action is enabled after the selected modeling node is successfully run, if the model is not already listed on the **Pipeline Comparison** tab.
- **Add child node** — Adds an applicable node after a node in the pipeline.
- **Add parent node** — Inserts an applicable node before a node in the pipeline.
- **Copilot suggestions** — Opens the SAS Viya Copilot side pane. This action is available for the **Data** node and any data preprocessing node. See [Chapter 7, “Using SAS Viya Copilot,” on page 161](#).
- **Copy** — Copies a node to the clipboard.
- **Copy node link** — Copies a link to a node to the clipboard. You can send the link to other users who have access to the project. When the users open the link and sign in, they are directed to the pipeline with the node already selected.
- **Create PDF report** — Exports a node’s results to a PDF file. See [“Create and Export a PDF Report” on page 150](#). This action is enabled after the selected node is successfully run.
- **Delete** — Deletes a node from the pipeline.
- **Download Score Code** — Downloads the score code that was generated by a supervised learning node to a file. This action is enabled after the supervised learning node is successfully run.
- **Force run** — Runs a node that has already completed a successful run. This action might be needed if there was an update applied outside of the SAS Viya environment (for example, a Python or R program). This action was added in release 2026.04.
- **Log** — Displays a log of a node’s processing. This action is enabled after the node is successfully run.
- **Manage Input Variables** — Opens the Manage Input Variables window for a **Scorecard** node.

- **Manage models** — Opens the model that is associated with a **Model Registration** node in SAS Model Manager. This action is enabled after the **Model Registration** node is successfully run and the associated model is registered in SAS Model Manager.
- **Move** — Moves an **Open Source Code**, **SAS Code**, or **Scorecard** miscellaneous node from preprocessing to Supervised Learning, or from Supervised Learning to preprocessing.
- **Open** — Opens the code editor for an **Open Source Code** or **SAS Code** node, or opens the interactive editor for a **Decision Tree** node. For a **Decision Tree** node, this action is enabled after the node is successfully run.
- **Paste** — Pastes a node that was copied to the clipboard into the pipeline. This action is enabled after you copy a node to the clipboard.
- **Publish model** — Publishes a model that is associated with a **Model Registration** node in SAS Model Manager. This action is enabled after the **Model Registration** node is successfully run.
- **Register model** — Registers a model that is associated with a **Model Registration** node in SAS Model Manager. This action is enabled only if the **Model Registration** node is successfully run and the model has not already been registered.
- **Rename** — Renames a node. The new name applies only to the selected node in this pipeline.
- **Results** — Opens the results window for a node. This action is enabled after the selected node is successfully run.
- **Run** — Runs a node. Any preceding nodes in the pipeline that do not have a status of **Successful** are run first. After the preceding nodes complete successfully, the selected node runs.
- **Save as** — Saves a node to The Exchange.
- **Select models** — Adds one or more models to an **Ensemble** node.

Champion Model Badges on Nodes

After a pipeline is run, the node that contains the champion model within the pipeline is indicated by the ★ icon. The node that contains the champion model for the project is indicated by the 🏆 icon. Sometimes this icon does not show when the node image is too small. In this case, you need to enlarge the pipeline image with the mouse scroll wheel.

Starting with 2026.05, the champion badges are also indicated on the **Indicators** line in the champion node tooltip. If the node is the pipeline champion, **Pipeline champion** is displayed on the **Indicators** line. If the node is the project champion, **Project champion** is displayed. Otherwise, **None** is displayed.

Comparing Pipelines

You can use the **Model Comparison** tab to compare pipelines with different models to see which model gives the optimal result. You can even “compare” a single pipeline with itself; this displays the results for the single model. To see a pipeline comparison, you must successfully run a **Model Comparison** node in at least one pipeline. Once you have successfully run a **Model Comparison** node, select the **Pipeline Comparison** tab to see the comparison. The **Pipeline Comparison** tab displays the champion model, the algorithm used, and error statistics. The selection statistic used to determine the champion model is specified in the **Rules** section within the Project Settings window. To edit the project settings, click  in the upper right corner of the window and click **Project settings**.

To see alternate statistics, click , and select **Manage columns**. Use the Manage Columns window to add or remove alternate statistics about the pipelines.

The **Pipeline Comparison** tab also displays the various results of the champion model. For more information about these results, see the **Results** section for the given champion model in the [SAS Viya: Machine Learning Reference Help](#) documentation.

Note: The **Pipeline Comparison** tab compares only the champion models for each pipeline by default. If you have multiple algorithms in a single pipeline, use the **Model Comparison** node to compare the performance of each of these individual models. For more information, see [Overview of Model Comparison](#) in the SAS Model Studio reference documentation. If you want to add models to the **Pipeline Comparison** tab, right-click the model node in your pipeline, and select **Add challenger model**.

Another feature available on the **Pipeline Comparison** tab is the Compare window. The Compare window enables users to examine the various accuracy statistics of each of the pipelines directly. To compare multiple pipelines:

- 1 Select at least two pipelines in the left-most column of the comparison table.
- 2 Click **Compare** above the table. The Compare window appears. The Compare window contains a table of fit, lift, and ROC statistics. The window also contains line graphs of the statistics, comparing the data roles for each pipeline.

Create and Export a PDF Report

To create a PDF report from any node except the Data node after you run a pipeline:

- 1 Right-click the node and select **Create PDF report**. The Export PDF window appears.
- 2 On the **Page Setup** tab, you can specify the following options:
 - **PAGE SETUP**
 - ☐ **Paper size**
 - ☐ **Orientation**
 - ☐ **Margins**
 - **OPTIONS**
 - ☐ **Include table of contents**
 - ☐ **Show page numbers**
 - ☐ **Include title page**
 - ☐ **Include details**
- 3 After specifying options on the **Page Setup** tab, click the **Select Objects** tab. The objects that you can include in your report vary between node types.
- 4 Click **Export** to generate the PDF report.

Insights Tab

You can use the **Insights** tab to see a summary report for a project. To see the report, you must run a **Model Comparison** node in at least one pipeline. Once you have successfully run a **Model Comparison** node, select the **Insights** tab to see the report. The report includes six panels with summary information about the project and champion models.

- **Project Summary** — The Project Summary provides information about the project champion model and the most important input variables.

Additional project level information is provided:

- ☐ **Project Target** — The target variable specified in the project.

- **Overall Average** (interval target) — The average of the target variable values in the project data.
- **Event Percentage** (class target) — The percentage of events in the project data.
- **Pipelines** — The total number of pipelines in the project.
- **Project Champion** — The champion model selected in the project.
- **Created By** — The user ID of the person who originally created the project.
- **Modified** — The date and time of the most recent modification made to the project.
- **Project Notes** — The Project Notes panel enables you to add notes to the project. This can be useful if multiple users are working in the same project. When you click inside the Project Notes panel, a rich-text editor enables you to add headers and text, change text attributes, and add links.

Note: The content of the Project Notes panel cannot exceed 4000 characters.

Note: To open a link in the editor content, press and hold Ctrl, and then click the link.

- **Most Common Variables Selected Across All Models** — A bar chart that shows the number of times that an input variable was determined to be an important variable for any model on the **Pipeline Comparison** tab. This includes the champion model for each pipeline and any challenger models added. Variable importance is calculated using a surrogate model, which is a one-level decision tree for each input variable where the target is the predicted class or value. Input variables with a positive importance value are determined to be important.
- **Assessment for All Models** — A bar chart that shows the values of the selection statistic for all models on the **Pipeline Comparison** tab. The models are grouped by pipeline.
- **Most Important Variables for Champion Model** — A bar chart that shows the important variables for the project champion model. The relative importance is determined from the actual model when the champion is a decision tree, gradient boosting, or forest model. Otherwise, relative importance is calculated using a one-level decision tree for each input variable to predict the predicted value as a global surrogate model.
- **Cumulative Lift for Champion Model** (class target) — A chart that displays the cumulative lift as a function of the depth for the project champion model. The data is sorted by the predicted probabilities and then divided into 20 quantiles where each quantile contains 5% of the data. The number of events in each quantile is calculated. The cumulative lift for a particular quantile is the ratio of the number of events across all quantiles up to and including the current quantile and the number of events that would be in the quantile at random.
- **Actual versus Predicted for Champion Model** (interval target) — A chart that displays the actual mean and the predicted mean of the target variable for the project champion model. The data is sorted by the predicted target values and

then divided into 20 quantiles where each quantile contains 5% of the data. The actual mean and the predicted mean of the target are plotted for each quantile.

You can download the data that is used to build each chart by clicking . A CSV file with the chart data is downloaded.

Several charts have detailed descriptions available that explain how the charts were generated. The descriptions also provide information about how to interpret the charts that is custom to your particular data and models. You can access these detailed descriptions in one of the following ways:

- Expand an individual chart by clicking . The detailed description is in the right panel.
- Click  on an individual chart. A window appears with the detailed description.

Note: The detailed descriptions are available only in the English language.

You can also export your insights report as a PDF:

- 1 Navigate to the **Insights** tab.
- 2 Click  in the upper right corner of the **Insights** tab. The Export PDF window appears.
- 3 On the **Page Setup** tab, specify the following options:
 - **Paper size** — You can specify **Letter**, **Legal**, **Tabloid**, **A3**, or **A4** as the paper size for the PDF. The default value is **Letter**.
 - **Orientation** — You can specify **Portrait** or **Landscape**. The default value is **Portrait**.
 - **Margins** — You can specify vertical (**Top**) and horizontal (**Right**) margins in inches or centimeters.
 - **Options** — Specify which of the following options you want to enable.
 - ☐ **Include table of contents**
 - ☐ **Show page numbers**
 - ☐ **Include title page**
 - ☐ **Include details**

All options are enabled by default.
- 4 On the **Select Objects** tab, specify which of the following objects you want to include in your report:
 - **Project Summary**
 - **Project Notes**
 - **Most Common Variables Selected Across All Models**
 - **Assessment for All Models**
 - **Most Important Variables for Champion Model**
 - **Cumulative Lift for Champion Model**

All objects are selected by default.

- 5 Click **Export** in the lower right corner of the Export PDF window.

The PDF is downloaded to the default directory specified for your browser.

Managing Models

Register Models

Note: If the project data source is currently not available, click **Import data** in the Choose Data window to import the data set. See [“Importing Data” on page 106](#).

You can register a model in SAS Model Manager on the SAS Model Studio **Pipeline Comparison** tab or by using a **Model Registration** node in your pipeline. To register a model on the **Pipeline Comparison** tab:

- 1 Select the model that you want to register, click  in the upper right corner of the tab, and then select **Register models**. The Register Models window appears.
- 2 If you are prompted to specify the repository in which the model is to be registered, select a repository or leave the default, and then click **OK**
- 3 Allow the registration to complete, and then click **Close** to close the Register Models window.

To register a model by using a **Model Registration** node in your pipeline, follow the steps in [“Registering a Model” in SAS Viya: Machine Learning Node Reference](#).

When a model is registered, the name under which the model is registered is created by concatenating the model name and pipeline name as follows:

model-name (pipeline-name)

The maximum length of the concatenated name is 100 characters. To allow for the space and parentheses, the combined length of *model-name* and *pipeline-name* cannot exceed 97 characters. In 2023.07 and prior releases, you cannot register a model if the length of the concatenated name exceeds 100 characters. Starting with 2023.08, if the concatenated name exceeds 100 characters, the name is automatically shortened to 100 characters or less.

If SAS Model Manager is installed on the system, this action registers the model in SAS Model Manager. SAS Model Manager is used to store and organize models in a common model repository. It allows for model governance and model change control over time. To view the selected models in SAS Model Manager, click  in the upper right corner of the **Pipeline Comparison** tab and select **Manage Models**. For more information, see [SAS Model Manager: User's Guide](#).

If the CASHostAccountRequired group has been created, then Read and Write permissions must be configured for the ModelStore caslib and the ModelPerformanceData caslib. This enables members of the CASHostAccountRequired group to register analytic store models. For more information about how to configure permissions, see [File System Directory Permissions](#).

Note: If you update and re-register a model during the same user session, you must refresh the model in **SAS Model Manager** to see the updates.

The Model Repository service is always available, so you can still register models if SAS Model Manager is not licensed. For more information, see [SAS Viya Platform: Models Administration](#).

Publish Models

You might want to publish a model so that the model can be executed in various run-time engines. To publish models, you must first create a publishing destination. The following types of publishing destinations are supported:

- AML
- AZURE
- CAS
- CASFULL
- CASMODEL
- DOCKER
- EVENT_QUE
- GCP
- GIT
- GITCAS
- GITMAS
- HADOOP
- INDATABASE
- LOCAL_AZURE
- MAS_SHORT
- OTHER
- SAS Micro Analytic Service
- TERADATA

Information about how to configure publishing destinations can be found in [SAS Viya Platform: Publishing Destinations](#).

In order to publish analytic stores to SAS Micro Analytic Service, a separate location must be created to store the analytic stores. For more information, see [Access to Analytic Store Model Files](#).

If the CASHostAccountRequired group has been created, then Read and Write permissions must be configured for the ModelStore caslib and the ModelPerformanceData caslib. This enables members of the CASHostAccountRequired group to publish analytic store models. For more information about how to configure permissions, see [File System Directory Permissions](#).

Note: You cannot publish to SAS Micro Analytic Service if you are using user-defined formats.

Once you have created a publishing destination, you can publish the model on the **Pipeline Comparison** tab or by using a **Model Registration** node in your pipeline.

To publish the model:

- 1 Do one of the following based on how you are publishing your model:
 - If you are publishing your model on the **Pipeline Comparison** tab, select the model that you want to publish, click  in the upper right corner of the **Pipeline Comparison** tab, and then select **Publish models**.

Note: You cannot publish more than one model at a time.

- If you are publishing your model by using a **Model Registration** node, right-click the **Model Registration** node, and then select **Publish model** in the node menu.

Note: **Publish model** is enabled only if the **Model Registration** node has been successfully run. Otherwise, it is disabled.

- 2 The Publish Models window appears. Select the model publish destination that you created.

Publish Models

Destination:
CAS (AACASDestination) [Details](#)

Tags:
Press Enter after each tag.

ITEMS TO PUBLISH

Forest

Published name:
Forest_5b4f2464d43144a6869ba53bae60c4c9

☐ Replace item with the same name

Publish **Cancel**

- 3 Click **Publish**. The model is published immediately.

When you publish a model, that model is also automatically registered to the Model Repository service.

Note: If you are publishing a model to CAS, you must publish the model to the same CAS server where your data is located. Copying data from one CAS server to another is not supported.

Export Models for Production

Some models in SAS Viya Machine Learning are packaged in a single downloadable DATA step code file. Other models are packaged in two parts, score code and a binary file, for efficiency.

Models can be exported from both the **Pipelines** tab and the **Pipeline Comparison** tab.

- On the **Pipelines** tab, select the pipeline that contains your target model. Right-click the champion node, and select **Download Score Code**.
- On the **Pipeline Comparison** tab, select one pipeline, click  in the upper right corner of the **Pipeline Comparison** tab, and then select **Download Score Code**.

Both methods listed above download a ZIP file to the client that contains the model score code. The model score code contains the code generated by the supervised learning node, as well as any data mining preprocessing nodes preceding it.

For models packaged in two parts, the ZIP file contains DS2 code and the analytic stores are saved in the MODELS caslib. The DS2, or DATA step 2, code is also referred to as analytic store code or EP score code. The analytic store score code is a representation of the model pipeline including any pre- or post-processing steps. The second part of the two-part model is the analytical store. Each analytic store is a binary file that contains the state of an analytic procedure after training. There can be multiple analytic stores in a single model. Together, the analytic store code and the analytic store represent your model, and can be used to score new data. The **Download Score Code** action saves the analytic stores in the MODELS caslib. In addition, the analytic store score code is compressed and downloaded to the client. The analytic store score code contains a comment that specifies names of binary files. These binary files are required for use with the score code, and can be found in the MODELS caslib.

You can identify the path associated with the MODELS caslib by using SAS Environment Manager. For more information, see [SAS Viya Administration: Using SAS Environment Manager](#).

In the CAS server's file system, navigate to the location of the MODELS caslib to find the analytic stores for your model. Each analytic store specified in the analytic store score code is in the MODELS directory with an extension of SASHDAT.

Score Holdout Data

You can score holdout data for all of the applicable models that are listed in the **Name** column on the **Pipeline Comparison** tab.

Note: Holdout data scoring is not supported for an **Open Source Code** node that is the supervised learning node or for any model with an upstream **Open Source Code** node that generates scoring code.

To score the holdout data:

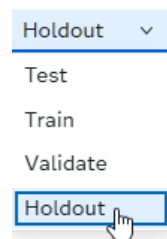
- 1 Click  in the upper right corner of the **Pipeline Comparison** tab.
- 2 Select **Score holdout data**.

Note: If there are no applicable models in the pipeline comparison, **Score holdout data** is disabled.

Note: The holdout data is scored for all of the applicable models regardless of whether they are selected.

- 3 The Choose Data window appears. Select the data set that contains the holdout data that you want scored.
- 4 Click **OK**. The Score Holdout Data window is displayed, and SAS Model Studio begins scoring the holdout data.
- 5 When the process is complete, click **Close**. The holdout data scoring results are displayed.

To see the results of this process at a later time, use the **Data** menu below **Pipeline Comparison** to select **Holdout**.



Download Score API

To download the model API:

- 1 Click **:** in the upper right corner of the **Pipeline Comparison** tab.
- 2 Select **Download score API**. The Scoring API window appears.
- 3 Select the **Download Type**. The following choices are available:
 - **SAS** — Downloads code to run in a SAS programming environment.
 - **Python** — Downloads code to run in a Python programming environment.
 - **REST** — Downloads a text file with example REST calls that you can use in an application.
- 4 After selecting the **Download type**, click **Download**. The model API downloads immediately.

Downloading Logs

To download project logs:

- 1 In the project, click  in the upper right corner (this is accessible on the Data, Pipelines, and **Pipeline Comparison** tabs).
- 2 Select **Project logs**. You can then select the **Project Partitioning** logs, the **Project Data Advisor** log, or the **Scoring and Assessing** model logs. For pipelines that have not yet run, only the **Project Data Advisor** log is available.
- 3 Once the log is selected, click **Download log**. The log downloads immediately, saved as a TXT file.

Using SAS Viya Copilot

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About SAS Viya Copilot for SAS Viya Machine Learning

What is SAS Viya Copilot for SAS Viya Machine Learning?

SAS Viya Copilot is a software as a service (SaaS), AI-powered assistant that uses large language models (LLMs) in order to provide a more intuitive way to work with SAS Viya products. SAS Viya Copilot for SAS Viya Machine Learning is an assistant that is tailored for users who are building machine learning model pipelines. It is available starting with 2025.12. It uses an LLM that has access to the official SAS

documentation for SAS Viya Machine Learning to help ensure that the responses are relevant and accurate for SAS Viya Machine Learning users.

For an overview of SAS Viya Copilot, see [SAS Viya Copilot: Foundation Overview](#). SAS Viya Copilot is not enabled by default. It must be enabled by your SAS Viya administrator in SAS Environment Manager. See “[Enable SAS Viya Copilot in SAS Environment Manager](#)” in [SAS Viya Copilot: Foundation Overview](#). When SAS Viya Copilot is enabled, all of its capabilities are available for your use.

How SAS Viya Copilot Can Assist You

There are many ways in which SAS Viya Copilot can assist you as you develop your pipelines in SAS Viya Machine Learning. Here are some of the common tasks with which it can provide assistance:

- getting specific information about SAS Viya Machine Learning
- adding a specific node to your pipeline
- determining the next node or nodes to add to your pipeline from the **Data** node or a data preprocessing node
- running a node
- getting information about a node, its parameters, or the algorithms that it uses
- improving nodes in your pipeline
- getting an analysis or summary of node results

IMPORTANT It is your responsibility to review any output generated with AI to ensure that the AI-generated content that you select is appropriate and accurate for its intended purpose.

For more information, see “[Getting Assistance from SAS Viya Copilot](#)” on page 165.

Third-Party Models

Model Information

Model Overview

For a list of the Microsoft Azure models that are used by SAS Viya Copilot, see “Models Used by SAS Viya Copilot” in [SAS Viya Copilot: Foundation Overview](#).

Input and Output Modalities

Input modalities and size limit for each modality as used by SAS:

- gpt-5.1: Text, 272,000 tokens
- gpt-5-nano: Text, 272,000 tokens
- gpt-4.1: Text, 1,047,576 tokens
- gpt-4.1-nano: Text, 1,047,576 tokens
- text-embedding-3-large: Text, 8,192 tokens

Output modalities and size limit for each modality as used by SAS:

- gpt-5.1: Text, 128,000 tokens
- gpt-5-nano: Text, 128,000 tokens
- gpt-4.1: Text, 32,768 tokens
- gpt-4.1-nano: Text, 32,768 tokens
- text-embedding-3-large: Embedded, Fixed-size vector (1536 dimensions)

Note: Because embedding models return a fixed-dimension vector rather than text tokens, there is no published output token size.

Prerequisites

- SAS Viya Machine Learning, version 2025.12 or later.

Using the Model

- Acceptable use policy: <https://www.sas.com/acceptableuse>.
- Intended use: This feature is intended to support the development of machine learning pipelines. It includes natural language processing capabilities for user interface interactions (for example., adding nodes and executing individual nodes or entire pipelines), AI-driven recommendations for pipeline and project enhancements—including suggested next steps and optimization of existing components—and comprehensive explanations of nodes, configuration options, and output results.

Type and Nature of AI Systems

The models support features in SAS software in a specific context but cannot be more broadly incorporated into customer AI systems.

Best Practices

- Users interact with the system through a conversational interface to assist in the development of machine learning pipelines. Through natural language input,

users can describe user interface actions such as adding nodes, executing individual nodes, or executing entire pipelines.

- The assistant can provide AI-generated recommendations for pipeline and project development, including suggested next steps and enhancements to existing components.
- Users can request generative AI-based explanations of nodes, configuration options, and resulting outputs.
- When conversing with the assistant, you can follow these best practices for writing your prompts:
 - Be as clear and specific as possible in the prompts.
 - Be intentional with the tone of your prompts because the AI-generated suggestions often match the tone of the original prompt.
 - Ensure that your prompts are safe and respectful.
 - Enter your prompts only in natural language English.
 - Avoid words or phrases in your prompts that are culturally specific, such as slang or idioms that are not easily understood by others.
 - If you are not getting good results, try rephrasing the prompt.
 - If you need more specific information, ask follow-up questions.

Known Issues or Limitations

- SAS does not guarantee that the AI responses are correct, or that the generated suggestions are optimal.
- LLMs are known to hallucinate.
- SAS Viya Copilot supports only English inputs and output.
- The Auto Machine Learning exploratory and node improvement processes are not exhaustive and might not produce optimal results.

SAS Viya Copilot Limitations

You can find a list of the SAS Viya Copilot limitations in [“SAS Viya Copilot Limitations” in SAS Viya Copilot: Foundation Overview](#).

How to Open the SAS Viya Copilot Side Pane

To open the SAS Viya Copilot side pane in SAS Viya Machine Learning, in the upper right corner of the window, click . The SAS Viya Copilot side pane appears.

Note: The  icon appears only when SAS Viya Copilot is enabled. If the icon does not appear, contact your SAS Viya administrator.

Note: Currently, SAS Viya Copilot for Visual Machine Learning supports natural-language English only. When you open the SAS Viya Copilot side pane, it appears in English regardless of your browser's locale setting.

Getting Assistance from SAS Viya Copilot

Submitting Your Prompts to SAS Viya Copilot

At the bottom of the SAS Viya Copilot side pane is a text field in which you enter your prompts:

➤

AI-generated content. Evaluate accuracy and suitability before use.

To submit a prompt, enter your prompt in the text field, and then click ➤ or press Enter. To get the best results, ensure that your prompts conform to the best practices listed in [“Best Practices” on page 163](#).

Note: SAS Viya Copilot supports English-only prompts at this time.

IMPORTANT It is your responsibility to review any output generated with AI to ensure that the AI-generated content that you select is appropriate and accurate for its intended purpose.

During your conversation, SAS Viya Copilot maintains a history of your prompts and its responses. You can use the scroll bar to review your conversation history.

TIP To clear your current conversation with SAS Viya Copilot and start a new one, click  in the top right corner of the SAS Viya Copilot side pane.

Getting Answers to General Questions

If you have general questions about SAS Viya Machine Learning or the nodes in your pipeline, you can prompt SAS Viya Copilot for an answer. These questions typically do not apply to a specific node in the pipeline. Enter your question in the text field, and then click  or press Enter. Here are some sample questions:

- What is a predictive model?
- What is autotuning?
- What is a target variable?

The questions must be within the scope of SAS Viya Machine Learning. The response includes summarized information related to your question. It might also include one or more links to more detailed documentation for your reference.

Adding a Specific Node to Your Pipeline

As you build your pipeline, if you know the node that you want to add next, SAS Viya Copilot can add it for you. If you know the purpose of the next node but are unsure about the node name, SAS Viya Copilot might be able to add an appropriate node for you.

Note: SAS Viya Copilot can add a node after an existing node in your pipeline. Currently, it does not support inserting a node before an existing node.

Note: A node cannot be added to a pipeline that is locked such as an MLPA pipeline. See [“About Automatically Generated Pipelines” on page 141](#).

In order for SAS Viya Copilot to add a node, the parent node must be selected or identified by name in the prompt, and the node that is to be added must be a valid

child of the parent node. If these conditions are true, enter your prompt, and then click ➤ or press Enter. Here are some sample prompts:

- For a node that is selected:
 - ☐ Add a Linear Regression node.
 - ☐ Add an Imputation node.
 - ☐ Add a node that reduces the number of inputs.
 - ☐ Add a node that fixes skewed data.
- For a node that is identified by name:
 - ☐ Add a Gradient Boosting node to the Data node.
 - ☐ Add a SAS Code node after the Gradient Boosting node.
 - ☐ Add an SVM node after the Transformations (1) node.

After you submit your prompt, SAS Viya Copilot adds the node to the pipeline. A response similar to the following appears in the side pane:

A new Linear Regression node "Linear Regression" was added after node "Data" in pipeline "Pipeline 1" with default properties.

The response includes a **Run node** button that you can click to run the newly added node. Alternatively, you can prompt SAS Viya Copilot to run the node as described in ["Running a Node" on page 172](#).

If you want to add a node and run it immediately afterward, you can combine both actions into a single prompt. For example, when the **Data** node is selected, this sample prompt adds a **Linear Regression** node after the **Data** node, and then runs it:

Add a Linear Regression node and run it.

After you submit the prompt, a response similar to the following appears in the side pane:

A new Linear Regression node "Linear Regression" was added after the Data node in Pipeline 1 with default properties. Now the run process for the Linear Regression node is starting. The node "Linear Regression" in the pipeline "Pipeline 1" has been started. Check the pipeline for status.

Getting Suggestions for the Next Node or Nodes to Add

About the Node Suggestion Capability

SAS Viya Copilot can provide suggestions for the next node to add after an applicable node (parent node). Applicable nodes are the **Data** node and all of the data preprocessing nodes except for risk modeling nodes **Pairwise Selection**, **Pricing Optimization**, and **Reject Inference**. If you are not sure which node to add after one of the applicable nodes, the node suggestion capability is very helpful.

In order for SAS Viya Copilot to suggest a node to add after an applicable node (parent node), the following conditions must be true:

- The pipeline must be unlocked. An MLPA pipeline might be in a locked state. See [“About Automatically Generated Pipelines” on page 141](#).
- The pipeline must not be read-only.
- The parent node must be one of the applicable nodes.
- The parent node must have already been successfully run.
- The parent node must be currently selected or identified by name in your prompt.

When these conditions are true, the SAS Viya Copilot node suggestion capability is enabled for the parent node that is selected or identified by name. Otherwise, it is disabled.

To get suggestions for the next node, you have the following choices:

- You can prompt SAS Viya Copilot to suggest the next node to add after an applicable parent node or you can right-click the applicable parent node and select **Copilot suggestions**.

Note: **Copilot suggestions** is available only when the conditions listed previously are true.

In either case, SAS Viya Copilot provides initial suggestions that are based on the node metadata. The parameters for the suggested nodes assume their default values. No tuning is performed. See [“Getting Initial Suggestions” on page 169](#).

- You can prompt SAS Viya Copilot to run the exploratory process in order to find a better node or series of connected nodes to add next based on your request. In this case, it trains several exploratory models in order to find the most suitable child node or nodes. Node parameter tuning might be performed during the exploratory process in order to improve performance, so some of the parameter

values for the suggested nodes might differ from their default values. See [“Getting Better Suggestions by Using the Exploratory Process” on page 169](#).

Getting Initial Suggestions

When you select the parent node and open the SAS Viya Copilot side pane, the button **Suggest the next node** appears in the side pane. You can click this button to get initial suggestions for the next node. You can also prompt SAS Viya Copilot to provide initial suggestions. In order to prompt for initial suggestions, you must select the parent node before you submit your prompt, or you must identify the parent node by name in your prompt. Enter your prompt, and then click ➤ or press Enter. Here are some sample prompts for initial suggestions:

- For a node that is selected:
 - ☐ Suggest the next node.
 - ☐ What node should I add next?
 - ☐ What do I do next?
 - ☐ What nodes can I add?
- For a node that is identified by name:
 - ☐ What node should I add after the Imputation node?
 - ☐ Suggest a node to add after the Transformations (1) node.

In its response, SAS Viya Copilot provides a list of possible nodes that you can add next. The list of suggested nodes can include modeling nodes, data preprocessing nodes, or both. In the response, a button is provided for each suggested node that you can click to add the node of your choice to your pipeline. You can add only one of the suggested nodes.

Note:

In order to add suggested nodes to your pipeline, you must be viewing the pipeline for which the suggestions were requested.

Click the button for the node that you want to add. When you click a button, that button is disabled, the remaining buttons are removed from the response, and the selected node is added to your pipeline. A **Run node** button is then provided in the response that you can click to run the newly added node.

Getting Better Suggestions by Using the Exploratory Process

The SAS Viya Copilot exploratory process can provide better suggestions. It involves running a series of trials during which various models are trained in order to determine the best child node or nodes to add next. During the exploratory

process, tuning might be performed in order to maximize performance of some models. The selection criteria for the project is used to determine model performance. The trial models are compared and ranked as the trials are completed. At the end of the process, the best models are presented as the suggested child nodes.

Note: SAS Viya Copilot cannot run the exploratory process on a locked pipeline such as an MLPA pipeline. See [“Automated Pipeline Creation” on page 137](#).

When you prompt SAS Viya Copilot for initial suggestions, it might include a **Start exploration** button in its response. If you are not satisfied with the initial suggestions, you can click this button to start the exploratory process to get better suggestions. Alternatively, you can skip the initial suggestions and prompt SAS Viya Copilot to run the exploratory process directly. In order to initiate the exploration process, you must select the parent node before you submit your prompt, or you must identify the parent node by name in your prompt. Here are some sample prompts for initiating the exploratory process:

- For a node that is selected:
 - ☐ Run the exploratory process.
 - ☐ Generate some automated models.
 - ☐ Start a model analysis.
- For a node that is identified by name:
 - ☐ Run the exploratory process on the Imputation (1) node.
 - ☐ Start a model analysis on the Transformations node.

Once the exploratory process is initiated, it runs several trials, which might take a few minutes to complete.

Note: You cannot interact with the pipeline while SAS Viya Copilot is running the exploratory process. However, you can interact with a different pipeline or create and interact with a new pipeline during this time.

As the process runs, the current results are displayed in the SAS Viya Copilot side pane. The results are updated periodically. The results list the suggestions that have been tried so far and the model comparison selection statistic for each trial model for the validation partition, if it exists. The model comparison statistic is specified in **Project settings** under **Rules**. (See [“Specifying Project Settings” on page 97](#).) If a validation partition does not exist, then training is used instead. If data sampling was used on the input data, that is also mentioned in the results. Here is an example of the suggestions and the model comparison statistics for each that are displayed as the exploratory process runs:

After running 16 exploratory trials, here are the best suggestions found so far:

- Gradient Boosting (ASE: 0.068)
- Forest (ASE: 0.081)
- Variable Selection, Feature Machine and GAM (ASE: 0.086)

Better suggestions might be found while the exploratory trials are run.

Running exploratory trials...

Stop running

In this example, the model comparison interval selection statistic average squared error (ASE) is displayed for each suggestion. The suggestions resulting from the exploratory process can include a modeling node only, or one or more data preprocessing nodes and a modeling node.

In the SAS Viya Copilot side pane, a **Stop running** button is provided that you can click to stop the process if the current results are satisfactory. Otherwise, you can let the process run to completion. When the process is stopped or completes, the final model comparison statistics are displayed in the side pane. Buttons are included that you can click to add a suggested node or a series of connected nodes.

Note:

In order to add suggested nodes to your pipeline, you must be viewing the pipeline for which the exploratory process was run.

You can click only one of the buttons. When you click a button, the buttons are removed from the response, and the suggested node or nodes are added to your pipeline.

Note: Some of the parameter values for the suggested nodes might have been changed from their default value as part of tuning that was performed during the exploratory process. When you add a suggested node, the changed parameter values are displayed in the side pane.

A **Run nodes** button is provided in the response that you can click to run the newly added node or nodes.

Getting Information about a Node

If you need more information about a node, you can prompt SAS Viya Copilot to provide a description of the node. When you select a node and open the SAS Viya Copilot side pane, a button is provided in the side pane that you can click to get information about the selected node. You can also prompt SAS Viya Copilot for information about any node in your pipeline. In order to prompt for information about a node, you must select the node before you submit your prompt, or you must identify it by name in your prompt. To submit your prompt, click ➤ or press Enter. Here are some sample prompts:

- For a node that is selected:
 - ☐ What is this node?
 - ☐ How is this node used?
- For a node that is identified by name:
 - ☐ What is the Linear Regression node?
 - ☐ How is the Imputation (1) node used?

SAS Viya Copilot provides a description of the selected node in the side pane. The description might also include links to more detailed documentation for your reference.

Getting Information about Node Parameters

As you adjust the parameters of the nodes in your pipeline, if you need information about a particular parameter and its possible values, you can prompt SAS Viya Copilot for information. To get the information, select the node, enter your prompt for information, and then click ➤ or press Enter. Here are some sample prompts:

- What does the Kernel property do?
- What values can I use for Penalty?
- What is LightGBM?

SAS Viya Copilot displays information about the parameter in the side pane. The information might include one or more links to more detailed documentation for your reference.

Running a Node

When SAS Viya Copilot adds one or more nodes to your pipeline, it provides a button in the side pane that you can click to run the newly added node or nodes. If

you need to run a node in your pipeline in other situations, you can prompt SAS Viya Copilot to run it for you. In order to prompt SAS Viya Copilot to run a node, you must select the node before you submit your prompt, or you must identify it by name in your prompt. Enter your prompt to run the node, and then click ► or press Enter. Here are some sample prompts:

- For a node that is selected:
 - ☐ Run node.
 - ☐ Run this node.
- For a node that is identified by name:
 - ☐ Run the Linear Regression (2) node.
 - ☐ Run Decision Tree.

Here is an example of the response in the SAS Viya Copilot side pane:

```
The node "Linear Regression (2)" in the pipeline "Pipeline 1" has been started.
Check the pipeline for status.
```

SAS Viya Copilot indicates in its response the name of the node that is running and the name of the pipeline that contains the node. Any preceding nodes that have not been run are run before the selected or specified node is run. SAS Viya Copilot does not indicate whether the node ran successfully. You must monitor the status of the node or nodes in the pipeline as they run to determine whether they run successfully.

Getting an Analysis or Summary of Node Results

After you run a node, you can prompt SAS Viya Copilot to provide an analysis or summary of the node results. To get the analysis, you must first select the node or you must identify the node by name in your prompt. Enter your prompt, and then click ► or press Enter. Here are some sample prompts for getting an analysis of a selected node:

- For a node that is selected:
 - ☐ Explain the results.
 - ☐ What do the results mean?
 - ☐ Describe this node.
- For a node that is identified by name:
 - Describe the Decision Tree (2) node results.

Note: If the selected or named node has not been run, SAS Viya Copilot indicates that you must run the node first. It provides a **Run node** button in the side pane that you can click to run it.

SAS Viya Copilot provides an analysis or summary of the results in the side pane. It includes a high-level overview of the modeling node or data preprocessing node, an

analysis of the key findings, and a conclusion. Additional information is provided based on the model type and node parameter settings.

Improving the Nodes in Your Pipeline

SAS Viya Copilot can help you improve certain nodes in your pipeline. This section describes the improvement process and how to prompt SAS Viya Copilot to run it on a node in your pipeline.

Nodes That Can Be Improved

The improvement process does not support all of the available nodes. The following table lists the nodes that are supported.

Table 7.1 *Nodes That Support the Improvement Process*

Node Type	Node
Data Mining Preprocessing	Feature Machine
	Imputation
	Replacement
	Text Mining
	Transformations
	Variable Selection
Supervised Learning	Bayesian Network
	Decision Tree
	Factorization Machine
	Forest
	GAM
	Gaussian Process Classification
	Gaussian Process Regression
	GLM

Node Type	Node
	Gradient Boosting (including LightGBM)
	Linear Regression
	Logistic Regression
	Neural Network
	Quantile Regression
	SVM

Note: If you prompt SAS Viya Copilot to improve a node that does not support the improvement process, it tells you that the improvement process does not support that node.

For the nodes listed in [Table 7.1 on page 174](#), the following exceptions apply:

- The improvement process cannot run on a **Decision Tree** node that is in interactive mode. In order to run the improvement process on a **Decision Tree** node in interactive mode, you must enable properties for the node by right-clicking it and selecting **Enable Properties**, and then rerun the node. However, this action clears the interactive edits that were made to the node. For more information, see [“Interactively Edit a Decision Tree” in SAS Viya: Machine Learning Node Reference](#).
- The improvement process cannot run on a node in a pipeline that was created from a SAS Visual Analytics report. (See [“Integrating SAS Model Studio with SAS Visual Analytics” on page 37](#).) In order to run the improvement process on a node in the pipeline in that case, you must enable properties for the node by right-clicking it and selecting **Enable Properties**, and then rerun the node. However, this action causes the model to retrain, which changes the results.
- The improvement process cannot run on a node in a pipeline that is locked. A pipeline that is created using MLPA is initially locked and identified as an automatically generated pipeline. In order to run the improvement process on a node in a locked pipeline, you must unlock the pipeline. However, unlocking the pipeline causes it to no longer be identified as an automatically generated pipeline. For more information, see [“Automated Pipeline Creation” on page 137](#).

How SAS Viya Copilot Improves a Node

The node improvement process is an iterative process that attempts to find the node parameter values that maximize the performance of the selected node. Each iteration is a trial of tuned parameter values. For a modeling node, in each trial, the

node's hyperparameter values are modified as determined by a tuning algorithm, and the model performance is measured.

For a data preprocessing node, in order to run the improvement process, a modeling node must exist downstream of the data preprocessing node in the pipeline. In each trial, the data preprocessing node's parameters are modified as determined by the tuning algorithm, and the performance of a downstream modeling node is measured. If there are multiple downstream modeling nodes in the pipeline, the improvement process selects a downstream modeling node to use. It indicates in the process output which modeling node it is using.

Note: The selected modeling node might change as the trials run.

For both modeling nodes and data preprocessing nodes, the performance of the modeling node is measured against the input data validation partition, if it exists. If a validation partition does not exist, training is used instead. Each trial is scored using the project model comparison selection statistic that is available in the modeling node results. The trials are then sorted according to the model assessment measure.

When the process completes or you click **Stop running**, if performance improvements were observed during the trials, the parameter settings that were used in the best trial are displayed in the response. An **Apply improvements** button is provided in the response that you can click to apply the modified parameter settings to the selected node.

How to Prompt SAS Viya Copilot to Run the Improvement Process

You can prompt SAS Viya Copilot to run the improvement process on a node if the following conditions are true:

- The node supports the improvement process. See [Table 7.1 on page 174](#).
- The node has been successfully run.
- The node is selected or is identified by name in the prompt.

If these conditions are true, enter your prompt, and then click ➤ or press Enter. Here are some sample prompts:

- For a node that is selected:
 - ☐ Improve this node.
 - ☐ Improve the node.
 - ☐ How can I improve this node?
- For a node that is identified by name:
 - Improve the Linear Regression (2) node.

When the prompt is submitted, SAS Viya Copilot initiates the improvement process. The process might take a few minutes to complete.

Note: You cannot interact with the pipeline while SAS Viya Copilot is running the node-improvement process. However, you can interact with a different pipeline or create and interact with a new pipeline during this time.

As the process runs, the progress is displayed in the side pane. If you want to stop the process, click **Stop running**. When the process completes or you click **Stop running**, if SAS Viya Copilot was able to improve the node, it provides details about how the improvement was achieved. If input data sampling was used, it might indicate that as well. It also provides the button **Apply improvements** that you can click to the apply the recommended changes to the node.

Note:

In order to apply the improvements, you must be viewing the pipeline for which the improvement process was run.

Troubleshooting

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Enable Debug Reporting

About Debug Reporting

By default, node logs summarize content in order to save space. To troubleshoot a problem, you might need to enable debug reporting. You can enable debug reporting for all new projects or you can enable debug reporting only for an existing project.

Enable Debug Reporting for All New Projects

To enable debug reporting for all new projects:

- 1 In the upper right corner of the window, click the user name, and then select **Settings**.
- 2 Under **Logging**, select **Enable debug reporting**.

- 3 Click **Close**. Debug reporting is now enabled for new projects.

The settings that you specify apply only to projects that you create after you enable debug reporting. The debug reporting settings for existing projects are not affected.

Enable Debug Reporting for an Existing Project

To enable debug reporting for an existing project:

- 1 Open the project, click Settings (⚙️) in the upper right corner of the window under the user name button, and then click **Project settings**.
- 2 Under **Logging**, select **Enable debug reporting**.
- 3 Click **Close**. Debug reporting is now enabled for the project.

Note: In order to enable debug reporting for nodes in an existing pipeline, you must complete the above steps and then rerun the pipeline. If you do not rerun the pipeline, debug reporting is not enabled.

View the Additional Information in the Node Log

After you enable debug reporting, you can view the additional information in the node log:

- 1 Right-click a node and select **Log**. When you view the node log with debug reporting enabled, a Log Size Warning window is displayed.
- 2 Click **Download log**, and then use a text editor to view the contents. Additional details that can help you correct problems are displayed in the log.

Troubleshooting

Pipelines Take a Long Time to Complete

Description

There are several factors that can influence the performance of a pipeline run. In some circumstances, if there are a lot of users running pipelines at the same time, node execution time can be affected. By default, a maximum of five users can run the modeling nodes simultaneously. For any additional users, the modeling nodes are placed in **Pending** state until a run completes and another node can be started.

Solution

To change the number of users that can concurrently run pipelines, the administrator can update the **Maximum Current Nodes** configuration property. The instructions for updating this property depend on the level of the SAS Viya platform that you are using.

- If you are using any release prior to 2022.11, follow these instructions: [“SAS Viya Platform 2022.10 or Prior Release” on page 181.](#)
- If you are using 2022.11 or later release, follow these instructions: [“SAS Viya Platform 2022.11 or Later Release ” on page 182.](#)

SAS Viya Platform 2022.10 or Prior Release

- 1 Click  in the upper left corner and select **Manage Environment**.
This opens SAS Environment Manager.
- 2 In the left side toolbar, click .
- 3 On the left side, click the **View** drop-down list and select **All services**.
- 4 In the list of services, select **Advanced Analytics Flows service**. You can find it faster by entering the service name in the **Filter** field.
On the right side of the page is a list of instances used to configure the service.
- 5 Scroll down to the sas.analytics.flows instance and click .

- 6 In the Edit `sas.analytics.flows` Configuration window, scroll down to **Maximum Concurrent Nodes**.

This is set to 5 by default. If you set this too high, many of the platform services can become highly stressed and fail. Start with setting this slightly higher, perhaps 6 or 7. If you do not see a significant impact on node execution time, you can repeat these steps to raise it a little higher.

- 7 Click **Save**.

SAS Viya Platform 2022.11 or Later Release

- 1 Click  in the upper left corner and select **Manage Environment**.

This opens SAS Environment Manager.

- 2 In the left side toolbar, click .

The Configuration page opens.

- 3 On the left side, click the **View** drop-down list and select **All services**.

- 4 In the list of services, select **Analytics Execution service**. You can find it faster by entering the service name in the **Filter** field.

On the right side of the page is a list of configuration instances used to configure the service.

- 5 Click .

If the `sas.analytics.execution` configuration instance is listed, click  to the right and proceed to step 6. Otherwise, follow these steps.

- a In the New `sas.analytics.execution` Configuration window, click the **New Configuration**.
- b In the Select Definition window, select `sas.analytics.execution` to create the configuration instance.

- 6 Scroll to Maximum Concurrent Nodes property.

This is set to 10 by default. If you set this too high, many of the platform services can become highly stressed and fail. Start with setting this slightly higher, perhaps 12 or 14. If you do not see a significant impact on node execution time, you can repeat these steps to raise it a little higher.

- 7 Click **Save**.

Cannot Open Project When Source Data Tables Are Not Loaded

Description

You attempt to open a project and receive this error message:

```
The project cannot be opened because the required data tables for the project are not loaded. Contact your system administrator.
```

This issue often occurs for projects that are shared as read-only. If the CAS server has been restarted, the source data tables cannot be reloaded for projects that are read-only.

Solution

The project owner can remove the read-only setting from the SAS Model Studio project page and then open the project. The source data tables are reloaded and the read-only setting can then be applied back to the project.

Project Cannot Be Created or Pipeline Cannot Be Opened

Description

A project cannot be created, or a pipeline cannot be opened in a project. Both of these issues can be caused by a corrupted pipeline template. This problem often occurs when the system is updated to stable release 2022.1.3.

Solution

The cluster (Kubernetes) administrator must examine the logs and restart the service as described below.

Check the sas-analytics-services logs for an error containing `com.sas.analytics.pipeline.services.InconsistentPipelineException`. The following is an example of this exception.

```
{ "version":1, "timeStamp":"2023-01-13T13:37:58.054Z", "level":"error",
  "source":"sas-analytics-services",
  "message":"The request to list pipelines failed for the query arguments
    \QueryOptions [paging=PagingParameters [start=0, limit=30] ,
    params={filter=eq($identical,id,\"default.textanalyticspipeline.template\")}}]\".\n
  com.sas.analytics.pipeline.services.InconsistentPipelineException:
    A component specified by flow connection link relation \"prev\"
    and URI [/analyticsComponents/components/5d0fcd58-b506-490f-844d-74756b77f169]
    was not found in the pipeline swim lanes.",
  "properties":{"logger":"com.sas.analytics.commons.utils.ExceptionMapper",
  "thread":"https-jsse-nio-0.0.0.0-8086-exec-19",
  "traceId":"7027b2a551fcef3f",
  "spanId":"fa85bc3bda844705",
  "__session":"dc885b74-65af-49ca-a697-e31c941517a7",
  "application":"analyticsPipelines",
  "__tenant":"intech",
  "username":"user1"}}}
```

Restart the service based on the template identified in the error message. The service could be one of these three:

- sas-text-analytics
- sas-data-mining
- sas-forecasting-pipelines

The message above identifies the `default.textanalyticspipeline.template`, so `sas-text-analytics` must be restarted in this case.

Projects Are Missing from the Project List

Description

When viewing the list of projects in SAS Model Studio, a shared project does not show up consistently in the list. The initial view does not show the project. When the screen is refreshed repeatedly, the project might appear in the list intermittently. However, even when the project appears in the list, attempts to open the project can result in an error.

This issue has been observed for these releases.

- 2024.08
- 2024.09
- 2024.10

The fix is available in subsequent releases.

Solution

The SAS administrator must follow these steps.

- 1 Update the deployment to obtain the fix for this issue.
- 2 Run the following command to obtain the access token for the user.


```
curl --location '<host-machine>:443/SASLogon/oauth/token' \
--header 'Authorization: Basic c2FzLmVjOg==' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Origin: <host-machine>:443' \
--data 'grant_type=password&username=<username>&password=<password>'
```
- 3 Re-index the SAS Model Studio projects, specifying the access token from the previous command.

```
curl --location --request POST '<host-machine>/searchIndex/index/typeJobs/ModelStudioProject' \
--header 'Authorization: Bearer <access-token>' \
--header 'Accept: application/vnd.sas.search.index.feed+json' \
--header 'Origin: <host-machine>:443'
```

Open Source Code Node Fails with JAVAOBJ Error

Description

The **Open Source Code** node fails to run and writes the following error to the log:

```
ERROR: JAVAOBJ cannot be used when SAS is in the lockdown state.
```

Starting in 2025.08, when the Compute server is in LOCKDOWN mode, the JAVAOBJ access method is blocked by default. When it is blocked, the **Open Source Code** fails to run, and the JAVAOBJ error shown previously appears in the log. The JAVAOBJ access method must be enabled in order to run the **Open Source Code** node.

Solution

Add the JAVAOBJ access method to the enabled list by following the instructions in the README file located at `$deploy/sas-bases/examples/sas-programming-environment/lockdown/README.md` (for Markdown format) or at `$deploy/sas-bases/docs/lockdown_settings_for_the_sas_programming_environment.htm` (for HTML format). Ensure that you modify the enable-lockdown-access-

methods.yaml file and also that you add a reference to that file in the transformers block of the base kustomization.yaml file (`$deploy/kustomization.yaml`).